Exercise list — Topic 5: Nonverifiable income, strategic default, termination, debt structure

Worked solutions

Statements are in `Listas/lista-topic5-nonverifiable-income-termination.md` (and its PDF); each exercise is cited by number and printed page and its full text is reproduced right before its solution. Exercises 14–17 are original exercises on the sovereign-debt model (Topic 5 slides 13–29), for which no course book has an exercise; they are marked "Original exercise". Short answers: `lista-topic5-nonverifiable-income-termination-solutions-short.pdf`. Editions are the local copies: Tirole, *The Theory of Corporate Finance* (2005); Amaro de Matos, *Theoretical Foundations of Corporate Finance* (2001); Berk = Berk, DeMarzo & Harford, *Fundamentals of Corporate Finance*, 2nd ed. (2011); Grinblatt–Titman, *Financial Markets and Corporate Strategy*, 2nd ed. (2002); Ross = Ross, Westerfield, Jordan & Lamb, *Fundamentos de Administração Financeira*, 9th ed. (2013).

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1. Control passing to creditors — Grinblatt & Titman 16.10, p. 593

Statement.

Source: Grinblatt & Titman, Ch. 16, Exercise 16.10, p. 593 (PDF 606) · Answer in book: no

16.10. In the event of bankruptcy, the control of a firm passes from the equity holders to the debt holders. Describe differences in the preferences of the equity holders and debt holders and how decisions following bankruptcy proceedings are likely to change.

What is asked. The exercise has two parts. *(preferences)*: explain how what shareholders want the firm to do differs from what creditors want, given the claims each one holds. *(decisions)*: explain which corporate decisions (risk, liquidation, investment, payouts, management) are likely to change once bankruptcy hands control of the firm to the debt holders.

Approach. Each claimant's preferences follow from the shape of its payoff as a function of firm value: equity is a call option on the firm (convex), debt is a short put plus a riskless bond (concave). Whoever holds control maximises the expected value of *its own* claim, not total value; this is the debt holder–equity holder conflict of G&T §16.2. In the lecture's language, default is the event that transfers control to creditors, which is what makes their claim enforceable when income is not verifiable. A two-state numerical example makes each bias concrete.

Notation.

- $F > 0$: face value of the debt due now; $V_L \geq 0$: value if the firm is liquidated now.
- $\tilde{V}$: random value of the firm if it keeps operating (going-concern value), $E[\tilde{V}]$ its mean; risk neutrality and a zero interest rate, as in the chapter's examples.
- $E(\cdot)$, $D(\cdot)$: payoffs of equity and debt for a given total value V.

Step 1 — Two claims, two payoff shapes. By absolute priority, at a value V the claims pay

$$D(V) = \min(V, F), \tag{1}$$
$$E(V) = \max(V - F, 0). \tag{2}$$

Equity (2) is a call option on the firm: convex, it gains from upside and is protected on the downside. Debt (1) is concave: it gains nothing above F and bears the losses below F. Each group, when in control, picks the action that maximises the expected value of its own claim, not $E[\tilde{V}]$ (G&T Result 16.2, p. 562).

Step 2 — Equity holders' preferences while they control the firm. Near distress, (2) makes shareholders (and the managers who represent them) prefer: (i) *risk* — asset substitution, even into negative-NPV gambles, because the downside falls on creditors; (ii) *continuation* over liquidation — reluctance to liquidate, since liquidation pays them $\max(V_L - F, 0) = 0$ when $F > V_L$, like exercising an out-of-the-money option; (iii) *no new equity-financed investment* — debt overhang, since the existing debt captures much of the payoff; (iv) *quick payoffs* — shortsighted investment (G&T §16.2, pp. 562–578). Illustration with the book's Eastern Airlines numbers (p. 576): $F = 500$; continuing gives $\tilde{V} = 600$ or 200 with probability $\frac{1}{2}$ each; liquidating now gives $V_L = 480$ ($ millions). Using (2) and (1),

$$\begin{aligned} E_{\text{cont}} &= \tfrac{1}{2}\max(600 - 500, 0) + \tfrac{1}{2}\max(200 - 500, 0) \qquad (3)\\ &= \tfrac{1}{2}(100) + \tfrac{1}{2}(0) = 50, \\ E_{\text{liq}} &= \max(480 - 500, 0) = 0, \qquad (4)\\ D_{\text{cont}} &= \tfrac{1}{2}\min(600, 500) + \tfrac{1}{2}\min(200, 500) \qquad (5)\\ &= \tfrac{1}{2}(500) + \tfrac{1}{2}(200) = 350, \\ D_{\text{liq}} &= \min(480, 500) = 480. \qquad (6) \end{aligned}$$

Shareholders want to continue ($50 > 0$), creditors want to liquidate ($480 > 350$), and the firm is worth $E[\tilde{V}] = \frac{1}{2}(600) + \frac{1}{2}(200) = 400$ if it continues against 480 if it liquidates: equity control destroys 80.

Step 3 — Creditors' preferences once they take control. In bankruptcy (or after a covenant default) control passes to the debt holders or their representatives (G&T p. 576). By (1) they are *conservative*: they value the downside fully and the upside above F not at all. They therefore prefer (i) safe strategies and lower risk; (ii) liquidation or asset sales whenever V_L exceeds the expected value of their claim under continuation, which is efficient in Step 2's example; (iii) cash collection and cost cutting over long-horizon, risky investment. The bias can now run the other way: with $F = V_L = 500$ and $\tilde{V} = 800$ or 300 (own numbers), continuing is worth $E[\tilde{V}] = \frac{1}{2}(800) + \frac{1}{2}(300) = 550 > 500$, but

$$\begin{aligned} D_{\text{cont}} &= \tfrac{1}{2}\min(800, 500) + \tfrac{1}{2}\min(300, 500) \\ &= \tfrac{1}{2}(500) + \tfrac{1}{2}(300) = 400 < 500 = D_{\text{liq}}, \end{aligned} \tag{7}$$

so creditors liquidate a firm worth more alive (the Emruss case, pp. 577–578, has the same logic: senior creditors liquidate because liquidation pays them in full).

Answer (preferences). Equity holders (convex claim $\max(V - F, 0)$) favour risk, continuation over liquidation, no new equity-financed investment and quick payoffs; debt holders (concave claim $\min(V, F)$) favour safety, liquidation whenever V_L exceeds the expected value of their claim under continuation, and cash collection. In the example, equity wants to continue ($50 > 0$), debt wants to liquidate ($480 > 350$).

Step 4 — How decisions change after bankruptcy. Relative to equity control: risk is cut, failing divisions are closed or sold, liquidation becomes more likely (sometimes too likely), dividends stop, and managers are often replaced. When Chapter 11 converts the debt into equity of the reorganised firm, the former creditors become residual claimants: debt overhang and the incentive to gamble disappear, and decisions move back toward maximising total value. Absolute-priority violations (p. 560) leave shareholders some bargaining power, so the shift of control is only partial.

Answer (decisions). After bankruptcy the firm takes less risk, sells or closes failing assets, liquidates more often (sometimes too often: with $\tilde{V} = 800$ or 300 and $V_L = F = 500$, creditors liquidate a firm worth $550 > 500$ alive), stops dividends and often replaces management; once the debt is converted into equity, decisions move back toward maximising total value.

Intuition. Each claimant behaves like the shape of its payoff: the convex claim gambles and delays liquidation, the concave claim plays safe and liquidates early. Shifting control to creditors in default is what makes their claim credible — the same role the termination threat plays in Topic 5 — but it replaces one bias by the opposite one unless the claims are restructured.

Common mistakes.

- Saying that creditors in control maximise firm value: they maximise $E[\min(\tilde{V}, F)]$ and ignore the upside above F, so they can liquidate a firm worth more alive.
- Treating liquidation as always efficient (or always inefficient): it is efficient only when $V_L > E[\tilde{V}]$; the bias depends on who holds control.
- Forgetting absolute-priority violations and debt-for-equity swaps, which change the payoff shapes and hence the post-bankruptcy decisions.

Ref: Grinblatt & Titman Ch. 16, §16.1 "Bankruptcy", pp. 559–561; §16.2 "Debt Holder–Equity Holder Conflicts", pp. 561–578 ("The Reluctance to Liquidate Problem", pp. 575–578); Ch. 18, p. 643 (creditor influence makes managers "too conservative").

2. Duration of debt: proving (5.11) — Amaro de Matos 5.2.5 #3, p. 148

Statement.

Source: Amaro de Matos, Ch. 5, Exercise 5.2.5 #3, p. 148 (PDF 166) · Answer in book: yes, p. 268

3. Prove equation (5.11).

Context needed to solve it *From Amaro de Matos §5.2.3 "The Choice of the Duration", pp. 139–141 (PDF 157–159):*

A model developed by Diamond (1991, 1993) illustrates in a very simple way the differences between short-term and long-term debt financing. The model is based on information asymmetry and credit rating.

Regaining the notation of Aghion and Bolton, consider an entrepreneur with a two-period project who needs funds for an initial investment $K = 1$. The project may be of one of two types: good or bad. At time $t = 2$ a good project has a payoff $X > 0$, whereas a bad project pays that amount with a probability π, and pays zero with probability $1 - \pi$. Let θ denote the type of project, θ_g referring to a good project and θ_b referring to a bad project. The quality of the project is known to the entrepreneur but not to the investor. Given a project, as of date $t = 0$ the investor assigns a probability f that $\theta = \theta_g$. This probability is referred to as a *credit rating* and reflects the lender's information about the project. It follows that, as of time 0, the probability q that a borrower pays the loan at date $t = 2$ is given by

$$q \equiv f + \pi(1 - f) = \pi + f(1 - \pi). \qquad (5.9)$$

At time $t = 1$ information about the project becomes available to the investor. This new information is not verifiable, so it makes no sense to write contracts contingent upon that information, since they cannot be enforced. However, the investors update their beliefs about the quality of the project and the credit rating will be either upgraded or downgraded.

Under this revision of the credit rating, let f^u denote the probability that $\theta = \theta_g$ given an upgrade and let f^d denote the probability that $\theta = \theta_g$ given a downgrade. In that case, the probability that an entrepreneur who receives a downgrade repays at time $t = 2$ is

$$q^d = \pi + f^d(1 - \pi) > f^d. \qquad (5.10)$$

Moreover, it is here assumed that

$$f^d < f < f^u,$$

meaning that the probability that the project is good is greater given an upgrade than given a downgrade. Furthermore, the simplifying assumption $f^u = 1$ is also made. Thus, given an upgrade at date $t = 1$, it is certain that the project is good. However, it is not certain that all projects of good quality are upgraded. There is room for one such project to be downgraded. Let e denote the probability of such an event.

All projects are also assumed to produce a nonassignable control rent C if the management has control at time $t = 2$. Also, if at time $t = 1$ the manager defaults on short-term debt, lenders then have the right to remove the management from control, selling the assets for a liquidation value $L < K = 1$. All projects may be liquidated at that point in time, under those conditions. If the project is liquidated, the entrepreneur is prevented from consumption of his/her present or future cash flows and control rents. In this case, an optimal contract is a debt contract enforced by the right to liquidate the project if the debt is not repaid in full.

Just as in the case of the borrowers, lenders are assumed be risk neutral. Moreover, lenders have alternative investments at $t = 0$ that provide a return R at $t = 2$. The condition $L < X/R$ ensures that a good project yields a higher return when not liquidated.

A project may be financed with either long-term or short-term debt. In the context of this model, long-term debt refers to a contract issued at $t = 0$ with no refinancing at date $t = 1$ and maturing at $t = 2$. If one such instrument provides $K = 1$ at time $t = 0$ with face value $\rho \leq X$, the lender expects to receive at maturity $\rho[\pi + f(1 - \pi)] = R^2$. Notice that if $\rho > X$, a borrower with credit rating f will not be able to provide the lender with the required rate of return R^2. Now, if the project is upgraded at time $t = 1$, the market value of the debt contract at that point in time becomes $V_l^u = \rho/R$. Similarly, if the project is downgraded, the market value of debt at time $t = 1$ becomes $V_l^d = q^d\rho/R$.

Short-term debt, on the other hand, is debt financed at $t = 0$ with face value r_1 and maturity at time $t = 1$. If short-term borrowers are upgraded at $t = 1$, their debt contract at that point in time is worth $V_s^u = r_1$ and they may refinance their project by issuing a new short-term debt contract with maturity at $t = 2$ and face value r_1R that will be paid with certainty and guarantees a return of R per unit invested

by the lender. If short-term borrowers are downgraded, however, in principle they may refinance their debt with a new short-term contract only if $q^d X/R > r_1$, since X is the maximum that a borrower can pay at $t = 2$. If that is the case, at $t = 1$ the market value of the short-term debt issued at $t = 0$ would be $V_s^d = r_1$ if the borrower pays, possibly through the refinancement mechanism. But what happens if $q^d X/R < r_1$? In that case there is still room for a refinancement provided that the liquidation value L is less than the expected value to the lender from a refinancement, $q^d X/R$. Under a downgrading, the time $t = 1$ value of the short-term debt is thus

$$V_s^d = \min\left[r_1, \max(L, q^d X/R)\right].$$

The face value of short-term debt is established such as to provide the lenders with an expected return of R per period. It then follows that

$$r_1 = \frac{R - [1 - (1 - e)f]V_s^d}{(1 - e)f}. \qquad (5.11)$$

What is asked. Derive the face value r_1 of one-period (short-term) debt in Diamond's (1991) rating model, i.e. show that competitive lenders who must earn R per period set r_1 as in (5.11).

Approach. Short-term debt is priced by a one-period break-even condition: what the lenders expect to hold at $t = 1$, averaged over the upgrade and downgrade states, must equal R per unit lent. The state probabilities come from the rating model ($f^u = 1$: only good projects are upgraded), and the claim's $t = 1$ value in each state is given in the text. This is the lecture's maturity question: short-term debt is repriced at $t = 1$ on the basis of nonverifiable (but observable) news.

Notation.

- $f \in (0, 1)$: date-0 probability that the project is good (credit rating); $f^u = 1$, $f^d < f$: posterior after an upgrade / a downgrade.
- $e \in (0, 1)$: probability that a good project is downgraded at $t = 1$; bad projects are always downgraded (since $f^u = 1$).
- r_1: face value of one-period debt due at $t = 1$; V_s^u, V_s^d: its $t = 1$ value after an upgrade / a downgrade.
- $R > 1$: gross one-period return of the lenders' alternative; L: liquidation value; X: payoff of a successful project; $\pi \in (0, 1)$; $q^d = \pi + f^d(1 - \pi)$: repayment probability after a downgrade (5.10).
- P_u, P_d: probabilities of an upgrade and a downgrade for a borrower of unknown type; $M \equiv \max(L, q^d X/R)$.

Step 1 — Probabilities of upgrade and downgrade. We need the weights of the two $t = 1$ states. By the law of total probability over the two types, with $\Pr(\text{up} \mid \text{good}) = 1 - e$ and $\Pr(\text{up} \mid \text{bad}) = 0$ (only good projects are upgraded, since $f^u = 1$):

$$\begin{aligned} P_u &= f\,(1 - e) + (1 - f)\,(0) \\ &= f(1 - e), \qquad (8) \\ P_d &= 1 - P_u = 1 - (1 - e)f. \qquad (9) \end{aligned}$$

The downgrade posterior is consistent with these by Bayes' rule, since a downgrade comes from a good project with probability fe and from a bad one with probability $(1 - f)(1)$:

$$f^d = \frac{fe}{fe + (1 - f)} = \frac{fe}{P_d}. \qquad (10)$$

Step 2 — Value of the claim at $t = 1$. After an upgrade the borrower is known to be good and refinances r_1 with new debt of face value $r_1 R$, repaid with certainty, so

$$V_s^u = r_1. \qquad (11)$$

After a downgrade the value is the one stated in the text, $V_s^d = \min[r_1, \max(L, q^d X/R)] = \min(r_1, M)$.

Step 3 — Lenders' break-even. Lenders give $K = 1$ at $t = 0$ and require an expected value R at $t = 1$ (one period at the per-period return R):

$$P_u V_s^u + P_d V_s^d = R. \tag{12}$$

Substituting (8), (9) and (11) into (12):

$$\begin{aligned} (f(1-e))(r_1) + (1-(1-e)f)V_s^d &= R, \\ (1-e)f\, r_1 &= R - [1-(1-e)f]V_s^d, \\ r_1 &= \frac{R - [1-(1-e)f]V_s^d}{(1-e)f}, \end{aligned} \tag{13}$$

where the second line subtracts $[1-(1-e)f]V_s^d$ from both sides and the third divides by $(1-e)f > 0$. This is (5.11). ■

Step 4 — Solving the fixed point. Since V_s^d depends on r_1, (13) is an equation in r_1; we solve it case by case to see when the debt is risky. Case A, $r_1 \le M$: then $V_s^d = r_1$ and (13) gives

$$\begin{aligned} (1-e)f\, r_1 + [1-(1-e)f](r_1) &= R, \\ r_1\big[(1-e)f + 1 - (1-e)f\big] &= R, \\ r_1 &= R, \end{aligned} \tag{14}$$

consistent if and only if $R \le M$: the debt is riskless. Case B, $R > M$: then $V_s^d = M$ and

$$r_1 = \frac{R - [1-(1-e)f](M)}{(1-e)f}. \tag{15}$$

It is consistent ($r_1 > M$) because

$$\begin{aligned} r_1 - M &= \frac{R - [1-(1-e)f]M - (1-e)f\, M}{(1-e)f} \\ &= \frac{R-M}{(1-e)f} > 0, \end{aligned}$$

and $r_1 > R$ because $r_1 - R = [1-(1-e)f](R-M)/[(1-e)f] > 0$ (subtract $R = (1-e)fR/[(1-e)f]$ from (15)). The premium $r_1 - R$ pays for the shortfall $R - M$ that lenders bear in the downgrade state.

Answer (5.11). $r_1 = \dfrac{R - [1-(1-e)f]\, V_s^d}{(1-e)f}$, from the break-even $f(1-e)\, r_1 + [1-(1-e)f]\, V_s^d = R$. Since $V_s^d = \min(r_1, M)$, it gives $r_1 = R$ (riskless debt) when $R \le M$ and $r_1 = [R - (1-(1-e)f)M]/[(1-e)f] > R$ otherwise.

Intuition. Short-term lenders are paid in full whenever the borrower is upgraded and receive at most the refinancing or liquidation value after a downgrade; the face value is set so that the probability-weighted average of the two equals the safe return. Because the downgrade state carries the loss, r_1 rises with the probability $1-(1-e)f$ of a downgrade.

Common mistakes.

- Requiring R^2 instead of R: short-term debt runs one period, so the lenders' target at $t = 1$ is R (long-term debt is the one priced against R^2).
- Writing the upgrade probability as f or $1-e$: it is $f(1-e)$, because bad projects are never upgraded ($f^u = 1$) and good ones are upgraded only with probability $1-e$.

- Treating V_s^d as a constant: it is $\min[r_1, \max(L, q^d X/R)]$, so (5.11) is a fixed point and must be checked for consistency.

Book answer. Matches the book's answer (p. 268), which writes the break-even condition $P_u V_s^u + P_d V_s^d = R$ with $P_u = f(1-e)$ and solves for r_1. Step 4 is an addition.

Ref: Amaro de Matos Ch. 5, §5.2.3 "The Choice of the Duration", pp. 139–141; Appendix C.5.2, p. 268.

3. Threat of termination: continuation probability — Amaro de Matos 5.2.5 #1, p. 148

Statement.

Source: Amaro de Matos, Ch. 5, Exercise 5.2.5 #1, p. 148 (PDF 166) · Answer in book: yes, p. 267

1. Show that in the optimal debt contract, the probability $\beta_\circ$ defined in (5.8) decreases as q increases as stated in the text.

Context needed to solve it *From Amaro de Matos §5.2.1 "Extending the Model", pp. 134–137 (PDF 152–155):*

As before, all the agents are risk neutral and the risk-free rate is assumed to be zero. The time schedule refers to a two-period model and is characterized as follows: At time $t = 0$ an amount K is invested in the project. The entrepreneur, as before, is assumed to have no initial wealth. In this specification of the model, the state of nature is characterized by the cash flow at $t = 1$. It may be $\theta_g = x$ with probability q and $\theta_b = 0$ with probability $1 - q$. The cash flow is observable for both the investor and the entrepreneur, but not for any third party, and is thus not verifiable. Therefore, as before, a contract between the investor and the entrepreneur is incomplete.

The manager is assumed to have the ability to divert cash flow without the control of the investor. In that case a zero payoff would be declared at $t = 1$ when the actual payoff was x. If the declared cash flow at $t = 1$ is x, the contract states a payment of R_x to the creditholder and, with probability β_x, the right of the creditholder to liquidate a part of the assets that is worth L_x. If the cash flow is null, the values are, respectively, $R_\circ$, $\beta_\circ$, and $L_\circ$.

Notice that taking $\beta_x = 0$, $\beta_\circ = 1$, and $L_x = L_\circ$, being equal to the value of the assets of the project at time $t = 1$, this reduces to a usual debt contract with face value $R_x < x$ and one-period maturity. The issue in designing a contract is to find the optimal values for the preceding parameters. Therefore, it is possible in this simple setting to analyze the conditions under which a debt contract would result as optimal.

In the second period, two things may happen. Either the manager continues the project or the project is liquidated, in the sense that it is separated from the manager. If the manager's activity continues, the cash flow at time $t = 2$ is y. In the case of no liquidation, the project thus has a net present value of $qx + y - K$.

The expected return to the firm is then written as

$$\pi_f = q[x - R_x + (1 - \beta_x)y] + (1 - q)[-R_\circ + (1 - \beta_\circ)y]. \qquad (5.3)$$

On the other hand, the investor expects to get

$$\pi_i = q[R_x + \beta_x L_x] + (1 - q)[R_\circ + \beta_\circ L_\circ] - K. \qquad (5.4)$$

Suppose for a moment that at time $t = 1$ the cash flow is x. The expected return to the firm would then be $x - R_x + (1 - \beta_x)y$, decomposed as follows: The cash flow x goes to the entrepreneur, who has to pay R_x to the investor. With probability β_x the project is liquidated at date $t = 1$, leading to a continuation value of zero, and with probability $1 - \beta_x$ the project continues until time $t = 2$, generating a cash flow y at that point. If, however, the manager hides from the outer investor the cash flow x, then only $R_\circ$ will be paid at time $t = 1$, and liquidation occurs with probability $\beta_\circ$, leaving to the firm a possible payoff S, not necessarily zero, as before. This happens because the liquidation value $L_\circ$ does not correspond to the true value of the assets in that state of nature. In that case the expected return to the firm is $x - R_\circ + \beta_\circ S + (1 - \beta_\circ)y$.

For the manager to be induced to pay the due R_x when the cash flow is x, one must therefore impose

$$x - R_x + (1 - \beta_x)y \geq x - R_\circ + \beta_\circ S + (1 - \beta_\circ)y. \qquad (5.5)$$

Noticing that $R_\circ = \beta_x = 0$ is optimal in the sense that these values maximize the profit to the firm and facilitates the verification of (5.5), the preceding inequality reads

$$R_x \leq \beta_\circ (y - S).$$

Among all these possible values for R_x that ensure a trustworthy behavior from the manager of the project, any investor would maximize its expected return by choosing the greatest value. Thus, at the optimum, this restriction is binding. Substituting the optimal values of the parameters in expression (5.4) for the expected result of the investor, one gets

$$\pi_i = \beta_\circ [q(y - S) + (1 - q)L_\circ] - K. \qquad (5.6)$$

Using the optimal values of the parameters, namely those minimizing the profit to the investor, in the profit of the firm (5.3), it follows that

$$\pi_f = (qx + y - K) - (1 - q)\beta_\circ (y - L_\circ). \qquad (5.7)$$

This result can be interpreted as the net present value of the project in the absence of liquidation minus a cost due to contract incompleteness.

For the profit in (5.6) to be positive, ensuring the minimum condition of feasibility of the contract, one must have

$$1 \geq \beta_\circ \geq \frac{K}{q(y - S) + (1 - q)L_\circ},$$

where the first inequality comes from the fact that $\beta_\circ$ is a probability and the second inequality comes from the condition $\pi_i \geq 0$. Notice that π_f is decreasing in $\beta_\circ$ whereas π_i is monotonically increasing in that parameter. Maximizing the expected return to the firm conditioned to a positive expected return to the investor defines the optimal value of $\beta_\circ$ as the lowest possible. Thus

$$\beta_\circ = \frac{K}{q(y - S) + (1 - q)L_\circ}.$$

Both S and $L_\circ$ (and thus also $\beta_\circ$) are still to be specified. There is no need to specify L_x, since the main feature of the optimal contract just described is that there is no liquidation in the good state. Liquidation occurs only, if at all, in the case of a poor performance. In order to understand what determines S and $L_\circ$ some additional assumptions are required.

It is assumed that the entrepreneur is not able to continue managing the project in the case of bankruptcy after a zero cash flow. This is the case of a *liquidity default*. In that case, the creditor may sell the position to a third party who obtains at $t = 2$ a cash flow αy, with $\alpha \in [0, 1]$, meaning that the incoming manager cannot get better results out of the assets than the manager who left the firm.

In the case of bankruptcy after a nonzero cash flow, the original manager will negotiate the buyback of the firm with the creditholder, sharing the second-period return y. In that situation, $S = y/2$. This is the case of a *strategic default*.

It is also assumed that, in the case of liquidity default, the buyer of the assets incurs an informational cost $c \in [0, \bar{c}]$. This cost is modeled as a uniformly distributed random variable. Therefore, by a negotiating process similar to the preceding, the remaining value of the assets, αy, is equally divided between the creditor and the buyer. The buyer then negotiates only if $c \leq \alpha y/2$. Thus, assuming that $\alpha y/2 \leq \bar{c}$, the probability that this happens is

$$\Pr\left(c \leq \frac{\alpha y}{2}\right) = \frac{\alpha y}{2\bar{c}}.$$

Therefore, the expected value of the liquidation payoff in the case of a zero cash flow at $t = 1$ is

$$L_\circ = \frac{\alpha^2 y^2}{4\bar{c}},$$

and the optimal $\beta_\circ$ reads

$$\beta_\circ = \frac{K}{q\frac{y}{2} + (1 - q)\frac{\alpha^2 y^2}{4\bar{c}}}. \qquad (5.8)$$

It follows[3] that the optimal value of $\beta_\circ$ decreases as q increases. Thus, the probability of liquidation of assets decreases as the probability of positive cash flow increases. Thus, the inefficiency due to contract incompleteness, expressed in (5.7) as $(1 - q)\beta_\circ (y - L_\circ)$, is lower for higher q, or, in other words, with lower risk of bankruptcy.

From Amaro de Matos, Notes to Chapter Five, note 3:

3. This is left as an exercise.

What is asked. Show that the optimal liquidation probability $\beta_\circ$ after a declared zero cash flow, given by (5.8), falls when the probability q of the good cash flow rises (i.e. sign the derivative $\partial\beta_\circ/\partial q$).

Approach. With nonverifiable income, repayment is enforced only by the *threat of termination*: after a zero report the investor liquidates with probability $\beta_\circ$, set at the lowest level that lets her break even. The denominator of (5.8) is her expected recovery per unit of $\beta_\circ$, a q-weighted average of the recovery after a *strategic default* ($y/2$) and after a *liquidity default* ($L_\circ$). Differentiating the ratio and signing $y/2 - L_\circ$ answers the question.

Notation.

- $q \in (0,1)$: probability of the cash flow x at $t=1$; $y>0$: second-period cash flow; $K>0$: investment.
- $\beta_\circ \in [0,1]$: probability of liquidation after a declared zero cash flow (the continuation probability is $1-\beta_\circ$).
- $S = y/2$: entrepreneur's share in a strategic-default buyback, so the investor gets $y - S = y/2$.
- $\alpha \in [0,1]$: efficiency of an outside buyer; $c \sim U[0,\bar{c}]$: buyer's information cost, with $\alpha y/2 \le \bar{c}$; $L_\circ = \alpha^2 y^2/(4\bar{c})$: expected liquidation payoff.
- $\Pi(q)$: investor's expected recovery per unit of $\beta_\circ$, the denominator of (5.8).

Step 1 — Write $\beta_\circ$ as a ratio. From (5.8),

$$\beta_\circ(q) = \frac{K}{\Pi(q)}, \tag{16}$$

$$\Pi(q) = q\,\frac{y}{2} + (1-q)\frac{\alpha^2 y^2}{4\bar{c}}. \tag{17}$$

Step 2 — Differentiate. The sign of the comparative static is the sign of a derivative, so we compute it. Writing (16) as $\beta_\circ = K\,\Pi(q)^{-1}$ and applying the chain rule (K, y, α, $\bar{c}$ do not depend on q):

$$\begin{aligned}\frac{\partial\beta_\circ}{\partial q} &= K\cdot(-1)\,\Pi(q)^{-2}\cdot\Pi'(q)\\ &= -\frac{K\,\Pi'(q)}{\Pi(q)^2}.\end{aligned} \tag{18}$$

(17) is linear in q; $\frac{d}{dq}\big[q\,\frac{y}{2}\big] = \frac{y}{2}$ and $\frac{d}{dq}\big[(1-q)\frac{\alpha^2y^2}{4\bar{c}}\big] = -\frac{\alpha^2y^2}{4\bar{c}}$, so

$$\Pi'(q) = \frac{y}{2} - \frac{\alpha^2 y^2}{4\bar{c}}. \tag{19}$$

Substituting (19) and (17) into (18):

$$\frac{\partial\beta_\circ}{\partial q} = \frac{-K\left(\dfrac{y}{2} - \dfrac{\alpha^2y^2}{4\bar{c}}\right)}{\left[q\,\dfrac{y}{2} + (1-q)\dfrac{\alpha^2y^2}{4\bar{c}}\right]^2}. \tag{20}$$

Step 3 — Sign. The denominator of (20) is positive and $K > 0$, so the sign is the opposite of that of $\Pi'(q)$. Factor $y/2$ out of (19):

$$\begin{aligned}\Pi'(q) &= \frac{y}{2}\left[1 - \frac{\alpha^2 y^2}{4\bar{c}} \cdot \frac{2}{y}\right] \\ &= \frac{y}{2}\left[1 - \frac{\alpha^2 y}{2\bar{c}}\right] \\ &= \frac{y}{2}\left[1 - \alpha \cdot \frac{\alpha y}{2\bar{c}}\right]. \end{aligned} \tag{21}$$

The assumption $\alpha y/2 \le \bar{c}$ gives $\alpha y/(2\bar{c}) \le 1$, and $\alpha \le 1$, so the bracket is ≥ 0; it is > 0 unless $\alpha = 1$ and $y/2 = \bar{c}$. Hence $\Pi'(q) \ge 0$ and

$$\frac{\partial \beta_\circ}{\partial q} \le 0, \tag{22}$$

strictly except in that knife-edge case, where $L_\circ = y/2$ and $\beta_\circ = K/(y/2)$ does not depend on q. ■ (Interior solution assumed: $\Pi(q) \ge K$, so that $\beta_\circ \le 1$.) The inefficiency $(1-q)\beta_\circ(y - L_\circ)$ in (5.7) is then also decreasing in q. By the product rule, with $L_\circ$ independent of q and $y - L_\circ > 0$ (because $L_\circ \le y/2$),

$$\frac{\partial}{\partial q}\Big[(1-q)\beta_\circ(y - L_\circ)\Big] = (y - L_\circ)\Big[-\beta_\circ + (1-q)\frac{\partial \beta_\circ}{\partial q}\Big] < 0, \tag{23}$$

since $\beta_\circ > 0$ and $\partial\beta_\circ/\partial q \le 0$ by (22).

Answer (sign). $\dfrac{\partial \beta_\circ}{\partial q} = -\dfrac{K\left(\frac{y}{2} - \frac{\alpha^2 y^2}{4\bar{c}}\right)}{\left[q\frac{y}{2} + (1-q)\frac{\alpha^2 y^2}{4\bar{c}}\right]^2} \le 0$ (strictly negative unless $\alpha = 1$ and $y/2 = \bar{c}$); the incompleteness cost $(1-q)\beta_\circ(y - L_\circ)$ also falls with q.

Intuition. The investor recovers $y/2$ when she threatens a firm that defaulted strategically (it had x and buys itself back) but only $L_\circ \le y/2$ when the default is a genuine liquidity default (an outside buyer has to be found). A higher q puts more weight on the first, richer recovery, so a smaller liquidation probability suffices to break even — as in the lecture's $1 - \gamma_0 = (I - A)/[pE(\tilde{R}_2) + (1-p)L]$ (slide 11).

Common mistakes.

- Letting $L_\circ$ depend on q: it is fixed by the buyer's bargaining and information cost, $L_\circ = \alpha^2 y^2/(4\bar{c})$, so only the weights q and $1 - q$ move.
- Signing the derivative without the assumption $\alpha y/2 \le \bar{c}$: it is what makes $\Pr(c \le \alpha y/2) = \alpha y/(2\bar{c})$ a probability and gives $L_\circ \le y/2$.
- Claiming a strict decrease always: at $\alpha = 1$, $y/2 = \bar{c}$ the two recoveries coincide and $\beta_\circ$ does not depend on q.

Book answer. Matches the book's answer (p. 267), which gives (20) with "≤ 0". The book's argument assumes $\alpha y/2 < \bar{c}$ (strict) and writes the step as "$\alpha^2 y/2 < \bar{c}$"; with the text's weak inequality $\alpha y/2 \le \bar{c}$ the derivative can be zero in the knife-edge case of Step 3.

Ref: Amaro de Matos Ch. 5, §5.2.1 "Extending the Model", pp. 134–137, eq. (5.8); Appendix C.5.2, p. 267.

4. Entrepreneur control and renegotiation — Amaro de Matos 5.1.6 #3, p. 134

Statement.

Source: Amaro de Matos, Ch. 5, Exercise 5.1.6 #3, p. 134 (PDF 152) · Answer in book: yes, p. 266

3. Under entrepreneur control, show that the renegotiation of the contract whenever $t_s < \hat{t}$ for $s = 0, 1$ leads to an increase in the expected payoff of the entrepreneur.

Context needed to solve it *From Amaro de Matos §5.1.1 "The Model", pp. 127–129 (PDF 145–147):*

The relationship between incomplete financial contracts and the allocation of control in a firm may be discussed within a simple analytical context, as in the paper of Aghion and Bolton (1992). This section discusses their model, where a bilateral contract is considered between an entrepreneur having no initial wealth and an investor who has the resources to invest the amount K at $t = 0$. This is a model with three points in time characterizing the following dynamic setting:

At $t = 1$, the state of nature θ is revealed. This state may be either good (θ_g) with probability q or bad (θ_b) with probability $1 - q$. Finally, at $t = 2$, the result of the investment is obtained, expressed as a rate of return r. After the realization of the state of nature at $t = 1$ and before the maturity of the contract at $t = 2$, an action a is undertaken by one of the agents, either the investor or the entrepreneur. The decision about who takes the action is to be contracted at $t = 0$. The agent who chooses the action is said to be the one who owns the project or who retains control.

The action a and the state of nature θ are determining elements of the stochastic return r. An action is said to be optimal if it maximizes the expected wealth conditional to a given state of nature. Let a_g denote the optimal action under the state of nature θ_g and let a_b denote the optimal action under the state of nature θ_b.

The model follows the idea of Grossman and Hart (1986) that incomplete contracts are characterized by the fact that the state of nature is impossible or very costly to describe ex-ante and, therefore, the ex-ante contract cannot be contingent on the state θ. Unlike Grossman and Hart (1986), however, it is assumed here that at $t = 1$ there is a publicly verifiable signal $s \in \{0, 1\}$ that is correlated with the state of nature θ and that the contract may be contingent on s. One then defines the two conditional probabilities

$$\beta_g = \text{Prob}(s = 1 \mid \theta = \theta_g) > 1/2$$
$$\beta_b = \text{Prob}(s = 1 \mid \theta = \theta_b) < 1/2.$$

Note that $\beta_g = 1$ and $\beta_b = 0$ imply that the public signal s is a perfect proxy for θ. In other words, one may infer θ directly from the observable signal s, and thus the contract is no longer incomplete. For a contract to be incomplete, either β_g is at a certain distance from 1 or β_b is away from zero, or both situations occur. One may then use the sum of these two distances, namely,

$$d = |\beta_g - 1| + \beta_b, \qquad (5.1)$$

as a measure of the degree of incompleteness of the contract.

The entrepreneur (E) and the investor (I) are assumed to be risk neutral in income and to have utility functions given, respectively, by

$$U_E(r, a) = g(r) + l(a, \theta)$$
$$U_I(r, a) = r - g(r),$$

where $l(a, \theta)$ is part of the total payoff of the entrepreneur that, although depending on the action undertaken and on the state of nature, is nonmonetary and not observable by third parties. Therefore, this term is taken as reflecting the *private benefits* for the entrepreneur.

It is also assumed that the stochastic return at $t = 2$ may take only two values: $r = 1$ or $r = 0$. As described previously, the stochastic return depends on both the state of nature θ and on the undertaken action a. Define y_j^i as

$$y_j^i = E\left[r \mid \theta = \theta_i;\ a = a_j\right].$$

In other words, y_j^i is the expected rate of return given that the state of nature is θ_i and the action undertaken is a_j. Under the same state of nature and action, let the private benefit to the entrepreneur be denoted by l_j^i. Then, for the project to be feasible from the point of view of the investor, one must have a positive net present value for the project,

$$q y_g^g + (1 - q) y_b^b > K.$$

Also, the optimality of a_g and a_b is assumed to be expressed as

$$y_g^g + l_g^g > y_b^g + l_b^g$$

$$y_b^b + l_b^b > y_g^b + l_g^b.$$

Finally, it is assumed that the entrepreneur is compensated with a positive monetary transfer schedule $g(r) = t(s, r)$, which is a function of the first-time-period public signal, revealing information about the state of nature and also the function of the return of the project at the end of the second period. Given that r can take only the values 0 and 1, with no loss of generality, this implies that the schedule $t(s, r)$ is composed of a fixed part and a variable part, proportional to r:

$$t(s, r) = t_s r + t'_s \geq 0, \qquad (5.2)$$

where t_s and t'_s are constants. All residual returns go to the investor. Hence, if each one of the agents could choose the action to be taken maximizing his/her utilitiy, then the entrepreneur would maximize $t_s y_j^i + t'_s + l_j^i$ and the investor would maximize $(1 - t_s)y_j^i - t'_s$. The problem to be discussed within this model is the following: How does the structure presented induce the agents to settle an ex-ante agreement on who should decide which action should be taken after the state of nature is revealed? Is there room for renegotiation of the contract once the state of the world is revealed, but before the action is undertaken?

From Amaro de Matos §5.1.2 "Entrepreneur Control", pp. 129–131 (PDF 147–149):

In this section, the case where the entrepreneur has full control is studied. This means that the entrepreneur chooses the action a. The action will therefore be chosen so as to maximize the entrepreneur's reward and will be given by

$$a^E(\theta_i, s) = \arg\max_{a_j} \left\{ t_s y_j^i + t'_s + l_j^i \right\},$$

for $i = g, b$ and $j = g, b$. In order to describe the nature of the different contracts that may arise in this context, we say that a contract is *efficient* if the action chosen is the optimal action for the realized state of nature. Also, we say that a contract is *feasible* if the investor expects to obtain at least as much as the amount invested, K.

First, consider the case where the private benefits (l) satisfy the inequalities $l_g^g > l_b^g$ and $l_b^b > l_g^b$. In that case, it is said that the private benefits (l) are comonotonic[1] with the total revenues ($y + l$). It is fairly simple to show that, in this situation, a contract with a schedule $t(s, r)$ given by a constant t, independent of the signal perceived and of the return obtained, is efficient. Under the assumption of comonotonic private benefits,

$$t + l_g^g > t + l_b^g$$
$$t + l_b^b > t + l_g^b,$$

thus ensuring that the revelation of a good state ($i = g$) implies $a^E = a_g$ and the revelation of a bad state ($i = b$) implies $a^E = a_b$. In other words, the optimal choice of the entrepreneur coincides with the optimal choice for the project and one may say that entrepreneur control is efficient.

In this case, for the contract also to be feasible, the constant fee schedule $t \geq 0$ must be chosen such that $q y_g^g + (1 - q)y_b^b - t = K$. One such constant always exists given the assumption $q y_g^g + (1 - q)y_b^b > K$ and it follows

Proposition 5.1.1 (Aghion and Bolton). *If $l_g^g > l_b^g$ and $l_b^b > l_g^b$, then entrepreneur control is always efficient and feasible.*

A different treatment is required when the assumptions of the preceding result are not satisfied. Without loss of generality, assume that in state θ_b the inequality $l_b^b < l_g^b$ holds.

Under such a condition and depending on the fee schedule designed (namely, on t_s), the action chosen by the entrepreneur in certain states may not be the one that maximizes the value of the project. Therefore, there is room for renegotiation of the contract, in order to guarantee that there are no ex-post inefficiencies. It is also clear that all the gains from renegotiation revert to the entrepreneur.

Since the issue of renegotiating or not depends on the design of the fee schedule, one first looks for sufficient conditions that guarantee the absence of renegotiation and feasibility of the contract. In this case one would like the contract to induce the entrepreneur to choose action a_b in state θ_b. In other words, the fee schedule and the private benefits to the entrepreneur should relate as

$$t_s y_b^b + l_b^b \geq t_s y_g^b + l_g^b$$

or $t_s \geq (l_g^b - l_b^b)/(y_b^b - y_g^b)$. Let this lower bound be denoted by

$$\hat{t} \equiv \frac{l_g^b - l_b^b}{y_b^b - y_g^b}.$$

For the contract with $t_s = \hat{t}$ to be feasible, then it is required that the expected payoff to the investor be greater than K or, denoting by π_1 the expected payoff,

$$\pi_1 \equiv q(1-\hat{t})y_g^g + (1-q)(1-\hat{t})y_b^b \geq K.$$

This case is feasible only if K is low enough as compared to the expected payoff (that is what the preceding condition means) and/or the fee schedule is high enough (that is the meaning of $t_s \geq \hat{t}$), transferring to the entrepreneur a relatively high part of the monetary proceedings of the project.

Consider now the case where $t_s < \hat{t}$ for $s = 0, 1$. In order to maximize the ex-ante expected payoff, the investor will only accept a contract with $t_s = 0$ whatever signal is observed. The schedule $t(r, s)$ in this contract becomes a simple fixed fee. This fact, together with the assumption that $l_b^b < l_g^b$, implies that the entrepreneur will always choose a_g, independently of the state of nature. Then, the ex-ante expected return to the investor is given by

$$\pi_2 = qy_g^g + (1-q)y_g^b.$$

For the contract to be feasible, one requires that $\pi_2 \geq K$. In this situation, and given that $y_b^b > y_g^b$, there is an incentive to renegotiate. The investor is willing to pay up to $qy_g^g + (1-q)y_b^b - \pi_2$ for the entrepreneur to choose action a_b in state θ_b. It turns out that this is an interesting offer to the entrepreneur who in this way increases its expected payoff. All the bargaining power being on the side of the entrepreneur, the ex-ante expected payoff to the investor remains π_2 after the renegotiation.

From Amaro de Matos, Notes to Chapter Five, note 1, p. 281 (PDF 299):

1. Take γ and δ as the expected values conditioned on θ and a of two random variables. If $\gamma_g^g > \gamma_b^g$ and $\gamma_b^b > \gamma_g^b$, then δ is said to be comonotonic with γ if $\delta_g^g > \delta_b^g$ and $\delta_b^b > \delta_g^b$.

What is asked. Under entrepreneur control and a contract with $t_s < \hat{t}$ (so the investor sets $t_s = 0$), show that renegotiating the contract after the state is revealed makes the entrepreneur better off in expectation than sticking to the original contract.

Approach. This is the Aghion–Bolton incomplete-contract model: the contract cannot depend on θ, and with $t_s = 0$ the entrepreneur picks the action with the larger private benefit, which is the wrong action in θ_b. Ex post both parties observe θ, so there is a surplus from switching to a_b that can be shared by a side payment (Coasean renegotiation, as in the lecture's discussion of renegotiation after default). Comparing her payoff with and without the side payment, given that she has all the bargaining power, gives the result.

Notation.

- $q \in (0, 1)$: probability of the good state θ_g; a_g, a_b: the actions optimal in θ_g, θ_b.
- $y_j^i \in [0, 1]$: expected return in state θ_i under action a_j; l_j^i: entrepreneur's private benefit.
- t_s, $t'_s \geq 0$: variable and fixed parts of the transfer $t(s, r) = t_s r + t'_s$; $\hat{t} = (l_g^b - l_b^b)/(y_b^b - y_g^b)$.
- $\pi_2 = qy_g^g + (1-q)y_g^b$: investor's expected return without renegotiation.
- T: transfer from the investor to the entrepreneur agreed in renegotiation; π_{wt}^E, π_w^E: entrepreneur's expected payoff without and with renegotiation.

Assumption. As in the text, $l_b^b < l_g^b$, $y_b^b > y_g^b$, and a_b is optimal in θ_b: $y_b^b + l_b^b > y_g^b + l_g^b$; in θ_g, $l_g^g > l_b^g$, so the entrepreneur picks a_g there ("the entrepreneur will always choose a_g"). The fixed fee is zero, as π_2 implies (a fixed fee paid in both scenarios would cancel in the comparison). Renegotiation happens after θ is revealed and before the action; the entrepreneur has all the bargaining power and offers to take a_b against a payment T (the text's "willing to pay up to").

Step 1 — Payoff without renegotiation. This is the benchmark. With $t_s = 0$ (and $t'_s = 0$) her reward $t_s y_j^i + t'_s + l_j^i$ reduces to l_j^i, so she takes a_g in both states ($l_g^g > l_b^g$ and $l_g^b > l_b^b$) and gets

$$\pi_{wt}^E = q\, l_g^g + (1-q)\, l_g^b. \tag{24}$$

Step 2 — Room for renegotiation. We find the largest side payment the investor accepts. In θ_g the action is already optimal; nothing to renegotiate. In θ_b switching from a_g to a_b raises the investor's return from y_g^b to y_b^b; paying T leaves her $y_b^b - T$, which beats y_g^b iff $T \le y_b^b - y_g^b$. With all the bargaining power, the entrepreneur asks for the maximum:

$$T = y_b^b - y_g^b. \tag{25}$$

Step 3 — The entrepreneur accepts. She switches to a_b if $T + l_b^b \ge l_g^b$. Substituting (25):

$$\begin{aligned} &(y_b^b - y_g^b) + l_b^b \ge l_g^b \\ \iff \quad &y_b^b + l_b^b \ge y_g^b + l_g^b, \end{aligned} \tag{26}$$

which holds strictly by optimality of a_b in θ_b.

Step 4 — Payoff with renegotiation. In θ_b she now gets $l_b^b + T$. Using (25),

$$\pi_w^E = q\, l_g^g + (1-q)\big[l_b^b + (y_b^b - y_g^b)\big]. \tag{27}$$

Subtracting (24) from (27) (the $q\, l_g^g$ terms cancel):

$$\begin{aligned} \pi_w^E - \pi_{wt}^E &= (1-q)\big[l_b^b + (y_b^b - y_g^b)\big] - (1-q)\, l_g^b \\ &= (1-q)\big[l_b^b + y_b^b - y_g^b - l_g^b\big] \\ &= (1-q)\big[(y_b^b + l_b^b) - (y_g^b + l_g^b)\big] > 0, \end{aligned} \tag{28}$$

by (26) and $q < 1$. The investor's payoff is unchanged:

$$q\, y_g^g + (1-q)\big[y_b^b - (y_b^b - y_g^b)\big] = q\, y_g^g + (1-q)\, y_g^b = \pi_2. \tag{29}$$

So renegotiation raises the entrepreneur's expected payoff by the full efficiency gain in the bad state and leaves the investor at π_2. ■

Answer (gain). $\pi_w^E - \pi_{wt}^E = (1-q)\big[(y_b^b + l_b^b) - (y_g^b + l_g^b)\big] > 0$, with side payment $T = y_b^b - y_g^b$ in θ_b; the investor's expected payoff stays at $\pi_2 = q y_g^g + (1-q) y_g^b$.

Intuition. With $t_s < \hat{t}$ the contract gives the entrepreneur no stake in the return, so in the bad state she picks the action with the larger private benefit, which destroys value. Ex post both parties see the loss and can split the gain by a side payment; with all bargaining power on her side, she captures the whole surplus. Feasibility is unaffected because the investor's payoff stays at π_2.

Common mistakes.

- Giving part of the renegotiation surplus to the investor: the text puts all bargaining power on the entrepreneur, so T is the investor's full willingness to pay and her payoff stays π_2.
- Renegotiating in θ_g too: there a_g is already optimal and there is no surplus.
- Skipping the entrepreneur's acceptance check $T + l_b^b \ge l_g^b$: it holds only because a_b is optimal in θ_b ($y_b^b + l_b^b > y_g^b + l_g^b$).

Book answer. Matches the book's answer (p. 266) in the result. The book's two middle lines carry sign typos: "$= q l_g^g + (1-q)(y_b^b - l_b^b) + (1-q) y_g^b$" should read $q l_g^g + (1-q)(y_b^b + l_b^b) - (1-q) y_g^b$, and "$\ge q l_g^g + (1-q)(y_g^b - l_g^b) + (1-q) y_g^b$" should read $q l_g^g + (1-q)(y_g^b + l_g^b) - (1-q) y_g^b$; the book also writes $\ge$, while the inequality is strict under the text's strict optimality assumption and $q < 1$.

Ref: Amaro de Matos Ch. 5, §5.1.1 "The Model" and §5.1.2 "Entrepreneur Control", pp. 127–131; Appendix C.5.1, p. 266.

5. Diamond (1991): short- vs. long-term debt, no liquidation — Amaro de Matos 5.2.5 #5, p. 148

Statement.

Source: Amaro de Matos, Ch. 5, Exercise 5.2.5 #5, p. 148 (PDF 166) · Answer in book: no (p. 268 only cites example 1 of Diamond 1991, pp. 716–717)

5. This exercise illustrates the case of Lemma 5.2.1 in the model of Diamond (1991). Take $L = R = 1$, $\pi = \frac{1}{2}$, $X = 1\frac{3}{4}$, $C = 0$, $f = \frac{1}{2}$, and $f^d = \frac{1}{3}$.

(a) What are the face values of a short-term and a long-term debt contract?

(b) Show that in this example there will be no liquidation.

(c) What is the payoff of a good project financed with long-term debt?

(d) What is the probability that a good project receives a downgrade at $t = 1$?

(e) What is the payoff of a good project financed with short-term debt?

Context needed to solve it *From Amaro de Matos §5.2.3 "The Choice of the Duration", pp. 139–141 (PDF 157–159):*

A model developed by Diamond (1991, 1993) illustrates in a very simple way the differences between short-term and long-term debt financing. The model is based on information asymmetry and credit rating.

Regaining the notation of Aghion and Bolton, consider an entrepreneur with a two-period project who needs funds for an initial investment $K = 1$. The project may be of one of two types: good or bad. At time $t = 2$ a good project has a payoff $X > 0$, whereas a bad project pays that amount with a probability π, and pays zero with probability $1 - \pi$. Let θ denote the type of project, θ_g referring to a good project and θ_b referring to a bad project. The quality of the project is known to the entrepreneur but not to the investor. Given a project, as of date $t = 0$ the investor assigns a probability f that $\theta = \theta_g$. This probability is referred to as a *credit rating* and reflects the lender's information about the project. It follows that, as of time 0, the probability q that a borrower pays the loan at date $t = 2$ is given by

$$q \equiv f + \pi(1 - f) = \pi + f(1 - \pi). \qquad (5.9)$$

At time $t = 1$ information about the project becomes available to the investor. This new information is not verifiable, so it makes no sense to write contracts contingent upon that information, since they cannot be enforced. However, the investors update their beliefs about the quality of the project and the credit rating will be either upgraded or downgraded.

Under this revision of the credit rating, let f^u denote the probability that $\theta = \theta_g$ given an upgrade and let f^d denote the probability that $\theta = \theta_g$ given a downgrade. In that case, the probability that an entrepreneur who receives a downgrade repays at time $t = 2$ is

$$q^d = \pi + f^d(1 - \pi) > f^d. \qquad (5.10)$$

Moreover, it is here assumed that

$$f^d < f < f^u,$$

meaning that the probability that the project is good is greater given an upgrade than given a downgrade. Furthermore, the simplifying assumption $f^u = 1$ is also made. Thus, given an upgrade at date $t = 1$, it is certain that the project is good. However, it is not certain that all projects of good quality are upgraded. There is room for one such project to be downgraded. Let e denote the probability of such an event.

All projects are also assumed to produce a nonassignable control rent C if the management has control at time $t = 2$. Also, if at time $t = 1$ the manager defaults on short-term debt, lenders then have the right to remove the management from control, selling the assets for a liquidation value $L < K = 1$. All projects may be liquidated at that point in time, under those conditions. If the project is liquidated, the entrepreneur is prevented from consumption of his/her present or future cash flows and control rents. In this case, an optimal contract is a debt contract enforced by the right to liquidate the project if the debt is not repaid in full.

Just as in the case of the borrowers, lenders are assumed be risk neutral. Moreover, lenders have alternative investments at $t = 0$ that provide a return R at $t = 2$. The condition $L < X/R$ ensures that a good project yields a higher return when not liquidated.

A project may be financed with either long-term or short-term debt. In the context of this model, long-term debt refers to a contract issued at $t = 0$ with no refinancing at date $t = 1$ and maturing at $t = 2$.

If one such instrument provides $K = 1$ at time $t = 0$ with face value $\rho \leq X$, the lender expects to receive at maturity $\rho[\pi + f(1-\pi)] = R^2$. Notice that if $\rho > X$, a borrower with credit rating f will not be able to provide the lender with the required rate of return R^2. Now, if the project is upgraded at time $t = 1$, the market value of the debt contract at that point in time becomes $V_l^u = \rho/R$. Similarly, if the project is downgraded, the market value of debt at time $t = 1$ becomes $V_l^d = q^d\rho/R$.

Short-term debt, on the other hand, is debt financed at $t = 0$ with face value r_1 and maturity at time $t = 1$. If short-term borrowers are upgraded at $t = 1$, their debt contract at that point in time is worth $V_s^u = r_1$ and they may refinance their project by issuing a new short-term debt contract with maturity at $t = 2$ and face value r_1R that will be paid with certainty and guarantees a return of R per unit invested by the lender. If short-term borrowers are downgraded, however, in principle they may refinance their debt with a new short-term contract only if $q^dX/R > r_1$, since X is the maximum that a borrower can pay at $t = 2$. If that is the case, at $t = 1$ the market value of the short-term debt issued at $t = 0$ would be $V_s^d = r_1$ if the borrower pays, possibly through the refinancement mechanism. But what happens if $q^dX/R < r_1$? In that case there is still room for a refinancement provided that the liquidation value L is less than the expected value to the lender from a refinancement, q^dX/R. Under a downgrading, the time $t = 1$ value of the short-term debt is thus

$$V_s^d = \min\left[r_1, \max(L, q^dX/R)\right].$$

The face value of short-term debt is established such as to provide the lenders with an expected return of R per period. It then follows that

$$r_1 = \frac{R - [1-(1-e)f]V_s^d}{(1-e)f}. \qquad (5.11)$$

From Amaro de Matos §5.2.3, pp. 141–142 (PDF 159–160):

Given this model, the point is to characterize under what conditions short-term or long-term maturities will be offered. The maturity preferred by the entrepreneurs of good projects will be the only one offered to all borrowers. This happens because if any other maturity is offered, it reveals a project of bad quality and lenders will not be willing to finance it.

The following result implies that, in the absence of liquidation given a downgrade, the entrepreneurs of good projects prefer short-term debt:

Lemma 5.2.1 (Diamond, 1991). *Comparing two debt contracts issued at $t = 0$, neither of which involves liquidation on a downgrade at $t = 1$, the one with the higher value at that point in time given a downgrade, V^d, is preferred by the entrepreneurs of good projects.*

Proof. The expected payoff to the entrepreneurs of good projects from debt not yielding liquidation is

$$e\left(X + C - \frac{RV^d}{q^d}\right) + (1-e)(X + C - r_1R).$$

Using expression (5.11) this payoff is rewritten as

$$X + C - \frac{R^2}{f} + RV^d e\left(\frac{1}{f^d} - \frac{1}{q^d}\right).$$

Given (5.10) this expression for the payoff is clearly an increasing function of V^d. ■

What is asked. (a) Compute the face value ρ of long-term debt (due at $t = 2$) and r_1 of short-term debt (due at $t = 1$) for the given numbers. (b) Show that no borrower is ever liquidated at $t = 1$. (c) Compute what a good project's entrepreneur gets with long-term debt. (d) Compute the probability e that a good project is downgraded. (e) Compute what a good project's entrepreneur gets with short-term debt (and compare with (c)).

Approach. Diamond's (1991) maturity model: lenders price debt competitively from the public rating, and at $t = 1$ nonverifiable news (an upgrade or a downgrade) reprices short-term debt. Long-term debt is priced once, $\rho q = R^2$; short-term debt by (5.11). Liquidation can only happen if a downgraded borrower cannot roll over, i.e. if $q^dX/R < r_1$ and L beats refinancing, so (b) is a check of that condition. Lemma 5.2.1 then predicts the answer to (c) vs. (e): without liquidation, good types prefer the debt with the higher downgrade value V^d, i.e. short-term debt, because it shifts repayment onto the (mostly bad) downgraded pool.

Notation.

- $L = R = 1$: liquidation value and gross per-period safe return; $\pi = \frac{1}{2}$: success probability of a bad project; $X = \frac{7}{4}$: payoff on success; $C = 0$: control rent.
- $f = \frac{1}{2}$, $f^u = 1$, $f^d = \frac{1}{3}$: prior and posterior probabilities of a good project; e: probability that a good project is downgraded.
- q, q^d: repayment probabilities at $t = 0$ and after a downgrade, (5.9)–(5.10).
- ρ: face value of long-term debt due at $t = 2$; r_1: face value of short-term debt due at $t = 1$; V^d: $t = 1$ value of the debt after a downgrade (V_l^d long, V_s^d short).
- U_l, U_s: expected payoff of a good project under long- and short-term debt.

Step 1 — Repayment probabilities and e. Every later step needs q, q^d and e, so we compute them first. From (5.9) and (5.10),

$$q = \pi + f(1-\pi) = \tfrac{1}{2} + \tfrac{1}{2}\left(\tfrac{1}{2}\right) = \tfrac{3}{4}, \tag{30}$$

$$q^d = \pi + f^d(1-\pi) = \tfrac{1}{2} + \tfrac{1}{3}\left(\tfrac{1}{2}\right) = \tfrac{2}{3}. \tag{31}$$

Bad projects are always downgraded ($f^u = 1$) and good ones with probability e, so a downgrade has probability $fe + (1-f)$ and Bayes' rule gives the downgrade posterior:

$$f^d = \frac{fe}{fe + (1-f)}. \tag{32}$$

Solving (32) for e (multiply out, collect the e terms, divide):

$$\begin{aligned} f^d\big[fe + (1-f)\big] &= fe, \\ f^d(1-f) &= fe - f^d fe = fe(1-f^d), \\ e &= \frac{f^d(1-f)}{f(1-f^d)} \\ &= \frac{\left(\frac{1}{3}\right)\left(\frac{1}{2}\right)}{\left(\frac{1}{2}\right)\left(\frac{2}{3}\right)} = \frac{1/6}{1/3} = \tfrac{1}{2}. \end{aligned} \tag{33}$$

Hence the probability of an upgrade is $(1-e)f = \left(\frac{1}{2}\right)\left(\frac{1}{2}\right) = \frac{1}{4}$ and of a downgrade $1 - \frac{1}{4} = \frac{3}{4}$.

Step 2 — (a) Face values. Each face value comes from the lenders' break-even condition at its own horizon. *Long-term debt*: the lender lends $K = 1$ for two periods and requires $\rho q = R^2$; substituting (30),

$$\begin{aligned} \rho\left(\tfrac{3}{4}\right) &= (1)^2, \\ \rho &= \frac{1}{3/4} = \tfrac{4}{3} \approx 1.3333 \leq X = 1.75. \end{aligned} \tag{34}$$

Short-term debt: guess $r_1 \leq \max(L, q^d X/R)$, so that $V_s^d = \min[r_1, \max(L, q^d X/R)] = r_1$ (the debt is repaid in full after a downgrade). Substituting $R = 1$, $(1-e)f = \frac{1}{4}$ and $V_s^d = r_1$ into (5.11):

$$\begin{aligned} r_1 &= \frac{1 - \left[1 - \frac{1}{4}\right](r_1)}{\frac{1}{4}}, \\ \tfrac{1}{4} r_1 &= 1 - \tfrac{3}{4} r_1, \\ \tfrac{1}{4} r_1 + \tfrac{3}{4} r_1 &= 1, \\ r_1 &= 1. \end{aligned} \tag{35}$$

Check of the guess: $q^d X/R = \left(\frac{2}{3}\right)\left(\frac{7}{4}\right)/1 = \frac{7}{6} \approx 1.1667$, so $\max(L, q^d X/R) = \frac{7}{6} \geq r_1 = 1$. At $t = 1$ the upgraded borrower rolls r_1 over into debt of face value $r_1 R = 1$, and the downgraded borrower into debt of face value $r_1 R/q^d = 1/\frac{2}{3} = \frac{3}{2}$.

Answer (a). Short-term debt: $r_1 = 1$ (riskless, $r_1 = R$). Long-term debt: $\rho = \frac{4}{3} \approx 1.3333$.

Step 3 — (b) No liquidation. Long-term debt has no payment due at $t = 1$, so there is no default and no liquidation. Short-term debt: after an upgrade the borrower refinances at face value 1; after a downgrade new lenders lend $r_1 = 1$ against face value $\frac{3}{2} \leq X = \frac{7}{4}$, and they break even because $q^d \cdot \frac{3}{2} = \left(\frac{2}{3}\right)\left(\frac{3}{2}\right) = 1 = Rr_1$. Equivalently the refinancing condition $q^d X/R = \frac{7}{6} > r_1 = 1$ holds. Every borrower repays r_1 at $t = 1$, so the liquidation right is never triggered. (Even after a default, lenders would refinance rather than liquidate, since $q^d X/R = \frac{7}{6} > L = 1$.) Both contracts are of the kind covered by Lemma 5.2.1.

Answer (b). No liquidation: long-term debt has no $t = 1$ payment, and with short-term debt every downgraded borrower rolls over $r_1 = 1$ into face value $\frac{3}{2} \leq X = \frac{7}{4}$ (equivalently $q^d X/R = \frac{7}{6} > r_1 = 1$).

Step 4 — (c) Good project, long-term debt. A good project pays X for sure, keeps its control rent C and repays ρ:

$$\begin{aligned} U_l &= X + C - \rho \\ &= \left(\tfrac{7}{4}\right) + (0) - \left(\tfrac{4}{3}\right) = \tfrac{21-16}{12} = \tfrac{5}{12} \approx 0.4167. \end{aligned} \tag{36}$$

Check with the Lemma's formula, using $V_l^d = q^d \rho/R = \left(\frac{2}{3}\right)\left(\frac{4}{3}\right) = \frac{8}{9}$, $R^2/f = 1/\frac{1}{2} = 2$ and $1/f^d - 1/q^d = 3 - \frac{3}{2} = \frac{3}{2}$:

$$\begin{aligned} X + C - \frac{R^2}{f} + RV^d e\left(\frac{1}{f^d} - \frac{1}{q^d}\right) &= \tfrac{7}{4} - 2 + (1)\left(\tfrac{8}{9}\right)\left(\tfrac{1}{2}\right)\left(\tfrac{3}{2}\right) \\ &= -\tfrac{1}{4} + \tfrac{2}{3} = \tfrac{-3+8}{12} = \tfrac{5}{12}. \end{aligned}$$

Answer (c). $U_l = X + C - \rho = \frac{5}{12} \approx 0.4167$.

Step 5 — (d) Probability that a good project is downgraded. From (33), $e = \frac{1}{2}$: half of the good projects are downgraded at $t = 1$ (and all bad ones are), which is what makes $f^d = \frac{1}{3}$. Check with (32): $fe/[fe + (1-f)] = \frac{1}{4}/(\frac{1}{4} + \frac{1}{2}) = \frac{1}{3}$.

Answer (d). $e = \frac{1}{2}$ (the overall downgrade probability is $1 - (1-e)f = \frac{3}{4}$).

Step 6 — (e) Good project, short-term debt. With probability $1 - e$ it is upgraded and repays $r_1 R = 1$ at $t = 2$; with probability e it is downgraded and repays $RV_s^d/q^d = \frac{3}{2}$ (with $V_s^d = r_1 = 1$). From the Lemma's payoff expression,

$$\begin{aligned} U_s &= e\left(X + C - \frac{RV_s^d}{q^d}\right) + (1-e)(X + C - r_1 R) \\ &= \tfrac{1}{2}\left(\tfrac{7}{4} + 0 - \tfrac{3}{2}\right) + \tfrac{1}{2}\left(\tfrac{7}{4} + 0 - 1\right) \\ &= \tfrac{1}{2}\left(\tfrac{1}{4}\right) + \tfrac{1}{2}\left(\tfrac{3}{4}\right) = \tfrac{1}{2}. \end{aligned} \tag{37}$$

Check: $X + C - R^2/f + RV_s^d e(1/f^d - 1/q^d) = -\frac{1}{4} + (1)(1)\left(\frac{1}{2}\right)\left(\frac{3}{2}\right) = -\frac{1}{4} + \frac{3}{4} = \frac{1}{2}$. So $U_s = \frac{1}{2} > U_l = \frac{5}{12}$ (gain $\frac{1}{2} - \frac{5}{12} = \frac{1}{12} \approx 0.0833$): good projects prefer short-term debt, as Lemma 5.2.1 predicts since $V_s^d = 1 > V_l^d = \frac{8}{9}$.

Answer (e). $U_s = \frac{1}{2}$, above $U_l = \frac{5}{12}$ by $\frac{1}{12} \approx 0.0833$: the good entrepreneur prefers short-term debt.

Intuition. Long-term debt locks in the date-0 pooled price $\rho = \frac{4}{3}$, which subsidises the bad projects. Short-term debt lets the price be reset at $t = 1$: the upgraded good projects (half of them) refinance at the riskless rate and pay only 1; the downgraded ones pay more ($\frac{3}{2}$), but no project is liquidated, so the reset is pure gain for the good type on average. Short maturity is attractive when refinancing risk is not a liquidation risk.

Common mistakes.

- Pricing long-term debt with R instead of R^2, or with q^d instead of q: at $t = 0$ the relevant repayment probability is the prior one, $q = \frac{3}{4}$, over two periods.
- Confusing e (probability that a *good* project is downgraded, $\frac{1}{2}$) with the probability of a downgrade for an unknown type ($\frac{3}{4}$).
- Charging the downgraded good borrower $r_1 R = 1$ instead of $RV_s^d/q^d = \frac{3}{2}$: after a downgrade new lenders price the rollover at the pooled repayment probability q^d.

Book answer. The book gives no worked answer: Appendix C (p. 268) only cites example 1 of Diamond (1991), pp. 716–717, which is not in `Livros/`. The solution above is derived here from §5.2.3 and Lemma 5.2.1. Note that the parameters set $L = 1 = K$, while the text assumes $L < K = 1$; this does not matter here because liquidation never occurs.

Ref: Amaro de Matos Ch. 5, §5.2.3 "The Choice of the Duration", pp. 139–142 (Lemma 5.2.1, p. 142); Appendix C.5.2, p. 268.

6. Optimal contracts in the Bolton–Scharfstein model — Tirole 7.7, p. 325

Statement.

Source: Tirole, Ch. 7, Exercise 7.7, p. 325 (PDF 338) · Answer in book: yes, pp. 603–604

Exercise 7.7 (optimal contracts in the Bolton–Scharfstein model). Redo the Bolton and Scharfstein analysis of Section 7.1.2, allowing for fully general contracts: the entrepreneur receives r_b^S in the case of date-0 success but no refinancing, R_b^{SS} in the cases of date-0 and date-1 success, and R_b^{FS} in the cases of date-1 success and date-0 failure (with $R_b^{SS}, R_b^{FS} \geqslant B/\Delta p$). (Under risk neutrality, there is no point rewarding failures unless it serves to deter predation. Hence, the exception R_b^{FS}.) Generalize the conditions (PD) and (IC) and show that $r_b^S = 0$ and that $R_b^{SS} \geqslant R_b^{FS}$ ($\geqslant B/\Delta p$).

Context needed to solve it *From Tirole §7.1.1.1 "Basics: Profit Destruction and Benchmarking Effects", p. 284 (PDF 297):*

The model is the basic, fixed-investment model of Section 3.2 except for the twist that a firm's profit depends on how successful the other firm is. Namely, while a firm makes no profit if it fails to develop the new technology, its profit when its succeeds in developing it depends on whether it faces a competitor, that is, whether the other firm also succeeds. Thus the firm's profit is as follows:

$$\text{profit} = \begin{cases} M & \text{if it is the only firm to succeed,} \\ D & \text{if both firms succeed,} \\ 0 & \text{if it fails,} \end{cases}$$

where

$$M \geqslant D \geqslant 0 \quad \text{and} \quad p_H M > I.$$

Here, M stands for "monopoly profit" and D for "duopoly profit." [...]

The familiar agency cost affects the development process. Each entrepreneur succeeds with probability p_H if she behaves (in which case she receives no private benefit) and with probability $p_L = p_H - \Delta p$ if she misbehaves (and thereby receives private benefit B), where $\Delta p > 0$.

Each entrepreneur needs to raise $I - A$ in order to finance her project, where I is the investment cost and A her initial net worth.

From Tirole §7.1.2.1 "Financial Muscle and Predation", p. 290 (PDF 303):

Simple-minded long-purse story: short-term financing arrangements. There are two dates, $t = 0, 1$. There is no discounting between the two periods.

Consider a duopoly. Firms $i = 1, 2$ are identical in all respects except the amount of wealth they have access to in order to finance investments. Firm 1 (the predator, the financially strong firm) has a large amount of wealth and never needs to go to the capital market to finance investment. Firm 2 (the prey, the financially weak firm) has just enough wealth to finance the date-0 investment.[16]

[...] The second period, date 1, is described exactly as in Section 7.1.1: for each firm, the investment cost is I. Entrepreneurs then engage in moral hazard. The probability of success of the date-1 project is p_H if the entrepreneur behaves and $p_L = p_H - \Delta p$ if she misbehaves (in which case she obtains private benefit B). A firm's date-1 profit is M if it alone succeeded, D if both firms succeeded, and 0 otherwise. [...]

[16] Alternatively, one could assume that it is able to secure financing for the date-0 investment, where the loan is to be repaid from date-0 profits: that is, there is no long-term financing arrangement.

From Tirole §7.1.2.1, "Strategic security design (Bolton and Scharfstein)", pp. 292–293 (PDF 305–306):

Let us now investigate these points in more detail. To do so, we need to build upon the simple-minded model by explicitly describing the date-0 actions and by allowing for long-term financing. To simplify the analysis, let us assume that

- $p_H = 1$;
- furthermore, date 0 is identical to date 1 except for the private benefit from misbehaving—the latter is equal to B_0 at date 0 and B at date 1;[19]
- the financially strong firm's act of predation in a given period[20] results in 0 profits (failure) for both firms in that period;
- investors in the financially weak firm observe only that firm's profit at date 0.[21]

As earlier, the financially strong firm is self-financed. The financially weak firm signs a long-term agreement with its investors that specifies:[22]

- the probabilities z^S and z^F of refinancing (date-1 investment) in the case of date-0 success (i.e., profit D) or failure (i.e., profit 0), respectively;
- a reward R_b^i in the case of date-1 success (which assumes a date-1 reinvestment in the first place), where $i \in \{S, F\}$ indexes the date-0 outcome—for all i, R_b^i must satisfy the date-1 incentive constraint $(\Delta p)R_b^i \geqslant B$.[23]

The timing is summarized in Figure 7.5.

Figure 7.5 (timeline, left to right; p. 293, PDF 306)

Date 0: step 1	Date 0: step 2	Date 0: step 3	Date 1: step 4	Date 1: step 5	Date 1: step 6
Financially weak firm signs initial contract. Financing of investment I. Choice of z^S, z^F, and R_b^i.	Date-0 efforts. Probability of success: $p_H = 1$ for high effort or p_L for low effort (private benefit B_0) if no predation; 0 if predation. Strong firm preys?	Profits: D if successful and no predation, 0 otherwise.	Reinvestment I by weak firm?	Date-1 effort(s). Probability of success: $p_H = 1$ for high effort, or p_L for low effort (private benefit B).	Date-1 profit(s).

Let us make the following assumptions.

Assumption 7.1. *Positive-NPV investment even under duopoly in the absence of predation:*

$$D - I > 0.$$

(Recall: the probability of success in the case of good behavior is here equal to 1.)

Assumption 7.2. *Dearth of pledgeable income in the case of continuation:*[24]

$$\left(D - \frac{B}{\Delta p}\right) - I < 0.$$

Preventing predation by firm 1 at date 0 benefits firm 2 in two ways. First, its date-0 income is increased by D, the duopoly profit. Second, it may avoid credit rationing, which from Assumption 7.2 is a threat, and from Assumption 7.1 undesirable.

Predation deterrence constraint. To avoid predation firm 2 must choose its financial contract so that firm 1's date-0 cost of predation, D, exceed its date-1 gain from monopolization. To compute the latter, note that it is not in firm 1's interest to prey in the last period. Hence, preventing refinancing by firm 2 raises firm 1's profit from D to M. And the probability that firm 2 receives refinancing falls from z^S in the absence of predation to z^F under predation (recall that the probability of refinancing is z^S if firm 2's date-0 profit is D and z^F if it is equal to 0). The predation deterrence constraint is therefore

$$D \geqslant (z^S - z^F)(M - D). \qquad \text{(PD)}$$

To deter predation, the weak firm's contract must make the continuation decision relatively insensitive to that firm's date-0 profit performance. Note that (PD) can be rewritten as

$$\frac{D}{M - D} \geqslant z^S - z^F. \qquad \text{(PD}')$$

Suppose that competition between the two firms reduces industry profit:

$$M \geqslant 2D.$$

The left-hand side of (PD′) can then take any value between 0 (extreme, Bertrand competition) and 1 (perfect tacit collusion or noncompeting goods). In the latter case, (PD′) really does not constrain the financial contract and there is little incentive to prey. By contrast, with Bertrand competition, predation can only be deterred by a performance-insensitive continuation rule (on the other hand, remaining in the market is also unattractive for firm 2).

Weak firm's date-0 incentive constraint. The weak firm's contract must also induce its entrepreneur to behave. Here, the entrepreneur's compensation is delayed. She receives R_b^i if there is reinvestment and firm 2 has profit D in the last period. Let

$$\mathcal{R}_b^S \equiv z^S R_b^S \quad \text{and} \quad \mathcal{R}_b^F \equiv z^F R_b^F$$

denote the expected continuation payoffs for the entrepreneur in the cases of date-0 success and failure, respectively. By misbehaving at date 0, the entrepreneur receives private benefit B_0, but reduces the probability of date-0 success by Δp (provided that the rival does not prey, i.e., if constraint (PD) is satisfied).

Hence, the incentive constraint is

$$\mathcal{R}_b^S - \mathcal{R}_b^F \geqslant \frac{B_0}{\Delta p}. \qquad \text{(IC)}$$

(a) *The no-predation benchmark.* Suppose, first, that the predator is unable to prey and so constraint (PD) is irrelevant. Let $U_b(z^S)$ denote the NPV:

$$U_b(z^S) \equiv D - I + z^S(D - I).$$

From Assumption 7.1, it increases in z^S.

The investors' breakeven constraint can be written as

$$U_b(z^S) - \mathcal{R}_b^S + A \geqslant 0. \qquad \text{(IR}_l\text{)}$$

Finally, the entrepreneur's incentive compatibility constraint is

$$\mathcal{R}_b^S - \mathcal{R}_b^F \geqslant \frac{B_0}{\Delta p}. \qquad \text{(IC)}$$

[. . .]

From Tirole §7.1.2.1, p. 294 (PDF 307):

(b) *Reintroducing the predation-deterrence constraint.* The best, predation-deterring financial contract is now obtained by maximizing firm 2's NPV subject to the predation-deterrence constraint (PD), the breakeven constraint (IR$_l$), and the incentive-compatibility (IC) constraint.

If the solution in the no-predation benchmark case satisfied (PD), then it is also the solution when predation if feasible. So, we will assume that (PD) is not satisfied by the benchmark solution. [. . .]

Footnotes 19–24, p. 292 (PDF 305):

[19] We will later assume that $B_0 < B$. This distinction between B_0 and B allows us to shorten the analysis by ensuring that date-0 incentives can be provided through the continuation policy. See below.

[20] Predation can, of course, occur at date 0 only. The predator can only lose by preying at date 1 because there is no future at that date.

[21] The important assumption is that a court cannot ascertain whether the absence of date-0 profit for firm 2 is due to a predatory act or to moral hazard (lack of luck is not a possible third explanation because we assumed that $p_H = 1$). [...]

[22] Exercise 7.7 shows that there is only a slight loss of generality in considering this class of contracts, and that the results are not qualitatively affected by our focus on this class.

[23] Note that if $(\Delta p)R_b^i < B$, then there is always renegotiation to an $R_b^i \geqslant B/\Delta p$ as long as $p_H(R - B/\Delta p) > p_L R$. But even in the absence of renegotiation, the analysis generalizes.

[24] Given that rent $p_H(B/\Delta p) = B/\Delta p$ must be given to the entrepreneurs, investors would not finance firm 2 at date 1 in the absence of retained earnings and long-term contracting obligations.

What is asked. Redo the Bolton–Scharfstein contract with three reward levels instead of one per date-0 outcome. *(PD)*: write the predation-deterrence constraint for this contract. *(IC)*: write the date-0 incentive constraint. *(r_b^S)*: show that the reward after date-0 success without refinancing can be set to zero. *(ordering)*: show that the reward after success–success is at least the reward after failure–success, which is at least $B/\Delta p$.

Approach. Bolton–Scharfstein is the lecture's threat-of-termination logic applied to a firm with a predatory rival: the refinancing probabilities z^S, z^F are the termination threat that disciplines date-0 effort, while predation deterrence requires that threat to be weak. With risk neutrality only *expected* rewards matter, so the proof is a reshuffling argument: move reward to where it relaxes the binding constraints and leaves the others unchanged.

Notation.

- $p_H = 1$, $p_L < 1$, $\Delta p = 1 - p_L$: success probabilities; B_0, B: private benefits from misbehaving at dates 0 and 1 ($B_0 < B$).
- D, M: duopoly and monopoly profits ($M \geq D > 0$); I: investment per date; A: firm 2's net worth.
- z^S, $z^F \in [0, 1]$: probabilities of date-1 refinancing after date-0 success (profit D) / failure (profit 0).
- $r_b^S \geq 0$: reward after date-0 success without refinancing; R_b^{SS}, $R_b^{FS} \geq B/\Delta p$: rewards after date-1 success following date-0 success / failure.
- $W^S \equiv z^S R_b^{SS} + (1 - z^S) r_b^S$: entrepreneur's expected reward after date-0 success; $U_b(z^S) = D - I + z^S(D - I)$: NPV.

Step 1 — Date-1 behaviour. Whenever firm 2 is refinanced, $R_b^{SS}, R_b^{FS} \geq B/\Delta p$ make good behaviour optimal at date 1, so firm 2 succeeds with probability $p_H = 1$ and the entrepreneur collects R_b^{SS} or R_b^{FS} for sure.

Step 2 — Generalised (PD). Firm 1's payoffs do not depend on how firm 2 pays its entrepreneur; they depend only on whether firm 2 is active at date 1, which by Step 1 means profit D for firm 1 instead of M. Preying costs firm 1 its date-0 profit D and lowers firm 2's refinancing probability from z^S to z^F, so firm 1's expected date-1 profit rises by $z^F D + (1 - z^F)M - [z^S D + (1 - z^S)M] = (z^S - z^F)(M - D)$. Preying is unprofitable iff the cost D is at least this gain. Hence (PD) is unchanged:

$$D \geq (z^S - z^F)(M - D). \tag{38}$$

Answer (PD). $D \geq (z^S - z^F)(M - D)$, the same as with simple contracts: the rewards $r_b^S, R_b^{SS}, R_b^{FS}$ do not enter it.

Step 3 — Generalised (IC). If the entrepreneur behaves at date 0 (and (PD) holds), date-0 success is certain and she gets W^S. If she misbehaves she gets B_0, succeeds with probability p_L

(then W^S), and fails with probability $1-p_L$ (then $z^F R_b^{FS}$). Behaving is optimal if

$$\begin{aligned} W^S &\geq B_0 + p_L W^S + (1-p_L) z^F R_b^{FS}, \\ W^S - p_L W^S - (1-p_L) z^F R_b^{FS} &\geq B_0, \\ (1-p_L)(W^S - z^F R_b^{FS}) &\geq B_0, \\ (\Delta p)\big[z^S R_b^{SS} + (1-z^S) r_b^S - z^F R_b^{FS}\big] &\geq B_0, \end{aligned} \tag{39}$$

using $\Delta p = p_H - p_L = 1 - p_L$ and the definition of W^S in the last line. With $r_b^S = 0$, $\mathcal{R}_b^S = z^S R_b^{SS}$ and $\mathcal{R}_b^F = z^F R_b^{FS}$ this is the text's (IC).

Answer (IC). $(\Delta p)\big[z^S R_b^{SS} + (1-z^S) r_b^S - z^F R_b^{FS}\big] \geq B_0$.

Step 4 — Generalised (IR_l) and the objective. On the equilibrium path date 0 succeeds; investors receive D, pay r_b^S if there is no refinancing, and if there is refinancing invest I, receive D and pay R_b^{SS}:

$$\begin{aligned} & D - (1-z^S) r_b^S + z^S (D - I - R_b^{SS}) \geq I - A \\ \iff \quad & (D-I) + z^S (D-I) - \big[z^S R_b^{SS} + (1-z^S) r_b^S\big] + A \geq 0 \\ & \iff \quad U_b(z^S) - W^S + A \geq 0, \end{aligned} \tag{40}$$

where the second line moves $I - A$ to the left and groups terms, and the third uses $U_b(z^S) = D - I + z^S(D-I)$ and the definition of W^S. The entrepreneur's utility is $W^S - A$; with (40) binding it equals $U_b(z^S)$, which is increasing in z^S (Assumption 7.1). The contract maximises $U_b(z^S)$ subject to (38), (39), (40) and the floors $R_b^{SS}, R_b^{FS} \geq B/\Delta p$.

Step 5 — $r_b^S = 0$. The pair (r_b^S, R_b^{SS}) enters (39), (40) and the objective only through W^S, and does not enter (38). Take a contract with $z^S \in (0,1)$ and $r_b^S > 0$, and replace it by

$$\tilde{r}_b^S = 0, \qquad \tilde{R}_b^{SS} = R_b^{SS} + \frac{(1-z^S) r_b^S}{z^S}. \tag{41}$$

Then $z^S \tilde{R}_b^{SS} + (1-z^S)(0) = z^S R_b^{SS} + (1-z^S) r_b^S = W^S$, and $\tilde{R}_b^{SS} > R_b^{SS} \geq B/\Delta p$, so every constraint and the objective are unchanged. If $z^S = 1$, r_b^S is never paid and can be set to 0. Finally $z^S = 0$ is never optimal: from a feasible contract with $z^S = 0$, moving to a small $z^S = \varepsilon > 0$ with $r_b^S = 0$ and $R_b^{SS} = W^S/\varepsilon$ keeps W^S (hence (39)), keeps (38) for $\varepsilon \leq D/(M-D)$, meets the floor $W^S/\varepsilon \geq B/\Delta p$ for small ε (as $W^S \geq B_0/\Delta p > 0$), relaxes (40) because $U_b(\varepsilon) > U_b(0)$, and raises the objective. Hence, without loss of generality, $r_b^S = 0$.

Answer (r_b^S). $r_b^S = 0$ without loss of generality: any $r_b^S > 0$ can be moved into R_b^{SS}, raising it by $(1-z^S) r_b^S / z^S$, with all constraints and the NPV unchanged.

Step 6 — $R_b^{SS} \geq R_b^{FS} \geq B/\Delta p$. R_b^{FS} is paid only off the equilibrium path: it appears in (39) with a negative sign and nowhere else. Lowering it to its floor $B/\Delta p$ relaxes (39) and leaves everything else unchanged, so the optimal contract can set

$$R_b^{FS} = \frac{B}{\Delta p} \leq R_b^{SS}. \tag{42}$$

The book adds that (39) binds when (38) binds (otherwise z^F could be raised to relax (PD)); then $R_b^{FS} > B/\Delta p$ would leave (39) slack, so (42) holds at every optimum. With (PD) binding, $z^F = z^S - D/(M-D)$ and $r_b^S = 0$, substituting into (39):

$$\begin{aligned} z^S R_b^{SS} - \left(z^S - \frac{D}{M-D}\right) R_b^{FS} &= \frac{B_0}{\Delta p}, \\ z^S (R_b^{SS} - R_b^{FS}) + \frac{D}{M-D} R_b^{FS} &= \frac{B_0}{\Delta p}, \end{aligned} \tag{43}$$

where the first line divides (39) (with equality) by Δp and the second expands $-(z^S - \frac{D}{M-D})R_b^{FS}$ and groups the z^S terms.

Answer (ordering). $R_b^{FS} = B/\Delta p$ (its floor) and $R_b^{SS} \geq B/\Delta p = R_b^{FS}$, so $R_b^{SS} \geq R_b^{FS} \geq B/\Delta p$.

Intuition. Under risk neutrality only expected rewards matter, so the entrepreneur should be paid where it helps incentives most — in the continuation branch after success, which also ties her pay to the date-1 output. Paying her after date-0 failure only weakens date-0 incentives; the contract does it only because predation deterrence forces refinancing after failure ($z^F > 0$), and then the date-1 incentive floor $B/\Delta p$ is the least it can pay.

Common mistakes.

- Setting $R_b^{FS} = 0$: once the firm is refinanced after failure, the date-1 incentive constraint still requires $R_b^{FS} \geq B/\Delta p$; the floor, not zero, is the minimum.
- Putting the rewards into (PD): firm 1 cares only about whether firm 2 is active at date 1, so (PD) depends on $z^S - z^F$ alone.
- Writing the IC with a generic p_H: here $p_H = 1$, so the date-0 success probability gap is $\Delta p = 1 - p_L$ and success is certain on the equilibrium path.

Book answer. Matches the book's answer (pp. 603–604): same (PD), (IC′) and (IR′), $r_b^S = 0$ by shifting reward into R_b^{SS}, and $R_b^{FS} = B/\Delta p$. The book argues $R_b^{SS} \geq R_b^{FS}$ via the binding (IC′) and leaves aside $z^S = 0$; Step 5 covers that case and Step 6 gives the direct argument.

Ref: Tirole Ch. 7, §7.1.2.1 "Financial Muscle and Predation" (Bolton–Scharfstein), pp. 290–294; Answers, pp. 603–604.

7. Liquidation incentives with a private signal — Tirole 3.14, pp. 149–150

Statement.

Source: Tirole, Ch. 3, Exercise 3.14, pp. 149–150 (PDF 162) · Answer in book: yes, pp. 577–578

Exercise 3.14 (liquidation incentives). This exercise extends the fixed-investment model of Section 3.2 by adding a signal on the profitability of the project that (a) accrues after effort has been chosen, and (b) is privately observed. (The following model is used as a building block in a broader context by Dessi (2005).)

An entrepreneur has cash A and wants to invest $I > A$ into a project. The project yields R (success) or 0 (failure) at the end. An intermediate signal reveals the probability γ that the project will succeed, with $\gamma = \bar{\gamma}$ or $\underline{\gamma}$ ($\bar{\gamma} = \underline{\gamma} + \Delta\gamma$ and $\Delta\gamma > 0$). The probability, p, that $\gamma = \bar{\gamma}$ depends on the entrepreneur's effort. If the entrepreneur behaves, then $p = p_H$ and the entrepreneur receives no private benefit. If the entrepreneur misbehaves, then $p = p_L$ and the entrepreneur receives private benefit B. Investors and entrepreneur are risk neutral and the latter is protected by limited liability. The competitive rate of return is equal to 0.

Introduce further an option to liquidate after the signal is realized but before the final profit accrues. Liquidation yields L, and L is entirely pledgeable to investors.

One will assume that

$$\bar{\gamma}R > L > \underline{\gamma}R,$$

so that it is efficient to liquidate if and only if the signal is bad; and that

$$p_H\bar{\gamma}R + (1 - p_H)L > I$$

(which will imply that the NPV is positive).

Figure 3.11 summarizes the timing.

Figure 3.11 (timeline, left to right; p. 150, PDF 163)

Step 1	Step 2	Step 3	Step 4
The entrepreneur borrows $I - A$ and invests in a project with cost I.	The entrepreneur behaves ($\Pr(\gamma = \bar{\gamma}) = p_H$; no private benefit), or misbehaves ($\Pr(\gamma = \bar{\gamma}) = p_L$; private benefit B).	State of the world $\gamma = \bar{\gamma}$ or $\underline{\gamma}$ realized.	Continuation: success (R) with probability γ, failure (0) with probability $1 - \gamma$. Or liquidation (yields L).

(i) Suppose first that γ is verifiable. Argue that the entrepreneur should be rewarded solely as a function of the realization of γ. What is the pledgeable income? Show that the project is financed if and only if $A \geqslant \bar{A}$, where

$$p_H \left(\bar{\gamma} R - \frac{B}{\Delta p} \right) + (1 - p_H) L = I - \bar{A}.$$

(ii) Suppose now that γ is observed only by the entrepreneur. This implies that the entrepreneur must be induced to tell the truth about γ. Without loss of generality, consider an incentive scheme in which the entrepreneur receives R_b in the case she announces $\gamma = \bar{\gamma}$ (and therefore the project continues) *and* the final profit is R, L_b if she announces $\gamma = \underline{\gamma}$ (and therefore the project is liquidated), and 0 otherwise.

Show that the project is funded if and only if

$$A \geqslant \bar{A} + \underline{\gamma} \frac{B}{(\Delta p)(\Delta \gamma)}.$$

Context needed to solve it *From Tirole §3.2.1 "The Fixed-Investment Model", pp. 115–116 (PDF 128–129):*

Variants of the following entrepreneurial model will be used in the following: an entrepreneur (also called the "insider" or the "borrower," "she") has a project. This project requires a fixed investment I. The entrepreneur initially has "assets" ("cash on hand" or "net worth") $A < I$. For the moment we interpret these assets as being cash or liquid securities that can be used toward covering the cost of investment. [...] The entrepreneur's cash can either be invested in the project or used for consumption. To implement the project the entrepreneur must borrow $I - A$ from lenders. [...]

Project. If undertaken, the project either succeeds, that is, yields verifiable income $R > 0$, or fails, that is, yields no income. The probability of success is denoted by p. The project is subject to moral hazard. The entrepreneur can "behave" ("work," "exert effort," "take no private benefit") or "misbehave" ("shirk," "take a private benefit") [...].

Behaving yields probability $p = p_H$ of success and no private benefit to the entrepreneur, and misbehaving results in probability $p = p_L < p_H$ of success and private benefit $B > 0$ (measured in units of account) to the entrepreneur. In the "effort interpretation," B can also be interpreted as a disutility of effort saved by the entrepreneur when shirking. Let $\Delta p \equiv p_H - p_L$.

Preferences and the loan agreement. Both the borrower and the potential lenders (or "investors") are risk neutral. For notational simplicity, there is no time preference; the rate of return expected by investors (which is also the riskless rate, due to risk neutrality) is taken to be 0. The borrower is protected by limited liability, and so her income cannot take negative values.

Lenders behave competitively in the sense that the loan, if any, makes zero profit. [...]

Let us turn to the loan contract. A contract first stipulates whether the project is financed. If so, it further specifies how the profit is shared between the lenders and the borrower. The borrower's limited liability will imply that both sides receive 0 in the case of failure [...]. In the case of success, the two parties share the profit R; R_b goes to the borrower and R_l to the lenders. To sum up, we posit an incentive scheme for the entrepreneur of the following form: R_b in the case of success, 0 in the case of failure.

[...] We summarize the timing in Figure 3.1.

Figure 3.1 (timeline, left to right): Loan agreement (sharing rule) → Investment → Moral hazard → Outcome.

We assume that the project is viable only in the absence of moral hazard. That is, the project has positive NPV if the entrepreneur behaves,

$$p_H R - I > 0, \qquad (3.1)$$

but negative NPV, even if one includes the borrower's private benefit, if she does not,

$$p_L R - I + B < 0. \qquad (3.2)$$

From Tirole §3.2.2 "The Lenders' Credit Analysis", pp. 116–117 (PDF 129–130):
[...] the borrower will therefore behave if the following "incentive compatibility constraint" is satisfied:

$$p_H R_b \geqslant p_L R_b + B \quad \text{or} \quad (\Delta p) R_b \geqslant B. \qquad (\text{IC}_b)$$

From this incentive compatibility constraint we infer that the highest income in the case of success that can be pledged to the lenders without jeopardizing the borrower's incentives is

$$R - \frac{B}{\Delta p}.$$

The *(expected) pledgeable income* is then

$$\mathcal{P} = p_H \left(R - \frac{B}{\Delta p} \right).$$

Because the lenders must break even in order to be willing to finance the project, a necessary condition for the borrower to receive a loan is that the expected pledgeable income exceed the lenders' initial outlay:

$$\mathcal{P} \equiv p_H \left(R - \frac{B}{\Delta p} \right) \geqslant I - A, \qquad (\text{IR}_l)$$

[...] Thus a necessary condition for financing to be arranged is

$$A \geqslant \bar{A} = p_H \frac{B}{\Delta p} - (p_H R - I). \qquad (3.3)$$

To make things interesting, we will assume that

$$\bar{A} > 0 \iff p_H R - I < p_H \frac{B}{\Delta p}, \qquad (3.4)$$

otherwise even a borrower with no wealth of her own would be able to obtain credit. [...]
Thus, *the borrower must have enough assets in order to be granted a loan.* Note that, if $A < \bar{A}$, the project has positive NPV and yet is not funded. [...] There is *credit rationing.* [...]
Conversely, if $A \geqslant \bar{A}$, the entrepreneur is able to secure financing, and so condition (3.3) is both a necessary and a sufficient condition for financing. The entrepreneur offers claim R_l to competitive investors so as not to leave them with a surplus:

$$p_H R_l = I - A.$$

Her stake,

$$R_b = R - R_l = R - \frac{I - A}{p_H} \geqslant R - \frac{I - \bar{A}}{p_H} = \frac{B}{\Delta p}, \qquad (3.5)$$

then induces her to behave.
[...] Using the breakeven condition for the lenders ($p_H R_l = I - A$), the borrower obtains net utility or payoff (where "net" means that we subtract the consumption utility, A, that the entrepreneur would get by not undertaking the project):

$$U_b = \begin{cases} 0 & \text{if } A < \bar{A}, \\ p_H R_b - A = p_H (R - R_l) - A = p_H R - I & \text{if } A \geqslant \bar{A}. \end{cases} \qquad (3.6)$$

As could have been expected from the zero-profit condition for the lenders, *the borrower receives the entire social surplus or net present value if the project is funded.*
So, the borrower's utility jumps up at $A = \bar{A}$. [...]

What is asked. (i) With a verifiable interim signal γ: explain why the entrepreneur's pay should depend only on γ, compute the pledgeable income, and show that the project is financed iff $A \geq \bar{A}$. (ii) With γ seen only by the entrepreneur: design the cheapest scheme (R_b, L_b) that makes her both behave and report γ truthfully, and show that financing now requires $A \geq \bar{A} + \underline{\gamma} B / [(\Delta p)(\Delta \gamma)]$.

Approach. Fixed-investment model of Tirole §3.2 (pledgeable income, $\mathcal{P} \geq I - A$) with an interim liquidation option. In (i) the signal is a sufficient statistic for effort, so pay on γ and liquidate after bad news. In (ii) the liquidation decision rests on information only the insider has — the Topic 5 problem of nonverifiable information — so the contract must add truth-telling constraints: the entrepreneur must not gain by hiding bad news to avoid termination. The extra rent this requires is the increase in the net-worth threshold.

Notation.

- $A < I$: entrepreneur's cash and the investment; R: final profit on success; L: liquidation value, fully pledgeable.
- $\gamma \in \{\underline{\gamma}, \bar{\gamma}\}$: success probability revealed by the signal; $\Delta\gamma = \bar{\gamma} - \underline{\gamma} > 0$; $\bar{\gamma}R > L > \underline{\gamma}R$.
- p_H, p_L, $\Delta p = p_H - p_L > 0$: probability of $\gamma = \bar{\gamma}$ if she behaves / misbehaves; B: private benefit.
- (i) $w(\gamma) \geq 0$: reward after signal γ; $\mathcal{P}$: pledgeable income; $\bar{A}$: financing threshold.
- (ii) R_b: reward if she announces $\bar{\gamma}$ and the profit is R; L_b: reward if she announces $\underline{\gamma}$ (liquidation); $\mathcal{P}'$: pledgeable income.

Step 1 — (i) Rewards depend on γ only. Effort affects only the distribution of γ; once γ is known, the final outcome is pure noise for the entrepreneur (γ is a sufficient statistic for effort). With risk neutrality, any scheme that pays according to γ and the final profit acts on effort and on the investors' cost only through the expected reward given γ, $w(\gamma)$. Paying $w(\gamma)$ directly leaves both unchanged, so rewards can be made contingent on γ alone.

Step 2 — (i) Liquidation policy. With γ verifiable, the contract liquidates if and only if $\gamma = \underline{\gamma}$: this raises investors' income by $L - \underline{\gamma}R > 0$ after a bad signal, and continuing after a good one gives $\bar{\gamma}R > L$. The rewards are transfers that do not depend on this choice, so it maximises both NPV and pledgeable income.

Step 3 — (i) Incentive constraint and pledgeable income. Pledgeable income is total expected income minus the smallest expected rent compatible with effort, so we first find that rent. Behaving is optimal if

$$\begin{aligned} p_H w(\bar{\gamma}) + (1-p_H)w(\underline{\gamma}) &\geq p_L w(\bar{\gamma}) + (1-p_L)w(\underline{\gamma}) + B, \\ (p_H - p_L)w(\bar{\gamma}) - \big[(1-p_L) - (1-p_H)\big]w(\underline{\gamma}) &\geq B, \\ (\Delta p)\big[w(\bar{\gamma}) - w(\underline{\gamma})\big] &\geq B. \end{aligned} \tag{44}$$

The expected rent $p_H w(\bar{\gamma}) + (1-p_H)w(\underline{\gamma})$ rises in both rewards, so the cheapest scheme sets $w(\underline{\gamma}) = 0$ (limited liability) and $w(\bar{\gamma}) = B/\Delta p$, an expected rent $p_H B/\Delta p$. With the liquidation policy of Step 2, total expected income is $p_H\bar{\gamma}R + (1-p_H)L$. Hence

$$\mathcal{P} = p_H\bar{\gamma}R + (1-p_H)L - p_H\frac{B}{\Delta p}. \tag{45}$$

Step 4 — (i) Financing condition. Investors lend if $\mathcal{P} \geq I - A$. Using (45) and the definition of $\bar{A}$, $p_H(\bar{\gamma}R - B/\Delta p) + (1-p_H)L = I - \bar{A}$:

$$\begin{aligned} p_H\left(\bar{\gamma}R - \frac{B}{\Delta p}\right) + (1-p_H)L \geq I - A \\ \iff \quad I - \bar{A} \geq I - A \\ \iff \quad A \geq \bar{A}. \end{aligned} \tag{46}$$

Answer (i). Pay on γ only, $w(\bar{\gamma}) = B/\Delta p$, $w(\underline{\gamma}) = 0$, liquidate iff $\gamma = \underline{\gamma}$; $\mathcal{P} = p_H(\bar{\gamma}R - B/\Delta p) + (1-p_H)L$; financed iff $A \geq \bar{A}$.

Step 5 — (ii) Truth-telling constraints. Now only the entrepreneur sees γ. Announcing $\bar{\gamma}$ gives continuation and an expected reward γR_b; announcing $\underline{\gamma}$ gives liquidation and L_b. Truth-telling in each state requires

$$\bar{\gamma} R_b \geq L_b, \tag{47}$$
$$L_b \geq \underline{\gamma} R_b. \tag{48}$$

Step 6 — (ii) Effort constraint. Given truthful reports, she gets $\bar{\gamma} R_b$ after a good signal and L_b after a bad one, so, as in (44),

$$(\Delta p)(\bar{\gamma} R_b - L_b) \geq B. \tag{49}$$

Step 7 — (ii) Cheapest scheme. Investors receive $p_H \bar{\gamma}(R - R_b) + (1 - p_H)(L - L_b)$, so they minimise the expected rent $p_H \bar{\gamma} R_b + (1 - p_H) L_b$. Both R_b and L_b should be as low as (48) and (49) allow; making both bind,

$$L_b = \underline{\gamma} R_b, \tag{50}$$
$$(\Delta p)(\bar{\gamma} R_b - (\underline{\gamma} R_b)) = B,$$
$$R_b = \frac{B}{(\Delta p)(\Delta \gamma)}, \tag{51}$$
$$L_b = \underline{\gamma} \frac{B}{(\Delta p)(\Delta \gamma)}.$$

(Formally: with $\delta \equiv \bar{\gamma} R_b - L_b \geq B/\Delta p$, (48) is $L_b \geq \underline{\gamma}\delta/\Delta\gamma$, so the rent $L_b + p_H \delta \geq (\underline{\gamma}/\Delta\gamma + p_H)\delta$ is minimised at $\delta = B/\Delta p$.) (47) holds since $\bar{\gamma} R_b - L_b = B/\Delta p > 0$.

Step 8 — (ii) Pledgeable income and financing condition. Substituting (50)–(51) into the rent, and writing $\bar{\gamma} = \underline{\gamma} + \Delta\gamma$ in the second line,

$$\begin{aligned} p_H \bar{\gamma} R_b + (1 - p_H) L_b &= \left[p_H \bar{\gamma} + (1 - p_H)\underline{\gamma}\right] \frac{B}{(\Delta p)(\Delta \gamma)} \\ &= \left[p_H \underline{\gamma} + p_H \Delta\gamma + \underline{\gamma} - p_H \underline{\gamma}\right] \frac{B}{(\Delta p)(\Delta \gamma)} \\ &= \left[\underline{\gamma} + p_H \Delta\gamma\right] \frac{B}{(\Delta p)(\Delta \gamma)} \\ &= p_H \frac{B}{\Delta p} + \underline{\gamma} \frac{B}{(\Delta p)(\Delta \gamma)}. \end{aligned} \tag{52}$$

So, subtracting this rent from total expected income and using (45),

$$\begin{aligned} \mathcal{P}' &= p_H \bar{\gamma} R + (1 - p_H) L - p_H \frac{B}{\Delta p} - \underline{\gamma} \frac{B}{(\Delta p)(\Delta \gamma)} \\ &= \mathcal{P} - \underline{\gamma} \frac{B}{(\Delta p)(\Delta \gamma)}, \end{aligned} \tag{53}$$

and $\mathcal{P}' \geq I - A$ is, using $\mathcal{P} = I - \bar{A}$ from (46),

$$\begin{aligned} I - \bar{A} - \underline{\gamma} \frac{B}{(\Delta p)(\Delta \gamma)} &\geq I - A \\ \iff \quad A &\geq \bar{A} + \underline{\gamma} \frac{B}{(\Delta p)(\Delta \gamma)}. \end{aligned} \tag{54}$$

■

(Continuing after both signals would avoid the truth-telling problem but costs the same rent, since then $(\Delta p)(\Delta\gamma) R_b \geq B$, and loses $(1 - p_H)(L - \underline{\gamma} R)$ of pledgeable income; so truthful liquidation is better. Implicitly $R_b \leq R$ and $L_b \leq L$.)

Answer (ii). $R_b = \dfrac{B}{(\Delta p)(\Delta\gamma)}$, $L_b = \underline{\gamma} R_b$; pledgeable income falls by $\underline{\gamma}B/[(\Delta p)(\Delta\gamma)]$, so the project is funded iff $A \geq \bar{A} + \underline{\gamma}\,\dfrac{B}{(\Delta p)(\Delta\gamma)}$.

Intuition. When the signal is private, the entrepreneur must be paid to admit bad news, because admitting it kills the project in which she has a stake: she gets $L_b = \underline{\gamma} R_b$, a "golden parachute" worth what she would expect by hiding the news. That raises the stake needed for effort from $B/\Delta p$ to $B/[(\Delta p)(\Delta\gamma)]$ and the expected rent by $\underline{\gamma}B/[(\Delta p)(\Delta\gamma)]$; the net-worth threshold rises one for one. It vanishes when $\underline{\gamma} = 0$ (nothing to hide) and grows as the signal becomes less informative ($\Delta\gamma \to 0$).

Common mistakes.

- Setting $L_b = 0$ in (ii): then an entrepreneur with bad news reports $\bar{\gamma}$ (she gets $\underline{\gamma} R_b > 0$ by continuing), the project is never liquidated and the L income is lost.
- Writing the effort constraint as $(\Delta p)\bar{\gamma} R_b \geq B$: after a bad signal she still gets L_b, so what effort buys is the *difference* $\bar{\gamma} R_b - L_b$.
- Counting L as pledgeable after a good signal or $\bar{\gamma} R$ after a bad one: the efficient policy continues iff $\gamma = \bar{\gamma}$ because $\bar{\gamma} R > L > \underline{\gamma} R$.

Book answer. Matches the book's answer (pp. 577–578): $L_b = \underline{\gamma} R_b$, $(\Delta p)(\bar{\gamma} R_b - L_b) = B$, and the threshold $\bar{A} + \underline{\gamma}B/[(\Delta p)(\Delta\gamma)]$.

Ref: Tirole Ch. 3, §3.2 "The Fixed-Investment Model" and §3.2.2 "The Lenders' Credit Analysis", pp. 115–117; Exercise 3.14, pp. 149–150; Answers, pp. 577–578.

8. Decreasing returns to scale and debt maturity — Tirole Exercise 5.7, p. 232

Statement.

Source: Tirole, Ch. 5, Exercise 5.7, p. 232 (PDF 245) · Answer in book: yes, p. 593

Exercise 5.7 (decreasing returns to scale). Extend the treatment of Section 5.6.1 to the case of decreasing returns to scale: the payoff in the case of continuation and success is $R(I)$, with $R(0) = 0$, $R' > 0$, $R'' < 0$, $R'(0) = \infty$, and $R'(\infty) = 0$. The rest is unchanged (the short-term income is rI, the reinvestment need is ρI, and the private benefit is BI).

(i) What are the first-order conditions yielding the optimal investment level I and cutoff ρ^*?

(ii) Assuming that $r > \rho^*$, and that $(R(I)/I - R'(I))$ is increasing in I (a condition satisfied, for example, by $R(I)$ quadratic), derive the impact of the strength of the balance sheet as measured, say, by A on debt maturity.

Context needed to solve it *From Tirole §3.4.1 "The Continuous-Investment Model", p. 127 (PDF 140):*

The continuous-investment model of this section is the polar opposite of the fixed-investment one. The fixed-investment model depicts a situation in which returns are sharply decreasing beyond some investment level. In contrast, we now assume that there are constant returns to scale in the investment technology. An investment $I \in [0, \infty)$ yields income RI, proportional to I, in the case of success, and 0 in the case of failure. The borrower's private benefit from misbehaving is also taken to be proportional to investment. As before, the borrower has a choice between behaving, in which case she derives no private benefit and the probability of success is p_H, and misbehaving, that is, enjoying private benefit BI, and reducing the probability of success to $p_L = p_H - \Delta p < p_H$. [...]

The borrower initially has cash A, and must therefore borrow $I - A$ to finance a project of size I. A loan agreement specifies that the lenders (who as before are assumed to make no profit) and the borrower receive 0 each in the case of failure, and R_l and R_b, respectively, in the case of success, where $R_l + R_b = RI$.

From Tirole §5.3.2 "Continuum of Liquidity Shocks", pp. 209–210 (PDF 222–223):

After the (endogenous size) investment I is sunk at date 0 and before the borrower works on the project, some exogenous shock occurs at date 1 that determines a per-unit-of-investment level $\rho \in [0, \infty)$ of "cost overruns." That is, a cash infusion equal to ρI is needed in order for the project to continue. If ρI is not

invested, the project is abandoned altogether and thus yields no income. As in Section 5.2, the fraction ρ is *a priori* distributed according to the continuous distribution $F(\rho)$ on $[0, \infty)$, with density $f(\rho)$. (As we already observed, the model of Section 3.4 is therefore a special case, with F being a spike at $\rho = 0$.)

Regardless of the required amount of the cash infusion, the project, if pursued, is still a project of size I, in that the income in the case of success is RI and the borrower's private benefit from misbehaving is BI. One cannot increase the size of the project after the initial stage.

The timing is summarized in Figure 5.4.

Figure 5.4 (timing, transcribed): **Date 0** — Loan agreement; Investment I. **Date 1** — Need for cash infusion ρI realized; then either Disbursement → Moral hazard, or No disbursement → Project is abandoned. **Date 2** — Outcome.

[...] It is easy to show that it is optimal to have a "cutoff rule" for infusing cash. There exists an optimal threshold ρ^* such that one should continue if and only if

$$\rho \leqslant \rho^*. \qquad (5.2)$$

The incentive constraint in the case of continuation is the same as in the absence of a liquidity shock (see Section 3.4):

$$(\Delta p) R_b \geqslant BI. \qquad (\mathrm{IC}_b)$$

The breakeven condition is slightly altered by the presence of liquidity shocks:

$$F(\rho^*)[p_H(RI - R_b)] \geqslant I - A + \int_0^{\rho^*} \rho I f(\rho)\, \mathrm{d}\rho. \qquad (\mathrm{IR}_l)$$

That is, the lenders receive a return only if the project is pursued, which has probability $F(\rho^*)$. The left-hand side of (IR_l) is the expected pledgeable income. Furthermore, there is a new term, representing the expected outlay on overruns, on the right-hand side. From these two constraints, we deduce the borrowing capacity (or, more precisely, the maximum investment that allows the lenders to break even):

$$I = k(\rho^*) A,$$

where

$$k(\rho^*) = \frac{1}{1 + \int_0^{\rho^*} \rho f(\rho)\, \mathrm{d}\rho - F(\rho^*)[p_H R - p_H B/\Delta p]} = \frac{1}{1 + \int_0^{\rho^*} \rho f(\rho)\, \mathrm{d}\rho - F(\rho^*)\rho_0} \qquad (5.3)$$

involves a straightforward modification relative to the no-liquidity-shock multiplier k. [...] Note that the borrower's borrowing capacity is maximal when the threshold ρ^* is equal to the expected per-unit pledgeable income $\rho_0 \equiv p_H(R - B/\Delta p)$.

Given that the competitive lenders make no profits, the borrower's net utility is as usual the social surplus brought about by the project, namely,

$$U_b = m(\rho^*) I = m(\rho^*) k(\rho^*) A, \qquad (5.4)$$

where

$$m(\rho^*) \equiv F(\rho^*) p_H R - 1 - \int_0^{\rho^*} \rho f(\rho)\, \mathrm{d}\rho$$

is the margin per unit of investment.

[...] Next, we observe that *this optimal threshold lies between the expected per-unit-of-investment pledgeable income and income*:

$$\rho_0 = p_H \left(R - \frac{B}{\Delta p} \right) < \rho^* < \rho_1 = p_H R. \qquad (5.8)$$

From Tirole §5.6.1 "Optimal Claim Maturity", p. 226 (PDF 239):

Let us return to the continuous-investment, continuous-shock version of Section 5.3.2, but with a short-term income (the analysis is not really new and is therefore only sketched): see Figure 5.10. Because the short-term income rI is fully pledgeable to the lenders, everything is *as if* the unit investment cost were equal to $1 - r$ instead of 1. The lenders' breakeven condition, that is, the equality between expected revenue and expected investment cost, becomes

$$rI + F(\rho^*)\rho_0 I = I - A + \left[\int_0^{\rho^*} \rho f(\rho)\,\mathrm{d}\rho\right] I$$

and so

$$k(\rho^*) = \frac{1}{1 + \int_0^{\rho^*} \rho f(\rho)\,\mathrm{d}\rho - [r + F(\rho^*)\rho_0]}. \qquad (5.3'')$$

The margin (expected profit of the firm per unit of investment) becomes

$$m(\rho^*) = [r + F(\rho^*)\rho_1] - \left[1 + \int_0^{\rho^*} \rho f(\rho)\,\mathrm{d}\rho\right]. \qquad (5.4'')$$

And thus the borrower's (gross) utility becomes

$$U_b = m(\rho^*)k(\rho^*)A = \frac{\rho_1 - c(\rho^*)}{c(\rho^*) - \rho_0}A,$$

where the expected unit cost of effective investment, $c(\rho^*)$, is given by

$$c(\rho^*) = \frac{1 - r + \int_0^{\rho^*} \rho f(\rho)\,\mathrm{d}\rho}{F(\rho^*)}. \qquad (5.5'')$$

So the optimal threshold is given by

$$\int_0^{\rho^*} F(\rho)\,\mathrm{d}\rho = 1 - r \qquad (5.6'')$$

and the borrower's utility by

$$U_b = \frac{\rho_1 - \rho^*}{\rho^* - \rho_0}A. \qquad (5.7'')$$

Figure 5.10 (timing, transcribed): **Date 0** — Loan agreement. Investment I. **Date 1** — Accrual of short-term income rI. Realization ρI of investment need. Then either Disbursement → Moral hazard, or No disbursement → Project is abandoned. **Date 2** — Date-2 income (RI or 0).

It is important to note that *the short-term income, even though it is deterministic and fully pledgeable, is not equivalent to an increase in the borrower's cash on hand A.* Such an increase in equity would result in a larger investment (as is the case here), but not in a modification of the continuation rule. By contrast, condition (5.6″) shows that the larger the short-term profit, the lower the optimal threshold ρ^*. [...]

(a) *Liquidity management.* Let us now turn to the implementation of the optimum and thus to the claim maturity. To this purpose we make the following assumption. (Footnote 57: Of course, it must also be the case that $\rho^* > \rho_0$ (otherwise, the borrower's borrowing capacity and utility would be infinite in this constant-returns-to-scale model). Because $\mathrm{d}\rho^*/\mathrm{d}r < -1$, we must thus also assume that r is not "too large.")

Free-cash-flow assumption: $r > \rho^*$. Under the free-cash-flow assumption, and given that the entrepreneur cannot steal the intermediate income, the entrepreneur would reinvest excessively if she were not asked to pay out money to investors at date 1. Namely, she would reinvest as long as $\rho \leqslant r$.

To obtain the optimal amount of reinvestment, an amount $P_1 \equiv (r - \rho^*)I$ must be pumped out of the firm, and the entrepreneur must be denied the right to dilute initial investors.

What is asked. (i) In the liquidity model with short-term income rI and a continuum of reinvestment shocks ρI, but with a concave success payoff $R(I)$, derive the first-order conditions that pin down the scale I and the continuation cutoff ρ^*. (ii) Under free cash flow ($r > \rho^*$), find how a stronger balance sheet (higher A) changes debt maturity, i.e. the short-term debt the firm must pay at date 1 per unit of investment.

Approach. Holmström–Tirole liquidity model (Tirole §5.3.2 and §5.6.1): the borrower maximises the NPV subject to the lenders' break-even, a Lagrangian problem in (I, ρ^*). Continuing after a shock ρ costs ρI and is worth ρ_1 in value but only ρ_0 in pledgeable income, so the optimal cutoff

lies between them. Under free cash flow the cutoff is implemented by short-term debt that pumps $(r-\rho^*)I$ out of the firm — the termination threat of the lecture, here a liquidity sweep — so maturity moves with ρ^*, and comparative statics in A answer (ii).

Notation.

- $I > 0$: investment (endogenous); $A \geq 0$: the entrepreneur's cash; lenders are competitive, rate of return 0.
- $R(I)$: date-2 income in case of continuation and success, $R(0) = 0$, $R' > 0$, $R'' < 0$.
- rI: short-term (date-1) income, verifiable and fully pledgeable, $r > 0$.
- $\rho \geq 0$: per-unit liquidity shock (reinvestment need ρI), c.d.f. F, density f on $[0, \infty)$; ρ^*: continuation cutoff (continue iff $\rho \leq \rho^*$).
- $p_H > p_L$: success probabilities when behaving / misbehaving, $\Delta p = p_H - p_L$; BI: private benefit from misbehaving; R_b: entrepreneur's stake in success.
- $\rho_1(I) \equiv p_H R(I)/I$ and $\rho_0(I) \equiv p_H[R(I)/I - B/\Delta p]$: per-unit expected income and per-unit pledgeable income at scale I; $\rho_1 - \rho_0 = p_H B/\Delta p$.
- $\mu \geq 0$: Lagrange multiplier of the lenders' breakeven constraint.

Step 1 — The program. We set up the borrower's problem. As in Section 5.6.1, the incentive constraint $(\Delta p)R_b \geq BI$ binds (any larger stake only lowers pledgeable income), so the income pledgeable in case of success is $R(I) - BI/\Delta p$. With competitive lenders the entrepreneur gets the NPV: short-term income, plus expected date-2 income when continuing (probability $F(\rho^*)$), minus the investment and the expected reinvestment,

$$U_b = rI + F(\rho^*)\, p_H R(I) - \Big[1 + \int_0^{\rho^*} \rho f(\rho)\, d\rho\Big] I, \tag{55}$$

subject to the lenders' breakeven constraint

$$rI + F(\rho^*)\, p_H \Big[R(I) - \frac{BI}{\Delta p}\Big] \geq I - A + \Big[\int_0^{\rho^*} \rho f(\rho)\, d\rho\Big] I. \tag{56}$$

The Lagrangian is $\mathcal{L} = U_b + \mu\,[\text{LHS} - \text{RHS of (56)}]$.

Step 2 — (i) First-order conditions. We differentiate $\mathcal{L}$ with respect to each choice variable. By the fundamental theorem of calculus, $\frac{d}{d\rho^*}\int_0^{\rho^*} \rho f\, d\rho = \rho^* f(\rho^*)$, and $\frac{d}{d\rho^*}F(\rho^*) = f(\rho^*)$. Term by term,

$$\frac{\partial U_b}{\partial \rho^*} = f(\rho^*)\, p_H R(I) - \rho^* f(\rho^*)\, I,$$
$$\frac{\partial[\text{LHS} - \text{RHS of (56)}]}{\partial \rho^*} = f(\rho^*)\, p_H \Big[R(I) - \frac{BI}{\Delta p}\Big] - \rho^* f(\rho^*)\, I,$$

so $\partial\mathcal{L}/\partial\rho^* = 0$ reads

$$0 = f(\rho^*)\big[p_H R(I) - \rho^* I\big] + \mu f(\rho^*)\Big[p_H R(I) - \frac{p_H BI}{\Delta p} - \rho^* I\Big].$$

Dividing by $f(\rho^*)I > 0$ and using $\rho_1(I) = p_H R(I)/I$ and $\rho_0(I) = p_H[R(I)/I - B/\Delta p]$,

$$0 = \big[\rho_1(I) - \rho^*\big] + \mu\big[\rho_0(I) - \rho^*\big]. \tag{57}$$

Collecting the ρ^* terms, $(1+\mu)\rho^* = \rho_1(I) + \mu\rho_0(I)$, so $\rho^* = \frac{\rho_1(I) + \mu\rho_0(I)}{1+\mu}$, a weighted average that lies between $\rho_0(I)$ and $\rho_1(I)$, the analogue of (5.8). With respect to I, the derivatives are

$$\frac{\partial U_b}{\partial I} = r + F(\rho^*) p_H R'(I) - 1 - \int_0^{\rho^*} \rho f\, d\rho,$$
$$\frac{\partial[\text{LHS} - \text{RHS of (56)}]}{\partial I} = r + F(\rho^*) p_H \Big(R'(I) - \frac{B}{\Delta p}\Big) - 1 - \int_0^{\rho^*} \rho f\, d\rho,$$

so $\partial\mathcal{L}/\partial I = 0$ is

$$0 = \Big[r + F(\rho^*)p_H R'(I) - 1 - \int_0^{\rho^*} \rho f\, d\rho\Big] + \mu\Big[r + F(\rho^*)p_H\Big(R'(I) - \frac{B}{\Delta p}\Big) - 1 - \int_0^{\rho^*} \rho f\, d\rho\Big]. \quad (58)$$

These two conditions, together with (56) holding with equality, determine (I, ρ^*, μ).

Step 3 — (i) Eliminating μ. The multiplier is not observable, so we combine the two conditions into one that involves only (I, ρ^*). Write the second bracket of (58) as the first minus $F(\rho^*)p_H B/\Delta p$ and collect:

$$(1+\mu)\Big[r + F(\rho^*)p_H R'(I) - 1 - \int_0^{\rho^*} \rho f\, d\rho\Big] = \mu F(\rho^*)\frac{p_H B}{\Delta p}. \quad (59)$$

From (57), $\mu(\rho^* - \rho_0) = \rho_1 - \rho^*$, so $\mu = \frac{\rho_1 - \rho^*}{\rho^* - \rho_0}$ and $1 + \mu = \frac{\rho^* - \rho_0 + \rho_1 - \rho^*}{\rho^* - \rho_0} = \frac{\rho_1 - \rho_0}{\rho^* - \rho_0}$, hence $\frac{\mu}{1+\mu} = \frac{\rho_1 - \rho^*}{\rho_1 - \rho_0}$. Since $\rho_1 - \rho_0 = p_H\frac{R(I)}{I} - p_H\Big(\frac{R(I)}{I} - \frac{B}{\Delta p}\Big) = \frac{p_H B}{\Delta p}$,

$$\frac{\mu}{1+\mu}\,\frac{p_H B}{\Delta p} = \frac{\rho_1(I) - \rho^*}{\rho_1(I) - \rho_0(I)}\,(\rho_1(I) - \rho_0(I)) = \rho_1(I) - \rho^*. \quad (60)$$

Dividing (59) by $1 + \mu$ and substituting (60):

$$r + F(\rho^*)p_H R'(I) - 1 - \int_0^{\rho^*} \rho f\, d\rho = F(\rho^*)\Big(\frac{p_H R(I)}{I} - \rho^*\Big).$$

Move the R terms to the left and the rest to the right:

$$\begin{aligned} F(\rho^*)\,p_H\Big[R'(I) - \frac{R(I)}{I}\Big] &= 1 - r + \int_0^{\rho^*} \rho f\, d\rho - \rho^* F(\rho^*) \\ &= 1 - r - \Big[\rho^* F(\rho^*) - \int_0^{\rho^*} \rho f\, d\rho\Big]. \end{aligned}$$

The bracket is $\int_0^{\rho^*}(\rho^* - \rho)f(\rho)\,d\rho$; integrating by parts with $F(0) = 0$, $\int_0^{\rho^*}(\rho^* - \rho)f\,d\rho = \big[(\rho^* - \rho)F(\rho)\big]_0^{\rho^*} + \int_0^{\rho^*} F(\rho)\,d\rho = \int_0^{\rho^*} F(\rho)\,d\rho$. Dividing by $F(\rho^*) > 0$,

$$p_H\Big[R'(I) - \frac{R(I)}{I}\Big] = \frac{1 - r - \int_0^{\rho^*} F(\rho)\,d\rho}{F(\rho^*)}. \quad (61)$$

The binding (56), solved for A and with $p_H[R(I) - BI/\Delta p] = I\rho_0(I)$, reads

$$\begin{aligned} A &= I - rI + \Big[\int_0^{\rho^*} \rho f(\rho)\,d\rho\Big]I - F(\rho^*)\,I\rho_0(I) \\ &= I\Big[1 - r + \int_0^{\rho^*} \rho f(\rho)\,d\rho - F(\rho^*)\rho_0(I)\Big]. \end{aligned} \quad (62)$$

Check: with constant returns, $R(I) = RI$, the left side of (61) is 0 and (61) becomes $\int_0^{\rho^*} F = 1 - r$, which is (5.6″). With strict concavity and $R(0) = 0$, $R(I)/I > R'(I)$, so the left side is negative and $\int_0^{\rho^*} F > 1 - r$: the marginal unit of scale is worth less than the average one, so at the margin the firm spends more on continuation and less on scale than (5.6″) would say.

Answer (i). The FOCs are (57), $\rho_1(I) - \rho^* + \mu[\rho_0(I) - \rho^*] = 0$, and (58); eliminating μ, the optimum (I, ρ^*) solves $p_H[R'(I) - R(I)/I] = [1 - r - \int_0^{\rho^*} F(\rho)\,d\rho]/F(\rho^*)$ together with the binding break-even (62), with $\rho_0(I) < \rho^* < \rho_1(I)$.

Step 4 — (ii) I and ρ^* move together. Condition (61) links the two choices; we find the sign of that link. Write $\varphi(I) \equiv p_H[R(I)/I - R'(I)] > 0$, increasing by assumption, and $h(\rho^*)$ for the

right side of (61), so (61) is $-\varphi(I) = h(\rho^*)$. The numerator of h is $N(\rho^*) = 1 - r - \int_0^{\rho^*} F$, with $N'(\rho^*) = -F(\rho^*)$; by the quotient rule,

$$\begin{aligned} h'(\rho^*) &= \frac{N'(\rho^*)F(\rho^*) - N(\rho^*)f(\rho^*)}{F(\rho^*)^2} \\ &= \frac{-F(\rho^*)^2 - f(\rho^*)\big[1 - r - \int_0^{\rho^*} F\big]}{F(\rho^*)^2} \\ &= -1 - \frac{f(\rho^*)}{F(\rho^*)}\, h(\rho^*). \end{aligned} \tag{63}$$

Assumption. Tirole's answer takes h decreasing. Since $h < 0$ at the optimum, (63) is negative iff $f(\rho^*)\big[\int_0^{\rho^*} F - (1-r)\big] < F(\rho^*)^2$. This holds, e.g., for F uniform on $[0, \bar\rho]$ ($f = 1/\bar\rho$, $F(\rho^*) = \rho^*/\bar\rho$, $\int_0^{\rho^*} F = \rho^{*2}/(2\bar\rho)$): multiplying by $\bar\rho^2$, the condition reads $\rho^{*2}/2 - \bar\rho(1-r) < \rho^{*2}$, always true for $r \le 1$. We assume it.

Then along (61) an increase in I lowers $-\varphi(I)$, which requires a lower h, i.e. a higher ρ^*. Totally differentiating $-\varphi(I) = h(\rho^*)$ gives $-\varphi'(I)\, dI = h'(\rho^*)\, d\rho^*$, so (61) defines $\rho^* = g(I)$ with

$$g'(I) = \frac{-\varphi'(I)}{h'(\rho^*)} > 0. \tag{64}$$

Step 5 — (ii) Effect of A. The binding break-even (62) tells how much net worth a given (I, ρ^*) requires; we differentiate its right side. With respect to ρ^*,

$$\begin{aligned} \frac{\partial A}{\partial \rho^*} &= I\big[\rho^* f(\rho^*) - f(\rho^*)\rho_0(I)\big] \\ &= I f(\rho^*)\big[\rho^* - \rho_0(I)\big] > 0, \end{aligned} \tag{65}$$

positive because $\rho^* > \rho_0(I)$ by (57). With respect to I (using $\frac{d}{dI}[I\rho_0(I)] = p_H[R'(I) - B/\Delta p]$),

$$\frac{\partial A}{\partial I} = 1 - r + \int_0^{\rho^*} \rho f\, d\rho - F(\rho^*) p_H \Big[R'(I) - \frac{B}{\Delta p}\Big].$$

The second bracket of (58) is $-\partial A/\partial I$ and the first is $-\partial A/\partial I + F(\rho^*)p_H B/\Delta p$, so (58) reads $-(1+\mu)\,\partial A/\partial I + F(\rho^*)p_H B/\Delta p = 0$, i.e.

$$\frac{\partial A}{\partial I} = \frac{F(\rho^*)\, p_H B}{(1+\mu)\Delta p} > 0. \tag{66}$$

Along the optimum, $\rho^* = g(I)$, so by (64), (65) and (66)

$$\frac{dA}{dI} = \frac{\partial A}{\partial I} + \frac{\partial A}{\partial \rho^*}\, g'(I) > 0, \tag{67}$$

hence $dI/dA = 1/(dA/dI) > 0$ and $d\rho^*/dA = g'(I)\, dI/dA > 0$: a weaker balance sheet lowers both the scale and the cutoff.

Step 6 — (ii) Debt maturity. Under the free-cash-flow assumption $r > \rho^*$ the optimum is implemented as in Section 5.6.1(a): the firm keeps $\rho^* I$ to meet shocks $\rho \le \rho^*$, must pay out $P_1 = (r - \rho^*)I$ at date 1 (short-term debt), is barred from diluting the initial investors, and the investors' long-term claim is $R(I) - BI/\Delta p$ in case of success. Per unit of investment the short-term debt is $r - \rho^*$, and by Step 5

$$\frac{d(r-\rho^*)}{dA} = -\frac{d\rho^*}{dA} < 0. \tag{68}$$

A firm with a weaker balance sheet (lower A) issues more short-term debt per unit of investment: its debt maturity is shorter. (The level $(r-\rho^*)I$ can move either way, because I falls too.)

Answer (ii). $dI/dA > 0$ and $d\rho^*/dA > 0$, so the short-term debt per unit of investment, $r - \rho^*$, falls with A: weaker balance sheets carry shorter maturity (more of the date-1 cash flow is paid out), provided $h'(\rho^*) < 0$ (true, e.g., for uniform shocks).

Intuition. A weak balance sheet makes pledgeable income scarce. The firm raises it by continuing less often, i.e. by keeping less liquidity in the firm, which means sweeping more of the date-1 cash flow out to creditors through short-term debt. Decreasing returns add one more channel: a lower scale raises $R(I)/I$ and hence per-unit pledgeable income, and the two margins (scale and continuation) are cut together.

Common mistakes.

- Carrying over the constant-returns result $\int_0^{\rho^*} F = 1 - r$, under which ρ^* does not depend on A: with $R'' < 0$ the left side of (61) is negative and moves with I, which is what makes ρ^* depend on A.
- Stating the FOCs without the multiplier on the break-even: the cutoff is a weighted average of ρ_1 (value) and ρ_0 (pledgeable income), not ρ_1 alone.
- Reading "shorter maturity" off the level $(r - \rho^*)I$: since I also falls with A, the unambiguous statement is per unit of investment.

Book answer. Matches the book's answer (p. 593): same condition (61) (the book writes its numerator as $1 - r - \int_0^{\rho^*}(\rho^* - \rho)f(\rho)d\rho$, equal to ours by integration by parts) and the same conclusion (lower $A \Rightarrow$ lower I and $\rho^* \Rightarrow$ more short-term debt). The book states without proof that the right side of (61) is decreasing in ρ^*; (63) shows this needs a condition on F (true for the uniform).

Ref: Tirole §5.3.2 "Continuum of Liquidity Shocks", pp. 209–210; §5.6.1 "Optimal Claim Maturity", p. 226; Exercise 5.7, p. 232; answer p. 593.

9. The priority game: uncoordinated lending and short-term bias — Tirole Exercise 5.9, p. 233

Statement.

Source: Tirole, Ch. 5, Exercise 5.9, p. 233 (PDF 246) · Answer in book: yes, p. 594

Exercise 5.9 (the priority game: uncoordinated lending leads to a short-term bias). This chapter, like Chapters 3 and 4, has assumed that the firm's balance sheet is transparent. In particular, each investor has perfect knowledge of loans made by other lenders and of the firm's obligations to them.

This exercise argues that uncoordinated lending leads to financing that is too oriented to the short term. In a nutshell, lenders, by cashing out early, exert a negative externality on other lenders. Because this externality is not internalized, the resulting financial structure contains too much short-term debt.

We consider a three-period model: $t = 0, 1, 2$. The entrepreneur has no cash ($A = 0$), is risk neutral, and is protected by limited liability. At date 0, a fixed investment I is made. The project yields a known return $r > 0$ at date 1, and an uncertain return (R or 0) at date 2. Because the point is quite general and does not require credit constraints, we assume away moral hazard; or, equivalently, the private benefit from misbehaving is 0. The probability of a date 2 success is

$$p + \tau(I_1),$$

where I_1 is the date-1 deepening investment, equal to r minus the level of short-term debt repaid to lenders and the date-1 payment to the entrepreneur (the firm does not return to the capital market at date 1), and τ is an increasing and concave function (with $\tau'(0) = \infty$). Assume that $\tau'(r)R < 1$.

We assume that the entrepreneur cannot engage in "fraud," that is, cannot fail to honor the short-term debt and, if the project succeeds at date 2, the long-term debt. By contrast, obligations to lenders, and in particular I_1, cannot be verified as the firm's balance sheet is opaque.

(i) Derive the first-best investment I_1^*. Show how this allocation can be implemented by a mixture of short- and long-term debt (note that in this model without moral hazard the structure of compensation for the entrepreneur exhibits a degree of indeterminacy).

(ii) Assume that $r - I_1^* < I$ (creditors must hold long-term debt). Suppose next that financing is not transparent. Start from the first-best solution, with a large number (a continuum of mass 1) of lenders, with the representative lender owning short-term claim r_l and contingent long-term claim R_l on the firm.

Show that the entrepreneur has an incentive to secretly collude with any lender to increase the latter's short-term claim in exchange for a smaller long-term claim.

Given the constraint that financing is provided by many lenders and that the latter do not observe each other's contracts, is the indeterminacy mentioned in question (i) resolved?

What is asked. (i) Find the first-best date-1 deepening investment I_1^* and show that a mix of short- and long-term debt implements it, with some indeterminacy in how the claims (and the entrepreneur's pay) are split. (ii) With an opaque balance sheet and many lenders, show that the entrepreneur gains by secretly swapping long-term claim for short-term claim with one lender, and say whether this pins down the split left open in (i).

Approach. Without moral hazard the borrower captures the NPV, so (i) is a plain optimisation plus counting equations against unknowns. In (ii) the date-1 payout to one lender is paid out of the reinvestment fund, lowering the success probability of *every* long-term claim: an externality that neither party to the side deal internalises (a common-agency, "priority" problem). Only collusion-proof structures survive, which is the lecture's point that maturity is chosen to discipline what the insider can do with unverifiable cash.

Notation.

- I: fixed date-0 investment; $A = 0$; no moral hazard (private benefit 0); risk-free rate 0.
- $r > 0$: known date-1 return; R or 0: date-2 return, success probability $p + \tau(I_1)$.
- $I_1 \geq 0$: date-1 deepening investment; τ increasing, concave, $\tau'(0) = \infty$, $\tau'(r)R < 1$.
- $r_l, r_b \geq 0$: date-1 payments to lenders (short-term debt) and to the entrepreneur; $R_l, R_b \geq 0$: success-contingent date-2 payments, $R_l + R_b = R$.
- I_1^*: first-best deepening investment; $\delta r_l, \delta R_l > 0$: sizes of a secret deal.

Step 1 — (i) First best. The first best maximises total surplus over I_1. The NPV is date-1 cash not reinvested plus expected date-2 income minus the initial investment:

$$\text{NPV}(I_1) = r - I_1 + [p + \tau(I_1)]R - I. \tag{69}$$

Its derivatives are $\text{NPV}'(I_1) = -1 + \tau'(I_1)R$ and $\text{NPV}''(I_1) = \tau''(I_1)R \leq 0$, so it is concave and the first-order condition $\text{NPV}'(I_1^*) = 0$ is sufficient:

$$-1 + \tau'(I_1^*)\,R = 0 \quad \Longleftrightarrow \quad \tau'(I_1^*)\,R = 1. \tag{70}$$

$\tau'(0) = \infty$ gives $I_1^* > 0$; $\tau'(r)R < 1 = \tau'(I_1^*)R$ and τ' decreasing give $I_1^* < r$, so $r - I_1^* > 0$ can be paid out at date 1.

Step 2 — (i) Implementation and indeterminacy. With transparent books the contract fixes the date-1 payouts, and hence $I_1 = r - r_l - r_b$. The first best requires

$$r_b + r_l + I_1^* = r, \tag{71}$$

$$R_b + R_l = R, \tag{72}$$

$$I = r_l + [p + \tau(I_1^*)]R_l, \tag{73}$$

the last being the lenders' breakeven condition. Four unknowns (r_b, r_l, R_b, R_l) and three equations: one degree of freedom. For instance (when $r - I_1^* < I$), maximal short-term debt $r_b = 0$, $r_l = r - I_1^*$ with $R_l = \frac{I-(r-I_1^*)}{p+\tau(I_1^*)}$; or pure long-term debt $r_l = 0$, $R_l = \frac{I}{p+\tau(I_1^*)}$ (if this is $\leq R$), with the entrepreneur taking $r_b = r - I_1^*$. Without moral hazard the entrepreneur gets the first-best NPV in all cases, so

the split of her pay between r_b and R_b, and of the lenders' claims between short and long term, is undetermined.

Answer (i). I_1^* solves $\tau'(I_1^*)R = 1$, with $0 < I_1^* < r$; any $(r_b, r_l, R_b, R_l) \geq 0$ with $r_b + r_l = r - I_1^*$, $R_b + R_l = R$ and $r_l + [p+\tau(I_1^*)]R_l = I$ implements it — one degree of freedom in the short-/long-term split.

Step 3 — (ii) The secret deal. With $r - I_1^* < I$, (73) and $r_l \leq r - I_1^*$ force $R_l > 0$, hence $R_b < R$ by (72). Starting from any first-best structure, the entrepreneur offers one lender $+\delta r_l$ in short-term claim against the transfer to herself of δR_l of his long-term claim. The lender is atomistic, takes I_1 as given, and accepts iff

$$\delta r_l \geq [p + \tau(I_1)]\, \delta R_l. \tag{74}$$

Date-1 cash is fixed and the other claims cannot be defrauded, so the extra payout comes out of deepening investment:

$$\delta I_1 = -\delta r_l. \tag{75}$$

The entrepreneur's utility is $U_b = r_b + [p + \tau(I_1)]R_b$; in the deal r_b is unchanged and she receives the long-term claim the lender gives up, $\delta R_b = \delta R_l$. To first order (product rule),

$$\delta U_b = \tau'(I_1)\,(\delta I_1)\, R_b + [p + \tau(I_1)]\,(\delta R_l).$$

Substituting (75) and (74) at equality ($\delta R_l = \delta r_l/[p + \tau(I_1)]$),

$$\begin{aligned} \delta U_b &= \tau'(I_1)\,(-\delta r_l)\, R_b + [p + \tau(I_1)]\Big(\frac{\delta r_l}{p + \tau(I_1)}\Big) \\ &= \big[1 - \tau'(I_1)R_b\big]\, \delta r_l. \end{aligned} \tag{76}$$

At $I_1 = I_1^*$, (70) gives $\tau'(I_1^*) = 1/R$, so $\tau'(I_1^*)R_b = R_b/R < 1$ and $\delta U_b = (1 - R_b/R)\, \delta r_l = \frac{R-R_b}{R}\, \delta r_l = (R_l/R)\, \delta r_l > 0$: the deal pays. The gain is exactly the loss $\tau'(I_1^*)R_l\, \delta r_l = (R_l/R)\delta r_l$ of the other long-term claimholders, whose success probability falls; total value is unchanged to first order because I_1^* maximises (69).

Step 4 — (ii) The indeterminacy is resolved. Lenders anticipate such deals, so only collusion-proof structures are equilibria. By (76) no deal pays iff

$$\tau'(I_1)\, R_b \geq 1. \tag{77}$$

The entrepreneur bears the NPV loss ex ante, and (69) rises with I_1 below I_1^*; since τ' falls with I_1, she sets I_1 as high as (77) allows, so (77) binds: $\tau'(I_1)R_b = 1$. Because $R_b < R$, $\tau'(I_1) > \tau'(I_1^*)$ and $I_1 < I_1^*$. A larger R_b permits a larger I_1, so R_b must be the largest compatible with breakeven, which requires paying her nothing at date 1, $r_b = 0$, and all date-1 free cash to lenders, $r_l = r - I_1$:

$$I = r - I_1 + [p + \tau(I_1)](R - R_b), \tag{78}$$

$$\tau'(I_1)\, R_b = 1. \tag{79}$$

These two equations pin down (I_1, R_b), hence $r_l = r - I_1$ and $R_l = R - R_b$. So yes, opacity resolves the indeterminacy: the entrepreneur is paid only long-term ($r_b = 0$), lenders hold more short-term debt ($r - I_1 > r - I_1^*$), and deepening investment is below the first best.

Answer (ii). At the first best a secret swap gives $\delta U_b = [1 - \tau'(I_1^*)R_b]\delta r_l = (R_l/R)\delta r_l > 0$. Yes, the indeterminacy is resolved: $r_b = 0$, $r_l = r - I_1$, and (I_1, R_b) solve $\tau'(I_1)R_b = 1$ and $I = r - I_1 + [p + \tau(I_1)](R - R_b)$, with $I_1 < I_1^*$ (a short-term bias).

Intuition. A lender who cashes out early is paid out of the firm's reinvestment fund, and the cost (a lower success probability) falls on every long-term claimant. Neither the colluding lender nor the entrepreneur internalises the loss of the other lenders, so with opaque books maturity is biased toward the short term. The entrepreneur can only commit not to strip the firm by holding a large long-term stake herself, and $\tau'(I_1)R_b = 1$ shows how large.

Common mistakes.

- Letting the colluding lender internalise the fall in I_1: he is atomistic, takes I_1 as given and accepts any deal with $\delta r_l \geq [p + \tau(I_1)]\delta R_l$.
- Counting the deal as value-creating: at I_1^* total value is unchanged to first order; the entrepreneur's gain is the other long-term holders' loss.
- Concluding that "more long-term debt" fixes the bias: what deters stripping is the entrepreneur's own long-term stake R_b, which is why $r_b = 0$.

Book answer. Matches the book's answer (p. 594): $\tau'(I_1^*)R = 1$; one degree of freedom; $\delta U_b = [1 - \tau'(I_1)R_b]\delta r_l > 0$; resolution with $r_b = 0$, $I = r - I_1 + [p + \tau(I_1)](R - R_b)$ and $\tau'(I_1)R_b = 1$, $I_1 < I_1^*$.

Ref: Tirole §5.6.1 "Optimal Claim Maturity", p. 226; Exercise 5.9, p. 233; answer p. 594.

10. One vs. two creditors: strategic and liquidity default — Amaro de Matos 5.2.5 #2, p. 148

Statement.

Source: Amaro de Matos, Ch. 5, Exercise 5.2.5 #2, p. 148 (PDF 166) · Answer in book: no (p. 268 only cites pages 10–13 of Bolton and Scharfstein 1996)

Show that two creditors are paid more in the case of strategic default than one creditor, but are paid less in the case of liquidity default.

Context needed to solve it *From Amaro de Matos §5.2.1 "Extending the Model", pp. 134–137 (PDF 152–155):*

As before, all the agents are risk neutral and the risk-free rate is assumed to be zero. The time schedule refers to a two-period model and is characterized as follows: At time $t = 0$ an amount K is invested in the project. The entrepreneur, as before, is assumed to have no initial wealth. In this specification of the model, the state of nature is characterized by the cash flow at $t = 1$. It may be $\theta_g = x$ with probability q and $\theta_b = 0$ with probability $1 - q$. The cash flow is observable for both the investor and the entrepreneur, but not for any third party, and is thus not verifiable. Therefore, as before, a contract between the investor and the entrepreneur is incomplete.

The manager is assumed to have the ability to divert cash flow without the control of the investor. In that case a zero payoff would be declared at $t = 1$ when the actual payoff was x. If the declared cash flow at $t = 1$ is x, the contract states a payment of R_x to the creditholder and, with probability β_x, the right of the creditholder to liquidate a part of the assets that is worth L_x. If the cash flow is null, the values are, respectively, $R_\circ$, $\beta_\circ$, and $L_\circ$.

Notice that taking $\beta_x = 0$, $\beta_\circ = 1$, and $L_x = L_\circ$, being equal to the value of the assets of the project at time $t = 1$, this reduces to a usual debt contract with face value $R_x < x$ and one-period maturity. The issue in designing a contract is to find the optimal values for the preceding parameters. Therefore, it is possible in this simple setting to analyze the conditions under which a debt contract would result as optimal.

In the second period, two things may happen. Either the manager continues the project or the project is liquidated, in the sense that it is separated from the manager. If the manager's activity continues, the cash flow at time $t = 2$ is y. In the case of no liquidation, the project thus has a net present value of $qx + y - K$.

The expected return to the firm is then written as

$$\pi_f = q[x - R_x + (1 - \beta_x)y] + (1 - q)[-R_\circ + (1 - \beta_\circ)y]. \qquad (5.3)$$

On the other hand, the investor expects to get

$$\pi_i = q[R_x + \beta_x L_x] + (1 - q)[R_\circ + \beta_\circ L_\circ] - K. \qquad (5.4)$$

Suppose for a moment that at time $t = 1$ the cash flow is x. The expected return to the firm would then be $x - R_x + (1 - \beta_x)y$, decomposed as follows: The cash flow x goes to the entrepreneur, who has to pay R_x to the investor. With probability β_x the project is liquidated at date $t = 1$, leading to a continuation value of zero, and with probability $1 - \beta_x$ the project continues until time $t = 2$, generating a cash flow y at that point. If, however, the manager hides from the outer investor the cash flow x, then only $R_\circ$ will be paid at time $t = 1$, and liquidation occurs with probability $\beta_\circ$, leaving to the firm a possible payoff S, not necessarily

zero, as before. This happens because the liquidation value $L_\circ$ does not correspond to the true value of the assets in that state of nature. In that case the expected return to the firm is $x - R_\circ + \beta_\circ S + (1-\beta_\circ)y$.

For the manager to be induced to pay the due R_x when the cash flow is x, one must therefore impose

$$x - R_x + (1-\beta_x)y \geq x - R_\circ + \beta_\circ S + (1-\beta_\circ)y. \qquad (5.5)$$

Noticing that $R_\circ = \beta_x = 0$ is optimal in the sense that these values maximize the profit to the firm and facilitates the verification of (5.5), the preceding inequality reads

$$R_x \leq \beta_\circ(y - S).$$

Among all these possible values for R_x that ensure a trustworthy behavior from the manager of the project, any investor would maximize its expected return by choosing the greatest value. Thus, at the optimum, this restriction is binding. Substituting the optimal values of the parameters in expression (5.4) for the expected result of the investor, one gets

$$\pi_i = \beta_\circ[q(y-S) + (1-q)L_\circ] - K. \qquad (5.6)$$

Using the optimal values of the parameters, namely those minimizing the profit to the investor, in the profit of the firm (5.3), it follows that

$$\pi_f = (qx + y - K) - (1-q)\beta_\circ(y - L_\circ). \qquad (5.7)$$

This result can be interpreted as the net present value of the project in the absence of liquidation minus a cost due to contract incompleteness.

For the profit in (5.6) to be positive, ensuring the minimum condition of feasibility of the contract, one must have

$$1 \geq \beta_\circ \geq \frac{K}{q(y-S) + (1-q)L_\circ},$$

where the first inequality comes from the fact that $\beta_\circ$ is a probability and the second inequality comes from the condition $\pi_i \geq 0$. Notice that π_f is decreasing in $\beta_\circ$ whereas π_i is monotonically increasing in that parameter. Maximizing the expected return to the firm conditioned to a positive expected return to the investor defines the optimal value of $\beta_\circ$ as the lowest possible. Thus

$$\beta_\circ = \frac{K}{q(y-S) + (1-q)L_\circ}.$$

Both S and $L_\circ$ (and thus also $\beta_\circ$) are still to be specified. There is no need to specify L_x, since the main feature of the optimal contract just described is that there is no liquidation in the good state. Liquidation occurs only, if at all, in the case of a poor performance. In order to understand what determines S and $L_\circ$ some additional assumptions are required.

It is assumed that the entrepreneur is not able to continue managing the project in the case of bankruptcy after a zero cash flow. This is the case of a *liquidity default*. In that case, the creditor may sell the position to a third party who obtains at $t = 2$ a cash flow αy, with $\alpha \in [0, 1]$, meaning that the incoming manager cannot get better results out of the assets than the manager who left the firm.

In the case of bankruptcy after a nonzero cash flow, the original manager will negotiate the buyback of the firm with the creditholder, sharing the second-period return y. In that situation, $S = y/2$. This is the case of a *strategic default*.

It is also assumed that, in the case of liquidity default, the buyer of the assets incurs an informational cost $c \in [0, \bar{c}]$. This cost is modeled as a uniformly distributed random variable. Therefore, by a negotiating process similar to the preceding, the remaining value of the assets, αy, is equally divided between the creditor and the buyer. The buyer then negotiates only if $c \leq \alpha y/2$. Thus, assuming that $\alpha y/2 \leq \bar{c}$, the probability that this happens is

$$\Pr\left(c \leq \frac{\alpha y}{2}\right) = \frac{\alpha y}{2\bar{c}}.$$

Therefore, the expected value of the liquidation payoff in the case of a zero cash flow at $t = 1$ is

$$L_\circ = \frac{\alpha^2 y^2}{4\bar{c}},$$

and the optimal $\beta_\circ$ reads

$$\beta_\circ = \frac{K}{q\frac{y}{2} + (1-q)\frac{\alpha^2 y^2}{4\bar{c}}}. \qquad (5.8)$$

[...]

From Amaro de Matos §5.2.2 "The Case of Many Creditholders", pp. 137–138 (PDF 155–156):

Bolton and Scharfstein (1996) bring to this discussion a related issue: The crucial element is the number, nature, and type of creditors—in other words, whether credit is private or public, whether it is granted by banks or obtained by issuing bonds in the marketplace, and also the nature of the governance structure among the creditors.

In this setting assume that at time $t = 0$ the amount K is invested to acquire the assets A and B. These are the assets that constitute the investment project. If there are two creditors, assume that each of the assets A and B provides collateral for each of them. Denoting by y^A and y^B the cash flows for assets A and B, as managed separately, one defines $\Delta = y - y^A - y^B$, which is assumed to be positive. This means that the cash flow generated by both assets is assumed to be greater than the sum of the cash flow generated by each of the assets individually. It is then possible to show that the expected value of liquidation at time $t = 1$ is given by

$$\overline{L}_\circ = \frac{\alpha^2 y^2}{4\bar{c}} - \frac{\alpha^2 \Delta^2}{36\bar{c}} < \frac{\alpha^2 y^2}{4\bar{c}} = L_\circ.$$

In other words, in the case of two creditors, the expected value of liquidation when there is a zero cash flow at $t = 1$ is lower than the value expected with only one creditor. Moreover, in the state with probability q,

$$S = \frac{y}{2} - \frac{\Delta}{6},$$

leading to an optimal value of $\beta_\circ$ with a denominator given by

$$q\left(\frac{y}{2} - \frac{\Delta}{6}\right) + (1-q)\left(\frac{\alpha^2 y^2}{4\bar{c}} - \frac{\alpha^2 \Delta^2}{36\bar{c}}\right),$$

which can be greater or less than the similar value found before with only one creditor. [...]

What is asked. *(strategic)*: show that when the manager defaults strategically and buys the firm back, the creditors collect more if the assets are split between two creditors than if one creditor holds them all. *(liquidity)*: show that when the default is a genuine liquidity default and the assets are sold to an outside buyer, two creditors collect less in expectation than one.

Approach. This is Bolton and Scharfstein (1996) inside the strategic-default model of §5.2.1: the investor's recovery is $y - S$ after a strategic default and $L_\circ$ after a liquidity default, and both come from bargaining. With two creditors each holds an asset the other party needs to realise the synergy Δ, so each has hold-up power. Splitting the surplus by the Shapley value (the three-player version of the equal split $S = y/2$) gives both formulas on p. 138; comparing them with $y/2$ and $\alpha^2 y^2/(4\bar{c})$ answers the two parts.

Notation.

- K: investment; q: probability of cash flow x at $t = 1$ (else 0); y: $t = 2$ cash flow if the manager continues.
- S: the manager's payoff when she strategically defaults and buys the firm back; $y - S$: what the creditor(s) get.
- $\alpha \in [0, 1]$: efficiency of an outside buyer (αy); $c \sim U[0, \bar{c}]$: buyer's information cost.
- $L_\circ$ / $\overline{L}_\circ$: expected liquidation payoff to creditors after a zero cash flow, with one / two creditors.
- $y^A, y^B \geq 0$: cash flows of assets A and B run separately; $\Delta = y - y^A - y^B \in (0, y]$: synergy.
- $\beta_\circ$: liquidation probability after a zero report, $\beta_\circ = K/[q(y - S) + (1-q)L_\circ]$ from (5.6).

Assumption. With two creditors, creditor 1 holds asset A and creditor 2 asset B. Multilateral bargaining follows the Shapley value, the three-player version of the equal split used for one creditor; disagreement payoffs are 0 (as behind $S = y/2$ in the one-creditor case). The book only cites Bolton

and Scharfstein (1996), pp. 10–13; the derivation below is ours, and it reproduces both formulas printed on p. 138.

Step 1 — One creditor (benchmark). We first record the one-creditor recoveries the comparison is made against. Manager and creditor split y equally: $S = y/2$, so in a strategic default the creditor gets

$$y - S = \frac{y}{2}. \tag{80}$$

After a zero cash flow the buyer gets $\alpha y/2$, buys iff $c \le \alpha y/2$ (probability $\alpha y/(2\bar{c})$), and the creditor gets $\alpha y/2$:

$$L_\circ = \Big(\frac{\alpha y}{2\bar{c}}\Big)\Big(\frac{\alpha y}{2}\Big) = \frac{\alpha^2 y^2}{4\bar{c}}. \tag{81}$$

Step 2 — Two creditors, strategic default. Players: manager M, creditors 1 and 2. Worth of a coalition: y if it contains M and both assets, y^A (y^B) if it contains M and only asset A (B), and 0 without M. Over the $3! = 6$ orderings, M's marginal contribution is 0 when she comes first (orderings $M12$, $M21$), y^A in ordering $1M2$ and y^B in $2M1$ (she joins a coalition holding one asset, worth 0 without her), and y when last ($12M$, $21M$). Averaging over the six orderings and using $y^A + y^B = y - \Delta$:

$$\begin{aligned} S &= \frac{0 + 0 + y^A + y^B + y + y}{6} = \frac{2y + y^A + y^B}{6} \\ &= \frac{2y + (y - \Delta)}{6} = \frac{3y - \Delta}{6} \\ &= \frac{y}{2} - \frac{\Delta}{6}, \end{aligned} \tag{82}$$

which is the book's S. The two creditors together get

$$\begin{aligned} y - S &= y - \Big(\frac{y}{2} - \frac{\Delta}{6}\Big) \\ &= \frac{y}{2} + \frac{\Delta}{6} \;>\; \frac{y}{2}. \end{aligned} \tag{83}$$

Comparing with (80): two creditors are paid more in a strategic default, by $\Delta/6$. Each creditor can hold up the synergy Δ, so the manager keeps less when she buys the firm back.

Answer (strategic). Two creditors get $y - S = \dfrac{y}{2} + \dfrac{\Delta}{6} > \dfrac{y}{2}$ (one creditor): more, by $\Delta/6$.

Step 3 — Two creditors, liquidity default. Replace M by the outside buyer, whose coalition worths are αy, αy^A, αy^B (the cost c is sunk and not bargained over, as in Step 1). By (82) the buyer gets $\alpha(y/2 - \Delta/6)$, buys iff $c \le \alpha(y/2 - \Delta/6)$, which for $c \sim U[0, \bar{c}]$ has probability $\alpha(y/2 - \Delta/6)/\bar{c}$ (assume $\alpha(y/2 - \Delta/6) \le \bar{c}$), and the creditors share $\alpha y - \alpha(y/2 - \Delta/6) = \alpha(y/2 + \Delta/6)$. Expected liquidation payoff = probability of a sale times the creditors' share:

$$\begin{aligned} \overline{L}_\circ &= \Big(\frac{\alpha(3y - \Delta)}{6\bar{c}}\Big)\Big(\frac{\alpha(3y + \Delta)}{6}\Big) \\ &= \frac{\alpha^2(3y - \Delta)(3y + \Delta)}{36\bar{c}} = \frac{\alpha^2(9y^2 - \Delta^2)}{36\bar{c}} \\ &= \frac{9\alpha^2 y^2}{36\bar{c}} - \frac{\alpha^2\Delta^2}{36\bar{c}} = \frac{\alpha^2 y^2}{4\bar{c}} - \frac{\alpha^2\Delta^2}{36\bar{c}}, \end{aligned} \tag{84}$$

the book's formula. By (81), $\overline{L}_\circ = L_\circ - \alpha^2\Delta^2/(36\bar{c}) < L_\circ$: two creditors are paid less in a liquidity default. The creditors' extra share does not compensate the lower probability that a buyer shows up, which falls from $\alpha y/(2\bar{c})$ to $\alpha(y/2 - \Delta/6)/\bar{c}$.

Answer (liquidity). Two creditors expect $\overline{L}_\circ = \dfrac{\alpha^2 y^2}{4\bar{c}} - \dfrac{\alpha^2\Delta^2}{36\bar{c}} < L_\circ = \dfrac{\alpha^2 y^2}{4\bar{c}}$: less, by $\alpha^2\Delta^2/(36\bar{c})$.

Step 4 — The $\beta_\circ$ denominator and the printed sign. By (5.6) the denominator of $\beta_\circ$ is $q(y-S)+(1-q)L$. With two creditors, substituting (83) and (84),

$$D_2 = q\Big(\frac{y}{2}+\frac{\Delta}{6}\Big) + (1-q)\Big(\frac{\alpha^2 y^2}{4\bar{c}} - \frac{\alpha^2\Delta^2}{36\bar{c}}\Big). \tag{85}$$

The book prints $q(\frac{y}{2}-\frac{\Delta}{6})$ in the first term (p. 138), i.e. qS instead of $q(y-S)$ (the two coincide only when $S=y/2$, the one-creditor case). That is a sign error: with the printed term the two-creditor denominator would be below the one-creditor one for every q, contradicting the same page's remarks that it "can be greater or less" and that two creditors are paid more in strategic default. With (85) and $D_1 = q\frac{y}{2} + (1-q)L_\circ$,

$$\begin{aligned} D_2 - D_1 &= q\Big(\frac{y}{2}+\frac{\Delta}{6}-\frac{y}{2}\Big) + (1-q)(\overline{L}_\circ - L_\circ) \\ &= q\,\frac{\Delta}{6} - (1-q)\,\frac{\alpha^2\Delta^2}{36\bar{c}}. \end{aligned} \tag{86}$$

It is positive (two creditors liquidate less often) iff, multiplying by $36\bar{c}/\Delta > 0$, $6\bar{c}\,q > (1-q)\alpha^2\Delta$, i.e. $q(6\bar{c}+\alpha^2\Delta) > \alpha^2\Delta$, i.e. $q > \bar{q} \equiv \frac{\alpha^2\Delta}{6\bar{c}+\alpha^2\Delta}$: when bankruptcy risk is low, as the book concludes.

Step 5 — Numerical illustration (ours). Take $y=10$, $y^A=4$, $y^B=3$ ($\Delta = 10-4-3=3$), $\alpha=0.8$, $\bar{c}=6$, $q=0.6$, $K=2$. Then $S = 10/2 - 3/6 = 5 - 0.5 = 4.5$ and $y-S = 5.5$ versus 5 with one creditor. In liquidity default the buyer gets $0.8(4.5)=3.6$, buys with probability $3.6/6 = 0.6$ (versus $4/6 = 0.666667$), and the creditors get $0.8(5.5) = 4.4$:

$$\overline{L}_\circ = (0.6)(4.4) = 2.64 \;<\; L_\circ = \frac{(0.64)(100)}{24} = 2.666667.$$

Denominators: $D_1 = (0.6)(5)+(0.4)(2.666667) = 4.066667$ and $D_2 = (0.6)(5.5)+(0.4)(2.64) = 4.356$, so $\beta_\circ$ falls from $2/4.066667 = 0.491803$ to $2/4.356 = 0.459137$. The misprinted term would give $(0.6)(4.5)+(0.4)(2.64) = 3.756$ and $\beta_\circ = 0.532481$. Here $\bar{q} = 1.92/37.92 = 0.050633$; at $q=0.02$, (86) gives $-0.016133 < 0$ and one creditor is better.

Intuition. Splitting the assets among creditors gives each a veto over the synergy Δ. This hold-up power makes strategic default costly for the manager, so more can be credibly repaid (good when default is mostly strategic, high q). The same hold-up power scares off outside buyers when the firm genuinely fails, lowering the liquidation value (bad when default is mostly for liquidity, low q).

Common mistakes.

- Comparing S instead of $y-S$: S is what the *manager* keeps; the creditors' recovery is $y-S$, which rises with two creditors.
- Keeping the buyer's probability of purchase fixed at $\alpha y/(2\bar{c})$: the buyer's share falls to $\alpha(y/2-\Delta/6)$, so fewer buyers show up, and that loss dominates.
- Copying the printed two-creditor denominator $q(y/2-\Delta/6)+\ldots$: its first term must be $q(y-S) = q(y/2+\Delta/6)$.

Book answer. The book (p. 268) only cites Bolton and Scharfstein (1996), pp. 10–13; derived here. Our S and $\overline{L}_\circ$ match p. 138. The printed two-creditor denominator on p. 138 has a sign error: its first term should be $q(y/2+\Delta/6) = q(y-S)$, not $q(y/2-\Delta/6)$.

Ref: Amaro de Matos §5.2.1 "Extending the Model", pp. 134–137; §5.2.2 "The Case of Many Creditholders", pp. 137–138; exercise p. 148.

11. When the good entrepreneur prefers liquidating debt — Amaro de Matos 5.2.5 #4, p. 148

Statement.

Source: Amaro de Matos, Ch. 5, Exercise 5.2.5 #4, p. 148 (PDF 166) · Answer in book: no (p. 268 only cites the appendix of Diamond 1991, pp. 732–733)

4. Prove that inequality (5.12) provides a sufficient condition for the good entrepreneur to prefer the debt structure that leads to liquidation under the conditions of the lemma to which it refers.

Context needed to solve it *From Amaro de Matos §5.2.3 "The Choice of the Duration", pp. 139–141 (PDF 157–159):*

A model developed by Diamond (1991, 1993) illustrates in a very simple way the differences between short-term and long-term debt financing. The model is based on information asymmetry and credit rating.

Regaining the notation of Aghion and Bolton, consider an entrepreneur with a two-period project who needs funds for an initial investment $K = 1$. The project may be of one of two types: good or bad. At time $t = 2$ a good project has a payoff $X > 0$, whereas a bad project pays that amount with a probability π, and pays zero with probability $1 - \pi$. Let θ denote the type of project, θ_g referring to a good project and θ_b referring to a bad project. The quality of the project is known to the entrepreneur but not to the investor. Given a project, as of date $t = 0$ the investor assigns a probability f that $\theta = \theta_g$. This probability is referred to as a *credit rating* and reflects the lender's information about the project. It follows that, as of time 0, the probability q that a borrower pays the loan at date $t = 2$ is given by

$$q \equiv f + \pi(1 - f) = \pi + f(1 - \pi). \qquad (5.9)$$

At time $t = 1$ information about the project becomes available to the investor. This new information is not verifiable, so it makes no sense to write contracts contingent upon that information, since they cannot be enforced. However, the investors update their beliefs about the quality of the project and the credit rating will be either upgraded or downgraded.

Under this revision of the credit rating, let f^u denote the probability that $\theta = \theta_g$ given an upgrade and let f^d denote the probability that $\theta = \theta_g$ given a downgrade. In that case, the probability that an entrepreneur who receives a downgrade repays at time $t = 2$ is

$$q^d = \pi + f^d(1 - \pi) > f^d. \qquad (5.10)$$

Moreover, it is here assumed that

$$f^d < f < f^u,$$

meaning that the probability that the project is good is greater given an upgrade than given a downgrade. Furthermore, the simplifying assumption $f^u = 1$ is also made. Thus, given an upgrade at date $t = 1$, it is certain that the project is good. However, it is not certain that all projects of good quality are upgraded. There is room for one such project to be downgraded. Let e denote the probability of such an event.

All projects are also assumed to produce a nonassignable control rent C if the management has control at time $t = 2$. Also, if at time $t = 1$ the manager defaults on short-term debt, lenders then have the right to remove the management from control, selling the assets for a liquidation value $L < K = 1$. All projects may be liquidated at that point in time, under those conditions. If the project is liquidated, the entrepreneur is prevented from consumption of his/her present or future cash flows and control rents. In this case, an optimal contract is a debt contract enforced by the right to liquidate the project if the debt is not repaid in full.

Just as in the case of the borrowers, lenders are assumed be risk neutral. Moreover, lenders have alternative investments at $t = 0$ that provide a return R at $t = 2$. The condition $L < X/R$ ensures that a good project yields a higher return when not liquidated.

A project may be financed with either long-term or short-term debt. In the context of this model, long-term debt refers to a contract issued at $t = 0$ with no refinancing at date $t = 1$ and maturing at $t = 2$. If one such instrument provides $K = 1$ at time $t = 0$ with face value $\rho \leq X$, the lender expects to receive at maturity $\rho[\pi + f(1 - \pi)] = R^2$. Notice that if $\rho > X$, a borrower with credit rating f will not be able to provide the lender with the required rate of return R^2. Now, if the project is upgraded at time $t = 1$, the market value of the debt contract at that point in time becomes $V_l^u = \rho/R$. Similarly, if the project is downgraded, the market value of debt at time $t = 1$ becomes $V_l^d = q^d\rho/R$.

Short-term debt, on the other hand, is debt financed at $t = 0$ with face value r_1 and maturity at time $t = 1$. If short-term borrowers are upgraded at $t = 1$, their debt contract at that point in time is worth

$V_s^u = r_1$ and they may refinance their project by issuing a new short-term debt contract with maturity at $t = 2$ and face value $r_1 R$ that will be paid with certainty and guarantees a return of R per unit invested by the lender. If short-term borrowers are downgraded, however, in principle they may refinance their debt with a new short-term contract only if $q^d X/R > r_1$, since X is the maximum that a borrower can pay at $t = 2$. If that is the case, at $t = 1$ the market value of the short-term debt issued at $t = 0$ would be $V_s^d = r_1$ if the borrower pays, possibly through the refinancement mechanism. But what happens if $q^d X/R < r_1$? In that case there is still room for a refinancement provided that the liquidation value L is less than the expected value to the lender from a refinancement, $q^d X/R$. Under a downgrading, the time $t = 1$ value of the short-term debt is thus

$$V_s^d = \min\left[r_1, \max(L, q^d X/R)\right].$$

The face value of short-term debt is established such as to provide the lenders with an expected return of R per period. It then follows that

$$r_1 = \frac{R - [1 - (1 - e)f]V_s^d}{(1 - e)f}. \qquad (5.11)$$

From Amaro de Matos §5.2.3, pp. 141–142 (PDF 159–160):

Given this model, the point is to characterize under what conditions short-term or long-term maturities will be offered. The maturity preferred by the entrepreneurs of good projects will be the only one offered to all borrowers. This happens because if any other maturity is offered, it reveals a project of bad quality and lenders will not be willing to finance it.

The following result implies that, in the absence of liquidation given a downgrade, the entrepreneurs of good projects prefer short-term debt:

Lemma 5.2.1 (Diamond, 1991). *Comparing two debt contracts issued at $t = 0$, neither of which involves liquidation on a downgrade at $t = 1$, the one with the higher value at that point in time given a downgrade, V^d, is preferred by the entrepreneurs of good projects.*

Proof. The expected payoff to the entrepreneurs of good projects from debt not yielding liquidation is

$$e\left(X + C - \frac{RV^d}{q^d}\right) + (1 - e)(X + C - r_1 R).$$

Using expression (5.11) this payoff is rewritten as

$$X + C - \frac{R^2}{f} + RV^d e\left(\frac{1}{f^d} - \frac{1}{q^d}\right).$$

Given (5.10) this expression for the payoff is clearly an increasing function of V^d. ■

From Amaro de Matos §5.2.3, p. 142 (PDF 160):

This implies that when there is no liquidation after a downgrade, short-term debt is preferred. The idea behind this result is simple. If there is no risk of liquidation, in order to reduce their expected financing costs, borrowers of the good type, who are less likely than average to be downgraded, prefer to extract larger payments from those who are downgraded. However, if there is the possibility of liquidation, the increase in the payment extracted from those who get a downgrade incorporates, for the good entrepreneurs, the risk of losing control rents. In that case the following result characterizes the trade-off incurred:

Lemma 5.2.2 (Diamond, 1991). *Consider a debt structure that does not imply liquidation and leads to a value of debt V^d, given a downgrade. Compare it with an alternative structure that implies liquidation. Entrepreneurs of good projects prefer the latter if and only if the control rent $C < \widehat{C}$, where the upper bound is given by*

$$\widehat{C} = \frac{LR}{f^d} - X - RV^d\left(\frac{1}{f^d} - \frac{1}{q^d}\right) > 0.$$

It follows that a sufficient condition for the good entrepreneur to prefer liquidation is that $C = 0$, representing the absence of control rents. It is also sufficient for the good entrepreneur to prefer liquidation if that state does not provide sufficiently high control rents C to the entrepreneur compared to the proceeds L to the investor or

$$\frac{q^d X + f^d C}{R} \leq L. \qquad (5.12)$$

What is asked. Prove that if $(q^d X + f^d C)/R \le L$, condition (5.12), then the entrepreneur of a good project prefers a debt structure that liquidates the firm after a downgrade to any structure that does not (the comparison of Lemma 5.2.2).

Approach. Diamond (1991): liquidation after a downgrade costs the good entrepreneur $X + C$ in that state but lets lenders recover L from the whole downgraded pool, most of which is bad, so the upgraded (good) firms pay less. Lemma 5.2.2 reduces the comparison to $C < \widehat{C}$. The proof therefore (1) derives both payoffs from the break-even (5.11) and Bayes' rule, (2) obtains $U^L - U^N = e(\widehat{C} - C)$, and (3) bounds $\widehat{C}$ from below using the fact that no non-liquidating debt can be worth more than $q^d X/R$ after a downgrade.

Notation.

- X: $t = 2$ payoff (good project surely, bad with probability π); $C \ge 0$: nonassignable control rent; L: liquidation value; R: gross return per period.
- f: initial rating; $f^u = 1$, f^d: posterior of "good" after an upgrade / downgrade; $e \in (0, 1)$: probability a good project is downgraded.
- P_d: probability of a downgrade for an unknown type; $q^d = \pi + f^d(1-\pi)$: repayment probability after a downgrade.
- V^d: $t = 1$ value of a non-liquidating debt after a downgrade; r_1: face value of short-term debt; U^N, U^L: good entrepreneur's payoffs under non-liquidating and liquidating debt.

Step 1 — Bayes. Both payoffs depend on the downgrade probability, which we link to f^d. Since $f^u = 1$, bad projects are always downgraded and good ones with probability e, so

$$P_d = fe + (1 - f) = 1 - (1 - e)f, \tag{87}$$

$$f^d = \frac{fe}{P_d} \quad \Longleftrightarrow \quad \frac{P_d}{f} = \frac{e}{f^d}. \tag{88}$$

Step 2 — Non-liquidating debt. We compute the good entrepreneur's payoff when a downgrade does not trigger liquidation. After a downgrade, the debt worth V^d at $t = 1$ requires a $t = 2$ payment RV^d/q^d (new or existing lenders break even at the pooled repayment probability q^d), which the good firm pays for sure; after an upgrade it pays $r_1 R$. (For long-term debt read $r_1 = \rho/R$: then the break-even $P_u(\rho/R) + P_d V_l^d = R$ is (5.11), since $P_u + P_d q^d = q$ and $\rho q = R^2$.) Rearranging (5.11), $(1 - e)f\, r_1 = R - P_d V^d$; multiplying by R/f and using (88),

$$\begin{aligned}(1 - e)\, r_1 R &= \frac{R\,(R - P_d V^d)}{f} \\ &= \frac{R^2}{f} - RV^d\,\frac{P_d}{f} \\ &= \frac{R^2}{f} - \frac{eRV^d}{f^d}.\end{aligned} \tag{89}$$

So, expanding and substituting (89),

$$\begin{aligned}U^N &= e\Big(X + C - \frac{RV^d}{q^d}\Big) + (1 - e)(X + C - r_1 R) \\ &= (X + C) - \frac{eRV^d}{q^d} - (1 - e)\, r_1 R \\ &= (X + C) - \frac{eRV^d}{q^d} - \frac{R^2}{f} + \frac{eRV^d}{f^d} \\ &= X + C - \frac{R^2}{f} + eRV^d\Big(\frac{1}{f^d} - \frac{1}{q^d}\Big),\end{aligned} \tag{90}$$

the expression in the proof of Lemma 5.2.1.

Step 3 — Liquidating debt. Now a downgraded firm is liquidated and the lenders get L, so (89) holds with $V^d = L$, and the good entrepreneur gets nothing after a downgrade:

$$\begin{aligned} U^L &= (1-e)(X+C) - (1-e)\, r_1 R \\ &= (1-e)(X+C) - \Big(\frac{R^2}{f} - \frac{eRL}{f^d}\Big) \\ &= (1-e)(X+C) - \frac{R^2}{f} + \frac{eRL}{f^d}. \end{aligned} \tag{91}$$

Step 4 — Lemma 5.2.2. We now compare the two. Subtracting (90) from (91), the R^2/f terms cancel and $(1-e)(X+C) - (X+C) = -e(X+C)$:

$$\begin{aligned} U^L - U^N &= -e(X+C) + \frac{eRL}{f^d} - eRV^d\Big(\frac{1}{f^d} - \frac{1}{q^d}\Big) \\ &= e\Big[\frac{LR}{f^d} - X - RV^d\Big(\frac{1}{f^d} - \frac{1}{q^d}\Big) - C\Big] \\ &= e\,(\widehat{C} - C), \end{aligned} \tag{92}$$

so (with $e > 0$) the good entrepreneur prefers liquidation iff $C < \widehat{C}$.

Step 5 — A bound on V^d. A downgraded firm can pay at most X at $t = 2$, and repays with probability q^d; so any non-liquidating debt has

$$V^d \le \frac{q^d X}{R}. \tag{93}$$

(Long-term: $V_l^d = q^d\rho/R$ with $\rho \le X$. Short-term without liquidation: $V_s^d = r_1 \le q^d X/R$, the refinancing condition.)

Step 6 — (5.12) is sufficient. Since $\frac{1}{f^d} - \frac{1}{q^d} > 0$ by (5.10), $\widehat{C}$ falls with V^d; substituting (93) into the definition of $\widehat{C}$,

$$\begin{aligned} \widehat{C} &\ge \frac{LR}{f^d} - X - R\Big(\frac{q^d X}{R}\Big)\Big(\frac{1}{f^d} - \frac{1}{q^d}\Big) \\ &= \frac{LR}{f^d} - X - \frac{q^d X}{f^d} + \frac{q^d X}{q^d} \\ &= \frac{LR}{f^d} - X - \frac{q^d X}{f^d} + X \\ &= \frac{LR - q^d X}{f^d}. \end{aligned} \tag{94}$$

Inequality (5.12), $(q^d X + f^d C)/R \le L$, i.e. $q^d X + f^d C \le LR$, is equivalent (subtract $q^d X$, divide by $f^d > 0$) to

$$C \le \frac{LR - q^d X}{f^d}. \tag{95}$$

Chaining (95) and (94): $C \le \widehat{C}$, and by (92) $U^L \ge U^N$. The preference is strict unless both (95) and (93) hold with equality (a knife-edge with $\rho = X$ or $r_1 = q^d X/R$). This proves the claim.

Answer (proof). For any non-liquidating structure, $V^d \le q^d X/R$ gives $\widehat{C} \ge (LR - q^d X)/f^d$; (5.12) is $C \le (LR - q^d X)/f^d$; hence $C \le \widehat{C}$ and $U^L - U^N = e(\widehat{C} - C) \ge 0$: the good entrepreneur prefers the liquidating structure.

Step 7 — Remark on $\widehat{C} > 0$. (94) also shows $\widehat{C} \ge 0$ whenever $LR \ge q^d X$, i.e. whenever lenders really prefer liquidating a downgraded firm to rolling it over — the condition for a liquidating

structure to exist. Without it, $\widehat{C}$ can be negative: with the numbers of Amaro de Matos exercise 5.2.5 #5 ($L = R = 1$, $X = \frac{7}{4}$, $f^d = \frac{1}{3}$, $q^d = \frac{2}{3}$) and short-term debt ($V^d = 1$), $\widehat{C} = 3 - 1.75 - (1)(1)(3 - \frac{3}{2}) = -0.25$; so the lemma's "$> 0$" presumes credible liquidation.

Intuition. Liquidation costs the good entrepreneur her firm (and C) in the downgrade state, but it shifts financing costs onto the pool of downgraded firms, most of whom are bad: lenders recover L from every downgraded firm, so the upgraded, i.e. good, firms pay less. (5.12) says the continuation value of a downgraded firm — what lenders could ever collect, $q^d X$, plus the control rent weighted by the share of good types, $f^d C$ — is below L; then the cross-subsidy gain outweighs the lost rent.

Common mistakes.

- Forgetting that a liquidated good entrepreneur loses both X and the control rent C in the downgrade state: U^L has $(1 - e)(X + C)$, not $X + C$.
- Signing $\frac{1}{f^d} - \frac{1}{q^d}$ wrongly: $q^d > f^d$ by (5.10), so it is positive and $\widehat{C}$ *decreases* in V^d; the bound $V^d \leq q^d X/R$ is therefore the worst case for liquidation.
- Dropping R when moving between (5.12) and $C \leq (LR - q^d X)/f^d$: (5.12) is stated per unit of R.

Book answer. The book (p. 268) only cites the appendix of Diamond (1991), pp. 732–733; proved here.

Ref: Amaro de Matos §5.2.3 "The Choice of the Duration", pp. 139–142 (Lemmas 5.2.1–5.2.2, (5.9)–(5.12)); exercise p. 148.

12. Diamond (1991): liquidation after a downgrade — Amaro de Matos 5.2.5 #6 (with #5), p. 148

Statement.

Source: Amaro de Matos, Ch. 5, Exercise 5.2.5 #6, p. 148 (PDF 166) · Answer in book: no (p. 268 only cites examples 2 and 3 of Diamond 1991, pp. 717–719)

6. This exercise illustrates the case of Lemma 5.2.2 in the model of Diamond (1991). Keep everything from the previous exercise except that here $X = 1.4$ and $C = 0.35$. Show that in this case, if the project is financed with short-term debt, it will be liquidated in the case of a downgrade at $t = 1$. Show also that this leads the entrepreneur of a good project to prefer long-term to short-term debt. Finally, show that this conclusion is reversed if the initial credit rating is $f = \frac{3}{4}$ instead of the initial $\frac{1}{2}$.

Context needed to solve it **Part 0.** *From Amaro de Matos, Ch. 5, Exercise 5.2.5 #5, p. 148 (PDF 166) — "the previous exercise":*

5. This exercise illustrates the case of Lemma 5.2.1 in the model of Diamond (1991). Take $L = R = 1$, $\pi = \frac{1}{2}$, $X = 1\frac{3}{4}$, $C = 0$, $f = \frac{1}{2}$, and $f^d = \frac{1}{3}$.

(a) What are the face values of a short-term and a long-term debt contract?

(b) Show that in this example there will be no liquidation.

(c) What is the payoff of a good project financed with long-term debt?

(d) What is the probability that a good project receives a downgrade at $t = 1$?

(e) What is the payoff of a good project financed with short-term debt?

From Amaro de Matos §5.2.3 "The Choice of the Duration", pp. 139–141 (PDF 157–159):

A model developed by Diamond (1991, 1993) illustrates in a very simple way the differences between short-term and long-term debt financing. The model is based on information asymmetry and credit rating.

Regaining the notation of Aghion and Bolton, consider an entrepreneur with a two-period project who needs funds for an initial investment $K = 1$. The project may be of one of two types: good or bad. At time $t = 2$ a good project has a payoff $X > 0$, whereas a bad project pays that amount with a probability π, and pays zero with probability $1 - \pi$. Let θ denote the type of project, θ_g referring to a good project and θ_b referring to a bad project. The quality of the project is known to the entrepreneur but not to the investor.

Given a project, as of date $t = 0$ the investor assigns a probability f that $\theta = \theta_g$. This probability is referred to as a *credit rating* and reflects the lender's information about the project. It follows that, as of time 0, the probability q that a borrower pays the loan at date $t = 2$ is given by

$$q \equiv f + \pi(1 - f) = \pi + f(1 - \pi). \qquad (5.9)$$

At time $t = 1$ information about the project becomes available to the investor. This new information is not verifiable, so it makes no sense to write contracts contingent upon that information, since they cannot be enforced. However, the investors update their beliefs about the quality of the project and the credit rating will be either upgraded or downgraded.

Under this revision of the credit rating, let f^u denote the probability that $\theta = \theta_g$ given an upgrade and let f^d denote the probability that $\theta = \theta_g$ given a downgrade. In that case, the probability that an entrepreneur who receives a downgrade repays at time $t = 2$ is

$$q^d = \pi + f^d(1 - \pi) > f^d. \qquad (5.10)$$

Moreover, it is here assumed that

$$f^d < f < f^u,$$

meaning that the probability that the project is good is greater given an upgrade than given a downgrade. Furthermore, the simplifying assumption $f^u = 1$ is also made. Thus, given an upgrade at date $t = 1$, it is certain that the project is good. However, it is not certain that all projects of good quality are upgraded. There is room for one such project to be downgraded. Let e denote the probability of such an event.

All projects are also assumed to produce a nonassignable control rent C if the management has control at time $t = 2$. Also, if at time $t = 1$ the manager defaults on short-term debt, lenders then have the right to remove the management from control, selling the assets for a liquidation value $L < K = 1$. All projects may be liquidated at that point in time, under those conditions. If the project is liquidated, the entrepreneur is prevented from consumption of his/her present or future cash flows and control rents. In this case, an optimal contract is a debt contract enforced by the right to liquidate the project if the debt is not repaid in full.

Just as in the case of the borrowers, lenders are assumed be risk neutral. Moreover, lenders have alternative investments at $t = 0$ that provide a return R at $t = 2$. The condition $L < X/R$ ensures that a good project yields a higher return when not liquidated.

A project may be financed with either long-term or short-term debt. In the context of this model, long-term debt refers to a contract issued at $t = 0$ with no refinancing at date $t = 1$ and maturing at $t = 2$. If one such instrument provides $K = 1$ at time $t = 0$ with face value $\rho \leq X$, the lender expects to receive at maturity $\rho[\pi + f(1 - \pi)] = R^2$. Notice that if $\rho > X$, a borrower with credit rating f will not be able to provide the lender with the required rate of return R^2. Now, if the project is upgraded at time $t = 1$, the market value of the debt contract at that point in time becomes $V_l^u = \rho/R$. Similarly, if the project is downgraded, the market value of debt at time $t = 1$ becomes $V_l^d = q^d\rho/R$.

Short-term debt, on the other hand, is debt financed at $t = 0$ with face value r_1 and maturity at time $t = 1$. If short-term borrowers are upgraded at $t = 1$, their debt contract at that point in time is worth $V_s^u = r_1$ and they may refinance their project by issuing a new short-term debt contract with maturity at $t = 2$ and face value r_1R that will be paid with certainty and guarantees a return of R per unit invested by the lender. If short-term borrowers are downgraded, however, in principle they may refinance their debt with a new short-term contract only if $q^dX/R > r_1$, since X is the maximum that a borrower can pay at $t = 2$. If that is the case, at $t = 1$ the market value of the short-term debt issued at $t = 0$ would be $V_s^d = r_1$ if the borrower pays, possibly through the refinancement mechanism. But what happens if $q^dX/R < r_1$? In that case there is still room for a refinancement provided that the liquidation value L is less than the expected value to the lender from a refinancement, q^dX/R. Under a downgrading, the time $t = 1$ value of the short-term debt is thus

$$V_s^d = \min\left[r_1, \max(L, q^dX/R)\right].$$

The face value of short-term debt is established such as to provide the lenders with an expected return of R per period. It then follows that

$$r_1 = \frac{R - [1 - (1 - e)f]V_s^d}{(1 - e)f}. \qquad (5.11)$$

From Amaro de Matos §5.2.3, pp. 141–142 (PDF 159–160):

Given this model, the point is to characterize under what conditions short-term or long-term maturities will be offered. The maturity preferred by the entrepreneurs of good projects will be the only one offered to

all borrowers. This happens because if any other maturity is offered, it reveals a project of bad quality and lenders will not be willing to finance it.

The following result implies that, in the absence of liquidation given a downgrade, the entrepreneurs of good projects prefer short-term debt:

Lemma 5.2.1 (Diamond, 1991). *Comparing two debt contracts issued at $t = 0$, neither of which involves liquidation on a downgrade at $t = 1$, the one with the higher value at that point in time given a downgrade, V^d, is preferred by the entrepreneurs of good projects.*

Proof. The expected payoff to the entrepreneurs of good projects from debt not yielding liquidation is

$$e\left(X + C - \frac{RV^d}{q^d}\right) + (1 - e)(X + C - r_1 R).$$

Using expression (5.11) this payoff is rewritten as

$$X + C - \frac{R^2}{f} + RV^d e\left(\frac{1}{f^d} - \frac{1}{q^d}\right).$$

Given (5.10) this expression for the payoff is clearly an increasing function of V^d. ■

From Amaro de Matos §5.2.3, p. 142 (PDF 160):

This implies that when there is no liquidation after a downgrade, short-term debt is preferred. The idea behind this result is simple. If there is no risk of liquidation, in order to reduce their expected financing costs, borrowers of the good type, who are less likely than average to be downgraded, prefer to extract larger payments from those who are downgraded. However, if there is the possibility of liquidation, the increase in the payment extracted from those who get a downgrade incorporates, for the good entrepreneurs, the risk of losing control rents. In that case the following result characterizes the trade-off incurred:

Lemma 5.2.2 (Diamond, 1991). *Consider a debt structure that does not imply liquidation and leads to a value of debt V^d, given a downgrade. Compare it with an alternative structure that implies liquidation. Entrepreneurs of good projects prefer the latter if and only if the control rent $C < \widehat{C}$, where the upper bound is given by*

$$\widehat{C} = \frac{LR}{f^d} - X - RV^d\left(\frac{1}{f^d} - \frac{1}{q^d}\right) > 0.$$

It follows that a sufficient condition for the good entrepreneur to prefer liquidation is that $C = 0$, representing the absence of control rents. It is also sufficient for the good entrepreneur to prefer liquidation if that state does not provide sufficiently high control rents C to the entrepreneur compared to the proceeds L to the investor or

$$\frac{q^d X + f^d C}{R} \leq L. \qquad (5.12)$$

Omitted: Proposition 5.2.1 (p. 142, PDF 160), which gives the credit-rating bound above which good entrepreneurs prefer short-term debt; applied to these numbers it states the exercise's conclusion.

What is asked. *Part 0* (exercise #5, needed as the base case): (a) face values of short- and long-term debt; (b) show there is no liquidation; (c) the good entrepreneur's payoff with long-term debt; (d) the probability e that a good project is downgraded; (e) her payoff with short-term debt. *(i)* With $X = 1.4$, $C = 0.35$, show that short-term debt now leads to liquidation after a downgrade. *(ii)* Show that the good entrepreneur then prefers long-term debt. *(iii)* Show that this preference reverses when the initial rating is $f = \frac{3}{4}$.

Approach. Diamond's (1991) maturity trade-off. Short-term debt is repriced at $t = 1$ on nonverifiable rating news: without liquidation that only shifts repayment onto the downgraded pool, which helps good types (Lemma 5.2.1); with liquidation it also costs the good type $X + C$ whenever she is downgraded (Lemma 5.2.2: liquidating debt wins iff $C < \widehat{C}$). A downgraded borrower is liquidated when the most she can promise, $q^d X/R$, is below both r_1 and L. The rating f matters through e: a higher f (with f^d fixed) means good types are rarely downgraded, so the liquidation cost is rarely paid.

Notation.

- X: $t = 2$ payoff (good project surely, bad with probability π); $C \geq 0$: control rent; L: liquidation value; R: gross return per period.
- f: initial rating; $f^u = 1$, f^d: posterior of "good" after an upgrade / downgrade; e: probability that a good project is downgraded; P_d: probability of a downgrade for an unknown type; $q^d = \pi + f^d(1-\pi)$; V^d: $t = 1$ value of the debt after a downgrade (V_s^d short, V_l^d long).
- ρ: face value of long-term debt; r_1: face value of short-term debt; q: $t = 0$ repayment probability (5.9); F^d: face value of the $t = 1$ refinancing after a downgrade; U_{LT}, U_{ST}: good entrepreneur's payoff with long- / short-term debt.
- Part 0 (#5): $L = R = 1$, $\pi = \frac{1}{2}$, $X = \frac{7}{4}$, $C = 0$, $f = \frac{1}{2}$, $f^d = \frac{1}{3}$. Exercise #6: $X = 1.4$, $C = 0.35$, then $f = \frac{3}{4}$.

Assumption. (a) $L = 1 = K$ contradicts the text's $L < K$; it is harmless here but makes short-term debt riskless ($r_1 = R$). (b) When f becomes $\frac{3}{4}$ we keep $f^d = \frac{1}{3}$, as the statement says ("keep everything") and as Proposition 5.2.1 does; then e adjusts. The book only cites Diamond (1991), examples 1–3; all numbers below are derived here.

Step 1 — Part 0 (a): face values. Each face value is set by the lenders' break-even at its horizon. By (5.9),

$$q = \pi + f(1-\pi) = \tfrac{1}{2} + (\tfrac{1}{2})(\tfrac{1}{2}) = \tfrac{3}{4}. \tag{96}$$

Long-term debt satisfies $\rho q = R^2$:

$$\rho = \frac{R^2}{q} = \frac{(1)}{(3/4)} = \frac{4}{3} \leq X. \tag{97}$$

For short-term debt we need e, P_d, q^d. Bad projects are always downgraded ($f^u = 1$) and good ones with probability e, so $P_d = fe + (1-f)$ and Bayes' rule gives

$$f^d = \frac{fe}{fe + (1-f)}. \tag{98}$$

Solving for e: $f^d[fe + (1-f)] = fe$, so $f^d(1-f) = fe(1-f^d)$ and

$$e = \frac{f^d(1-f)}{f(1-f^d)} = \frac{(1/3)(1/2)}{(1/2)(2/3)} = \frac{1/6}{1/3} = \frac{1}{2}, \tag{99}$$

$$P_d = 1 - (1-e)f = 1 - (\tfrac{1}{2})(\tfrac{1}{2}) = \tfrac{3}{4}, \tag{100}$$

$$q^d = \pi + f^d(1-\pi) = \tfrac{1}{2} + (\tfrac{1}{3})(\tfrac{1}{2}) = \tfrac{2}{3}. \tag{101}$$

A downgraded firm can roll over up to $q^d X/R = (2/3)(7/4) = 7/6 > L$. Conjecture no liquidation, $V_s^d = r_1$; then (5.11), multiplied by $(1-e)f$, gives

$$\begin{aligned} (1-e)f\, r_1 &= R - P_d\, r_1, \\ r_1\big[(1-e)f + P_d\big] &= R, \\ r_1 &= R = 1, \end{aligned} \tag{102}$$

since $(1-e)f + P_d = 1$. Indeed $r_1 = 1 \leq 7/6$, so $V_s^d = \min[1, \max(1, 7/6)] = 1 = r_1$: consistent.

Answer 0(a). $r_1 = 1$, $\rho = \frac{4}{3}$.

Step 2 — Part 0 (b): no liquidation. Long-term debt has no $t = 1$ payment. A downgraded short-term borrower refinances $r_1 = 1$ with face

$$F^d = \frac{Rr_1}{q^d} = \frac{(1)(1)}{(2/3)} = \frac{3}{2} \leq X = \frac{7}{4}, \tag{103}$$

and the lenders would not liquidate anyway since $q^d X/R = 7/6 > L = 1$.

Answer 0(b). No liquidation: the downgraded borrower refinances at face $F^d = \frac{3}{2} \le X$.

Step 3 — Part 0 (c)–(e). (c) The good project always pays ρ: $U_{LT} = X + C - \rho = \frac{7}{4} - \frac{4}{3} = \frac{21-16}{12} = \frac{5}{12} = 0.416667$. (d) By (99), $e = \frac{1}{2}$. (e) Upgraded firms roll over at $r_1 R = 1$, downgraded ones pay F^d:

$$\begin{aligned} U_{ST} &= e(X + C - F^d) + (1-e)(X + C - r_1 R) \\ &= (\tfrac{1}{2})(\tfrac{7}{4} - \tfrac{3}{2}) + (\tfrac{1}{2})(\tfrac{7}{4} - 1) \\ &= (\tfrac{1}{2})(\tfrac{1}{4}) + (\tfrac{1}{2})(\tfrac{3}{4}) = \tfrac{1}{8} + \tfrac{3}{8} = \tfrac{1}{2}. \end{aligned} \tag{104}$$

Short-term debt is preferred ($\frac{1}{2} > \frac{5}{12}$), as Lemma 5.2.1 predicts: by the Lemma's payoff expression, with $R^2/f = 2$ and $e(\frac{1}{f^d} - \frac{1}{q^d}) = \frac{1}{2}(3 - \frac{3}{2}) = \frac{3}{4}$,

$$U = X + C - \frac{R^2}{f} + eRV^d\Big(\frac{1}{f^d} - \frac{1}{q^d}\Big) = \tfrac{7}{4} + 0 - 2 + \tfrac{3}{4}V^d = -\tfrac{1}{4} + \tfrac{3}{4}V^d, \tag{105}$$

with $V_s^d = 1$ (giving $-\frac{1}{4} + \frac{3}{4} = \frac{1}{2}$) $> V_l^d = q^d\rho/R = \frac{8}{9}$ (giving $-\frac{1}{4} + \frac{2}{3} = \frac{5}{12}$).

Answer 0(c)–(e). (c) $U_{LT} = \frac{5}{12} \approx 0.416667$; (d) $e = \frac{1}{2}$; (e) $U_{ST} = \frac{1}{2}$.

Step 4 — #6, $f = \frac{1}{2}$: short-term debt is liquidated after a downgrade. Now $X = 1.4$, $C = 0.35$; q, ρ, e, P_d, q^d are unchanged, and $\rho = \frac{4}{3} \le 1.4$ keeps long-term debt feasible. A downgraded firm can now roll over at most

$$\frac{q^d X}{R} = (\tfrac{2}{3})(1.4) = \tfrac{14}{15} = 0.933333 \;<\; L = 1. \tag{106}$$

If it were not liquidated, (102) would give $r_1 = 1 > 14/15$: it could not refinance. So it defaults, and the lenders, offered at most $14/15$ by continuation against $L = 1$ from liquidation, liquidate. Hence $V_s^d = \max(L, q^d X/R) = 1$ and by (5.11)

$$r_1 = \frac{R - P_d V_s^d}{(1-e)f} = \frac{1 - (3/4)(1)}{(1/4)} = 1, \tag{107}$$

consistent with $V_s^d = \min[1, \max(1, 14/15)] = 1$.

Answer (i). $q^d X/R = \frac{14}{15} < r_1 = 1$ and $< L = 1$: a downgraded short-term borrower cannot refinance and is liquidated.

Step 5 — #6, $f = \frac{1}{2}$: long-term debt is preferred. The good firm keeps control only after an upgrade:

$$U_{ST} = (1-e)(X + C - r_1 R) = (\tfrac{1}{2})(1.4 + 0.35 - 1) = (\tfrac{1}{2})(0.75) = 0.375, \tag{108}$$

$$U_{LT} = X + C - \rho = 1.75 - \tfrac{4}{3} = 0.416667. \tag{109}$$

$U_{LT} > U_{ST}$. To check this against Lemma 5.2.2, write both payoffs with (5.11) and Bayes ($P_d/f = e/f^d$, from (98)). The liquidating short-term debt has $V_s^d = L$, so $(1-e)r_1 R = R(R - P_d L)/f = R^2/f - eRL/f^d$ and $U_{ST} = (1-e)(X+C) - R^2/f + eRL/f^d$; the long-term debt has the Lemma 5.2.1 payoff $U_{LT} = X + C - R^2/f + eRV_l^d(1/f^d - 1/q^d)$. Subtracting,

$$U_{ST} - U_{LT} = -e(X + C) + \frac{eRL}{f^d} - eRV_l^d\Big(\frac{1}{f^d} - \frac{1}{q^d}\Big) = e(\widehat{C} - C), \tag{110}$$

with $\widehat{C}$ as in Lemma 5.2.2. With $V^d = V_l^d = \frac{8}{9}$: $\widehat{C} = \frac{(1)(1)}{1/3} - 1.4 - (1)(\frac{8}{9})(3 - \frac{3}{2}) = 1.6 - \frac{4}{3} = 0.266667 < C = 0.35$, and (110) gives $U_{ST} - U_{LT} = (0.5)(0.266667 - 0.35) = -0.041667$. (Condition (5.12) fails: $(q^d X + f^d C)/R = 0.933333 + 0.116667 = 1.05 > L$.)

Answer (ii). $U_{LT} = 0.416667 > U_{ST} = 0.375$ (gap $-0.041667 = e(\widehat{C}-C)$ with $\widehat{C} = 0.266667 < C$): long-term debt is preferred.

Step 6 — #6, $f = \frac{3}{4}$: the conclusion reverses. With $f^d = \frac{1}{3}$ kept, (99)–(100) and (96)–(97) give

$$e = \frac{(1/3)(1/4)}{(3/4)(2/3)} = \frac{1/12}{1/2} = \frac{1}{6}, \quad P_d = 1 - (\tfrac{5}{6})(\tfrac{3}{4}) = 1 - \tfrac{5}{8} = \tfrac{3}{8},$$
$$q = \tfrac{1}{2} + (\tfrac{3}{4})(\tfrac{1}{2}) = \tfrac{7}{8}, \quad \rho = \frac{(1)}{(7/8)} = \tfrac{8}{7} = 1.142857.$$

$q^d = \frac{2}{3}$ is unchanged, so (106) still holds and short-term debt is still liquidated after a downgrade; with $(1-e)f = (\frac{5}{6})(\frac{3}{4}) = \frac{5}{8}$, (5.11) gives $r_1 = \frac{1-(3/8)(1)}{(5/8)} = \frac{5/8}{5/8} = 1$. Then

$$U_{ST} = (1-e)(X + C - r_1 R) = (\tfrac{5}{6})(0.75) = 0.625, \tag{111}$$
$$U_{LT} = X + C - \rho = 1.75 - \tfrac{8}{7} = 0.607143, \tag{112}$$

so now short-term (liquidating) debt is preferred. Check: $V_l^d = q^d\rho/R = (\frac{2}{3})(\frac{8}{7}) = \frac{16}{21}$, $\widehat{C} = 1.6 - (\frac{16}{21})(\frac{3}{2}) = 1.6 - \frac{8}{7} = 0.457143 > C = 0.35$, and by (110) $e(\widehat{C} - C) = \frac{1}{6}(0.107143) = 0.017857 = U_{ST} - U_{LT}$. Proposition 5.2.1 (p. 142) agrees: its bound is $\frac{\pi[R^2 - LR + f^d(C+X-R^2)]}{(1-\pi)[LR - f^d(C+X)]} = \frac{(1/2)[1-1+(1/3)(0.75)]}{(1/2)[1-(1/3)(1.75)]} = \frac{(1/2)(0.25)}{(1/2)(0.416667)} = 0.6$, and $f = \frac{1}{2} < 0.6 < \frac{3}{4}$.

Answer (iii). At $f = \frac{3}{4}$ ($e = \frac{1}{6}$): $U_{ST} = 0.625 > U_{LT} = 0.607143$ (gap 0.017857): short-term, liquidating debt is now preferred.

Intuition. With a low rating the good firm is downgraded often ($e = \frac{1}{2}$); liquidation then costs it the control rent too often, and long-term debt, which never triggers liquidation, wins. With a high rating, good firms are rarely downgraded ($e = \frac{1}{6}$), while the downgraded pool is mostly bad firms; liquidating that pool lets good firms borrow cheaply after the upgrade, which now outweighs the rare loss of control. (If instead $e = \frac{1}{2}$ were kept at $f = \frac{3}{4}$, then $f^d = \frac{3}{5}$, $q^d X/R = 1.12 > L$, there is no liquidation, and short-term debt still wins by Lemma 5.2.1, 0.625 vs 0.607143.)

Common mistakes.

- Keeping $e = \frac{1}{2}$ when f rises to $\frac{3}{4}$: the statement keeps everything else, including $f^d = \frac{1}{3}$, so e must be recomputed by Bayes ($e = \frac{1}{6}$).
- Giving the liquidated good entrepreneur something after a downgrade: liquidation removes X and C, so $U_{ST} = (1-e)(X + C - r_1 R)$ only.
- Checking only $q^d X/R < r_1$ for liquidation: lenders liquidate because, in addition, L beats the best refinancing offer, $L = 1 > q^d X/R = \frac{14}{15}$.

Book answer. The book (p. 268) only cites examples 1–3 of Diamond (1991), pp. 716–719; derived here. The three claims of #6 are confirmed (liquidation after downgrade; long-term preferred at $f = \frac{1}{2}$; reversal at $f = \frac{3}{4}$), consistent with Proposition 5.2.1.

Ref: Amaro de Matos §5.2.3 "The Choice of the Duration", pp. 139–142 ((5.9)–(5.12), Lemmas 5.2.1–5.2.2, Proposition 5.2.1); exercises p. 148.

13. A borrower-friendly bankruptcy court — Tirole Exercise 4.9, pp. 190–191

Statement.

Source: Tirole, Ch. 4, Exercise 4.9, pp. 190–191 (PDF 203) · Answer in book: yes, p. 586

Exercise 4.9 (borrower-friendly bankruptcy court). Consider the timing described in Figure 4.7. Figure 4.7 (timeline, left to right; p. 191, PDF 204)

Date 0	Date 1		Date 2
Investment, contract. Entrepreneur has cash A, must borrow $I - A$.	Short-term profit realized. Liquidate → Liquidation value L (known); or Continue →	Moral hazard: Effort decision (high or low).	Final profit: Success (profit R) or failure (profit 0)?

The project, if financed, yields random and verifiable short-term profit $r \in [0, \bar{r}]$ (with a continuous density and *ex ante* mean $E[r]$). After r is realized and cashed in, the firm either liquidates (sells its assets), yielding some known liquidation value $L > 0$, or continues. Note that (the random) r and (the deterministic) L are not subject to moral hazard. If the firm continues, its prospects improve with r (so r is "good news" about the future). Namely, the probability of success is $p_H(r)$ if the entrepreneur works between dates 1 and 2 and $p_L(r)$ if the entrepreneur shirks. Assume that $p'_H > 0$, $p'_L > 0$, and

$$p_H(r) - p_L(r) \equiv \Delta p$$

is independent of r (so shirking reduces the probability of success by a fixed amount independent of prospects). As usual, one will want to induce the entrepreneur to work if continuation obtains. It is convenient to use the notation

$$\rho_1(r) \equiv p_H(r)R \quad \text{and} \quad \rho_0(r) \equiv p_H(r)\left[R - \frac{B}{\Delta p}\right].$$

Investors are competitive and demand an expected rate of return equal to 0. Assume

$$\rho_1(r) > L \quad \text{for all } r \qquad (1)$$

and

$$E[r] + L > I - A > E[r + \rho_0(r)]. \qquad (2)$$

(i) Argue informally that, in the optimal contract for the borrower, the short-term profit and the liquidation value (if the firm is liquidated) ought to be given to investors.

Argue that, in the case of continuation, $R_b = B/\Delta p$. (If you are unable to show why, take this fact for granted in the rest of the question.)

Interpret conditions (1) and (2).

(ii) Write the borrower's optimization program.

Assume (without loss of generality) that the firm continues if and only if $r \geqslant r^*$ for some $r^* \in (0, r)$. Exhibit the equation defining r^*.

(iii) Argue that this optimal contract can be implemented using, inter alia, a short-term debt contract at level $d = r^*$. Interpret "liquidation" as a "bankruptcy."

How does short-term debt vary with the borrower's initial equity? Explain.

(iv) Suppose that, when the decision to liquidate is taken, the firm must go to a bankruptcy court. The judge mechanically splits the bankruptcy proceeds L equally between investors and the borrower.

Define $\hat{r}$ by

$$\rho_0(\hat{r}) \equiv \tfrac{1}{2}L.$$

Assume first that

$$r^* > \hat{r}$$

(where r^* is the value found in question (ii)).

Show that the borrower-friendly court actually prevents the borrower from having access to financing. (Note: a diagram may help.)

(v) Continuing on question (iv), show that when

$$r^* < \hat{r},$$

the borrower-friendly court either prevents financing or increases the probability of bankruptcy, and in all cases hurts the borrower and not the lenders.

Context needed to solve it The statement defines its own model; "as usual" refers to the fixed-investment model's moral-hazard setup (private benefit B, limited liability, incentive constraint), copied here.

From Tirole §3.2.1 "The Fixed-Investment Model", pp. 115–116 (PDF 128–129):

Variants of the following entrepreneurial model will be used in the following: an entrepreneur (also called the "insider" or the "borrower," "she") has a project. This project requires a fixed investment I. The entrepreneur initially has "assets" ("cash on hand" or "net worth") $A < I$. For the moment we interpret these assets as being cash or liquid securities that can be used toward covering the cost of investment. [...] The entrepreneur's cash can either be invested in the project or used for consumption. To implement the project the entrepreneur must borrow $I - A$ from lenders. [...]

Project. If undertaken, the project either succeeds, that is, yields verifiable income $R > 0$, or fails, that is, yields no income. The probability of success is denoted by p. The project is subject to moral hazard. The entrepreneur can "behave" ("work," "exert effort," "take no private benefit") or "misbehave" ("shirk," "take a private benefit") [...].

Behaving yields probability $p = p_H$ of success and no private benefit to the entrepreneur, and misbehaving results in probability $p = p_L < p_H$ of success and private benefit $B > 0$ (measured in units of account) to the entrepreneur. In the "effort interpretation," B can also be interpreted as a disutility of effort saved by the entrepreneur when shirking. Let $\Delta p \equiv p_H - p_L$.

Preferences and the loan agreement. Both the borrower and the potential lenders (or "investors") are risk neutral. For notational simplicity, there is no time preference; the rate of return expected by investors (which is also the riskless rate, due to risk neutrality) is taken to be 0. The borrower is protected by limited liability, and so her income cannot take negative values.

Lenders behave competitively in the sense that the loan, if any, makes zero profit. [...]

Let us turn to the loan contract. A contract first stipulates whether the project is financed. If so, it further specifies how the profit is shared between the lenders and the borrower. The borrower's limited liability will imply that both sides receive 0 in the case of failure [...]. In the case of success, the two parties share the profit R; R_b goes to the borrower and R_l to the lenders. To sum up, we posit an incentive scheme for the entrepreneur of the following form: R_b in the case of success, 0 in the case of failure.

From Tirole §3.2.2 "The Lenders' Credit Analysis", pp. 116–117 (PDF 129–130):

[...] the borrower will therefore behave if the following "incentive compatibility constraint" is satisfied:

$$p_H R_b \geqslant p_L R_b + B \quad \text{or} \quad (\Delta p) R_b \geqslant B. \qquad (\mathrm{IC}_b)$$

From this incentive compatibility constraint we infer that the highest income in the case of success that can be pledged to the lenders without jeopardizing the borrower's incentives is

$$R - \frac{B}{\Delta p}.$$

The *(expected) pledgeable income* is then

$$\mathcal{P} = p_H \left(R - \frac{B}{\Delta p} \right).$$

What is asked. (i) Explain why the date-1 profit r and the liquidation value L should go to investors, why the borrower's stake on continuation is exactly $B/\Delta p$, and what conditions (1) and (2) mean. (ii) Write the borrower's optimisation over the continuation rule and give the equation for the cutoff r^*. (iii) Show that short-term debt $d = r^*$ implements the optimum (liquidation = bankruptcy) and say how d moves with the borrower's equity A. (iv) If a court gives the borrower half of L in bankruptcy and $r^* > \hat{r}$, show that the borrower can no longer be financed. (v) If $r^* < \hat{r}$, show that the court either blocks financing or makes bankruptcy more likely, hurting the borrower and not the lenders.

Approach. Tirole's fixed-investment model with an interim, verifiable signal r and a liquidation option. Continuation is always efficient but pledges only $\rho_0(r) < \rho_1(r)$, while liquidation pledges all of L; when pledgeable income is scarce, the optimal contract buys it with a *threat of termination* after bad news, which short-term debt implements (default ⇒ creditors take control and liquidate). Anything that lowers what investors get in bankruptcy lowers pledgeable income ex ante; competitive lenders still break even, so the cost falls on the borrower.

Notation.

- I, $A < I$: investment and cash; $r \in [0, \bar{r}]$: verifiable date-1 profit, density f, c.d.f. F, mean $E[r]$.
- $L > 0$: liquidation value; R: date-2 success profit; $p_H(r)$, $p_L(r)$: success probabilities (work / shirk), $p'_H > 0$, Δp constant; B: private benefit.
- $\rho_1(r) = p_H(r)R$, $\rho_0(r) = p_H(r)[R - B/\Delta p]$: expected and pledgeable continuation income, both increasing in r.
- r^*: continuation cutoff; d: short-term debt; $\hat{r}$: $\rho_0(\hat{r}) = L/2$; r^{**}: cutoff under the borrower-friendly court.
- $\mathcal{P}(x)$: investors' expected income with cutoff x; NPV(x): total surplus with cutoff x.

Step 1 — (i) Who gets r and L; $R_b = B/\Delta p$; conditions (1)–(2). r and L are verifiable and not subject to moral hazard, so pledging them to investors costs no incentives. Any part of them left to the borrower lowers pledgeable income, which (Step 2) must then be recovered by liquidating more often, an efficiency loss by (1). For the same reason, in continuation the incentive constraint $[p_H(r) - p_L(r)]R_b \geq B$, i.e. $(\Delta p)R_b \geq B$, should bind: $R_b = B/\Delta p$. Allowing $R_b(r) \geq B/\Delta p$ in the program of Step 2 leaves the NPV unchanged (a transfer) but lowers the left side of the breakeven constraint by $\int_{r^*}^{\bar{r}} p_H(r)[R_b(r) - B/\Delta p]f\,dr$, forcing a higher r^*.

Interpretation. (1) $\rho_1(r) > L$: continuation is always efficient. (2), right inequality $I - A > E[r + \rho_0(r)]$: always continuing leaves too little pledgeable income to finance the project. (2), left inequality $E[r] + L > I - A$: always liquidating would raise enough. So financing requires some inefficient liquidation.

Answer (i). Give r and L to investors (they cost no incentives, and anything left to the borrower must be recovered by liquidating more often); $R_b = B/\Delta p$ on continuation. (1): continuation is always efficient; (2): the project cannot be financed without some liquidation, but can be with enough of it.

Step 2 — (ii) Program and cutoff. With cutoff x (continue iff $r \geq x$), the borrower maximises the NPV (lenders break even, so she receives the whole surplus): date-1 profit, plus $\rho_1(r)$ on continuation states, plus L on liquidation states (probability $F(x)$):

$$\max_x \ \text{NPV}(x) = E[r] + \int_x^{\bar{r}} \rho_1(r)f(r)\,dr + F(x)L - I \tag{113}$$

$$\text{s.t.} \quad \mathcal{P}(x) \equiv E[r] + \int_x^{\bar{r}} \rho_0(r)f(r)\,dr + F(x)L \ \geq \ I - A. \tag{114}$$

(A cutoff rule is without loss: continuing at r gains $\rho_1(r) - L > 0$ in value and costs $L - \rho_0(r)$ in pledgeable income; the gain/cost ratio rises with r, so continuation goes to the highest r's.) Differentiating by Leibniz's rule, $\frac{d}{dx}\int_x^{\bar{r}} g(r)f(r)\,dr = -g(x)f(x)$ and $\frac{d}{dx}F(x)L = f(x)L$, so

$$\begin{aligned} \text{NPV}'(x) &= -\rho_1(x)f(x) + f(x)L \\ &= -f(x)\big[\rho_1(x) - L\big] < 0, \end{aligned} \tag{115}$$

$$\mathcal{P}'(x) = f(x)\big[L - \rho_0(x)\big]. \tag{116}$$

By (115) the borrower wants the lowest x satisfying (114), so r^* is the lowest root of

$$E[r] + \int_{r^*}^{\bar{r}} \rho_0(r)f(r)\,dr + F(r^*)L = I - A. \tag{117}$$

By (2), $\mathcal{P}(0) = E[r + \rho_0(r)] < I - A < E[r] + L = \mathcal{P}(\bar{r})$, so $r^* \in (0, \bar{r})$; since ρ_0 is increasing, $\mathcal{P}$ rises while $\rho_0 < L$ and falls after, so at the lowest root $\rho_0(r^*) \leq L$. (The statement's "$r^* \in (0, r)$" should read $(0, \bar{r})$.)

Answer (ii). $\max_x E[r] + \int_x^{\bar{r}} \rho_1 f\,dr + F(x)L - I$ s.t. $E[r] + \int_x^{\bar{r}} \rho_0 f\,dr + F(x)L \geq I - A$; r^* is the lowest root of $E[r] + \int_{r^*}^{\bar{r}} \rho_0(r)f(r)\,dr + F(r^*)L = I - A$.

Step 3 — (iii) Short-term debt and equity. Let the firm owe short-term debt $d = r^*$ at date 1 (plus, inter alia, the residual $r - r^*$ to investors and a long-term claim $R - B/\Delta p$ on success). If $r \geq r^*$, it pays d out of r and continues. If $r < r^*$, it defaults; creditors take control and liquidate ("bankruptcy"), collecting $r + L$. They do not renegotiate, because continuing would give them at most $\rho_0(r) < \rho_0(r^*) \leq L$. Default cannot be strategic since r is verifiable. Differentiating (117) with respect to A and using (116),

$$f(r^*)\big[L - \rho_0(r^*)\big]\,\frac{dr^*}{dA} = -1$$
$$\frac{dr^*}{dA} = -\frac{1}{f(r^*)\,[L - \rho_0(r^*)]} < 0. \tag{118}$$

Less equity requires more pledgeable income, obtained by liquidating more often: short-term debt rises.

Answer (iii). Short-term debt $d = r^*$: the firm continues iff it can pay d; otherwise creditors take control and liquidate. $dd/dA = dr^*/dA = -1/\{f(r^*)[L - \rho_0(r^*)]\} < 0$: less equity, more short-term debt.

Step 4 — (iv) Court splits L, case $r^* > \hat{r}$. Investors now get $L/2$ on liquidation. Whatever the continuation policy, their income at state r (besides r) is at most $\max\{L/2, \rho_0(r)\}$, so the best they can be promised is

$$\mathcal{P}^c_{\max} = E[r] + \int_0^{\bar{r}} \max\{L/2, \rho_0(r)\} f(r)\, dr. \tag{119}$$

Compare state by state with the contract of (ii), which gives L on $[0, r^*)$ and $\rho_0(r)$ on $[r^*, \bar{r}]$. For $r < \hat{r}$: $\max = L/2 < L$. For $\hat{r} \leq r < r^*$: $\max = \rho_0(r) < \rho_0(r^*) \leq L$. For $r \geq r^* > \hat{r}$: $\max = \rho_0(r)$, the same. Hence $\mathcal{P}^c_{\max} < \mathcal{P}(r^*) = I - A$ by (117): no contract lets investors break even, and the borrower loses access to financing.

Answer (iv). With $r^* > \hat{r}$, even the best policy under the court yields $\mathcal{P}^c_{\max} < I - A$: no financing.

Step 5 — (v) Case $r^* < \hat{r}$. With cutoff x, investors get $\mathcal{P}^c(x) = E[r] + \int_x^{\bar{r}} \rho_0 f\, dr + F(x)L/2$, with $\mathcal{P}^{c\prime}(x) = f(x)[L/2 - \rho_0(x)]$, positive below $\hat{r}$ and negative above. At the old cutoff,

$$\mathcal{P}^c(r^*) = \mathcal{P}(r^*) - F(r^*)\frac{L}{2} = I - A - F(r^*)\frac{L}{2} < I - A. \tag{120}$$

So if $\mathcal{P}^c(\hat{r}) < I - A$ there is no financing; otherwise the new cutoff $r^{**} \in (r^*, \hat{r}]$ solves $\mathcal{P}^c(r^{**}) = I - A$: bankruptcy becomes more likely. Lenders break even in every case (or do not lend), so they are not hurt. The borrower receives the NPV, and by (115) $\text{NPV}(r^{**}) < \text{NPV}(r^*)$: she is hurt (or loses financing altogether). Revealed preference says the same: under an enforcing court she could have promised investors only $L/2$ in bankruptcy, and chose not to.

Answer (v). Either $\mathcal{P}^c(\hat{r}) < I - A$ (no financing) or the cutoff rises to $r^{**} \in (r^*, \hat{r}]$ (more bankruptcy); lenders always break even, the borrower's NPV falls.

Step 6 — Numerical illustration (ours). $r \sim U[0,1]$ ($E[r] = 0.5$), $p_H(r) = 0.3 + 0.5r$, $R = 2$, $B/\Delta p = 1.6$ (e.g. $B = 0.4$, $\Delta p = 0.25$), so $\rho_1(r) = (0.3 + 0.5r)(2) = 0.6 + r$, $\rho_0(r) = (0.3 + 0.5r)(2 - 1.6) = 0.12 + 0.2r$, and $E[r + \rho_0] = 0.5 + 0.12 + 0.2(0.5) = 0.72$. With $f = 1$ on $[0,1]$,

$$\mathcal{P}(x) = 0.5 + \int_x^1 (0.12 + 0.2r)\, dr + Lx = 0.5 + 0.12(1 - x) + 0.1(1 - x^2) + Lx$$
$$= 0.72 + (L - 0.12)x - 0.1x^2.$$

Case (v): $L = 0.5$, $I - A = 0.75$ (conditions (1)–(2) hold: $0.6 > 0.5$, $1 > 0.75 > 0.72$). (117) is $0.72 + 0.38x - 0.1x^2 = 0.75$, i.e. $0.1x^2 - 0.38x + 0.03 = 0$, with discriminant $0.38^2 - 4(0.1)(0.03) =$

$0.1444 - 0.012 = 0.1324$; the lowest root is

$$r^* = \frac{0.38 - \sqrt{0.1324}}{0.2} = \frac{0.38 - 0.363868}{0.2} = 0.080659,$$

with $\rho_0(r^*) = 0.12 + 0.2(0.080659) = 0.136132$ and, by (118) with $f = 1$, $dr^*/dA = -1/(0.5 - 0.136132) = -1/0.363868 = -2.748249$. Here $\hat{r}$ solves $0.12 + 0.2\hat{r} = 0.25$, so $\hat{r} = (0.25 - 0.12)/0.2 = 0.65 > r^*$. Under the court $\mathcal{P}^c(x) = 0.72 + (0.25 - 0.12)x - 0.1x^2 = 0.72 + 0.13x - 0.1x^2$, so $\mathcal{P}^c(\hat{r}) = 0.72 + 0.0845 - 0.04225 = 0.76225 \geq 0.75$, and $\mathcal{P}^c(r^{**}) = 0.75$ gives $0.1x^2 - 0.13x + 0.03 = 0$, discriminant $0.0169 - 0.012 = 0.0049$, $r^{**} = (0.13 - \sqrt{0.0049})/0.2 = (0.13 - 0.07)/0.2 = 0.3$. The borrower's loss is $\int_{r^*}^{r^{**}} (\rho_1 - L)\, dr = \int_{r^*}^{r^{**}} (0.1 + r)\, dr = 0.1(0.3 - 0.080659) + (0.09 - 0.080659^2)/2 = 0.021934 + 0.041747 = 0.063681$. *Case (iv):* $L = 0.3$, $I - A = 0.78$: (117) is $0.1x^2 - 0.18x + 0.06 = 0$, discriminant $0.0324 - 0.024 = 0.0084$, so $r^* = (0.18 - 0.091652)/0.2 = 0.441742$, with $\rho_0(r^*) = 0.208348 \leq L$; $\hat{r} = (0.15 - 0.12)/0.2 = 0.15 < r^*$. (119) gives $0.5 + \int_0^{0.15} 0.15\, dr + \int_{0.15}^1 (0.12 + 0.2r)\, dr = 0.5 + (0.15)(0.15) + 0.12(0.85) + 0.1(1 - 0.0225) = 0.5 + 0.0225 + 0.102 + 0.09775 = 0.72225 < 0.78$: no financing.

Intuition. Bankruptcy is the borrower's commitment device: handing L to investors after bad news is what makes the loan pay. A court that gives the borrower half of L ex post destroys pledgeable income ex ante. Competitive lenders still break even, so the whole cost lands on the borrower: more frequent (inefficient) bankruptcy, or no loan at all.

Common mistakes.

- Taking the highest root of the break-even equation: liquidation is inefficient by (1), so the borrower wants the *lowest* cutoff that still lets investors break even.
- Concluding that the court helps borrowers or hurts lenders: lenders anticipate the split and break even (or do not lend); the whole cost is borne by the borrower.
- Thinking the default could be strategic here: r is verifiable, so default happens only when $r < d$ and creditors liquidate because $\rho_0(r) < L$.

Book answer. Matches the book's answer (pp. 586–587): same program, r^* the lowest value satisfying breakeven, $d = r^*$, $dr^*/dA < 0$, no financing when $r^* > \hat{r}$, and when $r^* < \hat{r}$ either no financing or bankruptcy over a larger interval $[0, r^{**})$, $r^{**} > r^*$, hurting the borrower only. (The book's wording "no bankruptcy occurs on some interval $[0, r^{**}]$" must mean bankruptcy occurs there.)

Ref: Tirole §3.2.1 "The Fixed-Investment Model" and §3.2.2 "The Lenders' Credit Analysis", pp. 115–117; Exercise 4.9, pp. 190–191; answer pp. 586–587.

14. A two-period sovereign bond: default set, price and spread — Original exercise

Statement.

Origin: original exercise on sovereign default and bond pricing, built on Prof. Ricca's Topic 5 slides 13–18, 21 and 28–29 (Aula/Topic_5.pdf) · Answer: in the solutions file

A small open economy lives for two periods, $t = 0, 1$. Its government maximizes $u(c_0) + \beta E_0[u(c_1)]$, with $\beta \in (0, 1)$, $u' > 0$ and $u'' < 0$. At $t = 0$ it has no assets ($B = 0$), observes output $y_0 = 1$, and chooses the face value B' of the one-period, non-state-contingent bonds it buys: a position $B' < 0$ is a debt of face value $b \equiv -B' > 0$ due at $t = 1$, and it raises $-q(B')B'$ units of the good at $t = 0$, where $q(B')$ is the unit price. At $t = 1$ output is $y_H = 1.2$ with probability 0.8 or $y_L = 0.8$ with probability 0.2. Period $t = 1$ is the last one: nothing can be borrowed or saved at $t = 1$, and nothing happens afterwards. After observing y, the government either repays, consuming $c_1 = y + B'$, or defaults, consuming $h(y) = (1 - \lambda)y$ with $\lambda = 0.1$ (the output cost of default; exclusion from credit markets costs nothing more, because the economy ends). Debt contracts cannot be enforced: the government defaults whenever default gives strictly higher utility than repaying, and repays when indifferent. Foreign lenders are risk-neutral and competitive, can invest at the risk-free rate $r = 0.05$, and know the distribution of y.

(a) Write the value of repaying, $V^c(B', y)$, and the value of defaulting, $V^d(y)$, at $t = 1$. Show that the government defaults if and only if $b > \lambda y$. Does the answer depend on the utility function u?

(b) Find the default set $D(B') = \{y \in \{y_L, y_H\} : V^d(y) > V^c(B', y)\}$ for every B', including $B' \geq 0$ (the country lends).

(c) Derive the default probability $\delta(B')$ and the bond price $q(B')$ implied by the lenders' zero-profit condition. Evaluate q at $B' = 0.05$, -0.05, -0.08, -0.10, -0.12 and -0.15.

(d) For each B' in (c) with $q(B') > 0$, compute the gross interest rate $1 + r(B') = 1/q(B')$ and the spread $r(B') - r$. At $B' = -0.10$, check that a lender who buys the issue earns, in expectation, exactly the risk-free return.

(e) The country raises $-q(B')B'$ at $t = 0$. (i) Find the bond position B^* that maximizes the amount raised, and that maximum amount. (ii) Show that any B' with $-0.10 \leq B' < -0.08$ is dominated: a risk-free bond raises the same amount at $t = 0$ and costs the country no more at $t = 1$ in either state (strictly less in at least one). (iii) Explain why the country never chooses $B' < B^*$.

Context needed to solve it *From Prof. Ricca's Topic 5 slides 13–18, 21 and 28–29 (Aula/Topic_5.pdf), restated for this exercise's two-period setting:*

- Small open economy with a non-storable, stochastic endowment $\tilde{y}_t$; preferences $E_0 \sum_t \beta^t u(c_t)$ with $\beta \in (0, 1)$, $u' > 0$, $u'' < 0$ (slide 13). It smooths consumption by borrowing from or lending to competitive, risk-neutral foreign investors; bonds have a one-period maturity and are not state-contingent; debt contracts are not enforceable, so the country can renege even when it has the means to repay; default costs an output loss and exclusion from capital markets; the country defaults whenever this option is more valuable than repaying (slide 14).
- A bond with face value B' costs $q(B', y)B'$ today; the interest rate is $1 + r(B', y) = B'/[q(B', y)B'] = 1/q(B', y)$ (slide 15). $B' > 0$: the country pays qB' today and receives B' next period (it lends); $B' < 0$: it receives $-qB'$ today and pays $-B'$ next period (it borrows) (slide 16).
- When the country repays, its resource constraint is $c = y + B - q(B', y)B'$; when it defaults it is excluded from foreign credit, re-enters with probability θ each period, and consumes $y^{def} = h(y) \leq y$ while excluded (slide 17). In this exercise $t = 1$ is the last period, so $B' = 0$ is forced at $t = 1$ and θ plays no role.
- Competition and risk neutrality: $q(B', y)B'(1+r) = B'[1-\delta(B', y)]$, i.e. $q(B', y) = [1-\delta(B', y)]/(1+r)$; for $B' > 0$ the default probability is zero (slide 18).
- Default set $D(B) = \{y \in Y : V^d(y) > V^c(B, y)\}$; the probability of default next period on debt B' is $\delta(B', y) = \Pr[y' \in D(B') \mid y]$ (slide 21).
- The amount raised is $-q(B')B'$; the maximum increase in consumption is given by the B^* that solves $\min_{B'} q(B')B'$, and it is never optimal to choose $B' < B^*$ because the country can raise the same amount at a higher price, i.e. with a smaller repayment next period (slides 28–29).

Original exercise (no course book has one on this model), built on Prof. Ricca's Topic 5 slides 13–18, 21 and 28–29. Notation as on the slides: B' are the *assets* bought for $t = 1$, so $B' < 0$ is a debt of face value $b = -B'$.

What is asked. (a) Write the two values at the last date and find the rule that tells when the government defaults on a debt b. (b) List, for every bond position, the output states in which it defaults. (c) Turn those sets into default probabilities and the price a competitive lender pays per unit of face value, at six given positions. (d) Turn prices into interest rates and spreads over the risk-free rate, and check the lender's break-even at $B' = -0.10$. (e) Find the position that raises the most cash at $t = 0$, show a band of risky positions is beaten by a risk-free bond, and explain why borrowing beyond B^* never pays.

Approach. At the last date there is no future, so V^d and V^c are single-period utilities and the comparison reduces to consumption: default is a *strategic* choice (nonverifiable repayment, slide 14), disciplined here only by the output cost $h(y) \leq y$. Lenders price the bond by zero profit, $q = (1 - \delta)/(1 + r)$ (slide 18), which turns the default set (slide 21) into a step-shaped price schedule; the amount raised, $-q(B')B'$, then has a maximum B^* (slides 28–29).

Notation.

- B': assets bought at $t=0$ (face value due at $t=1$); $b=-B'$: debt; $y\in\{y_L,y_H\}=\{0.8,1.2\}$ with $\Pr(y_H)=0.8$, $\Pr(y_L)=0.2$.
- $h(y)=(1-\lambda)y$, $\lambda=0.1$: consumption after default; λy: output lost by defaulting.
- $V^c(B',y)$, $V^d(y)$: values of repaying and of defaulting at $t=1$; $D(B')$: default set; $\delta(B')$: default probability.
- $r=0.05$: risk-free rate; $q(B')$: unit bond price; $1+r(B')=1/q(B')$: gross interest rate; $r(B')-r$: spread.

Step 1 — (a) The default rule. At $t=1$ nothing can be borrowed, so repaying leaves $c_1=y+B'$ and defaulting leaves $h(y)$:

$$V^c(B',y)=u(y+B')=u(y-b), \tag{121}$$
$$V^d(y)=u\big(h(y)\big)=u\big((1-\lambda)y\big). \tag{122}$$

Default happens iff $V^d(y)>V^c(B',y)$. Since u is strictly increasing, $u(x_1)>u(x_2)\iff x_1>x_2$, so

$$\begin{aligned} V^d(y)>V^c(B',y) &\iff (1-\lambda)y>y-b\\ &\iff b>y-(1-\lambda)y\\ &\iff b>\lambda y. \end{aligned} \tag{123}$$

Answer (a). $V^c(B',y)=u(y+B')$, $V^d(y)=u((1-\lambda)y)$; default iff $b=-B'>\lambda y$ (the debt exceeds the output lost by defaulting). The rule does not depend on u (only on $u'>0$): at the last date default compares consumption levels.

Step 2 — (b) Default sets. The two thresholds of (123) are

$$\lambda y_L=(0.1)(0.8)=0.08, \qquad \lambda y_H=(0.1)(1.2)=0.12.$$

For $B'\geq 0$ we have $b\leq 0<\lambda y$ in both states (the country is owed money and gains nothing by defaulting). Ties go to repayment: at $b=0.08$, y_L, repaying leaves $0.8-0.08=0.72=(0.9)(0.8)$; at $b=0.12$, y_H, repaying leaves $1.2-0.12=1.08=(0.9)(1.2)$. Hence

$$D(B')=\begin{cases}\emptyset & B'\geq -0.08,\\ \{y_L\} & -0.12\leq B'<-0.08,\\ \{y_L,y_H\} & B'<-0.12.\end{cases} \tag{124}$$

Answer (b). $D(B')=\emptyset$ for $B'\geq -0.08$ (all lending positions included); $D(B')=\{y_L\}$ for $-0.12\leq B'<-0.08$; $D(B')=\{y_L,y_H\}$ for $B'<-0.12$.

Step 3 — (c) Default probability and price. $\delta(B')=\Pr[y\in D(B')]$ adds the probabilities of the default states in (124): 0, 0.2 or $0.2+0.8=1$. Zero profit for a lender buying face value B' at $q(B')B'$ (slide 18) gives

$$q(B')B'(1+r)=B'[1-\delta(B')]$$
$$q(B')=\frac{1-\delta(B')}{1+r}=\begin{cases}\dfrac{1}{1.05}=0.952381 & B'\geq -0.08,\\ \dfrac{1-0.2}{1.05}=\dfrac{0.8}{1.05}=0.761905 & -0.12\leq B'<-0.08,\\ \dfrac{1-1}{1.05}=0 & B'<-0.12.\end{cases} \tag{125}$$

At the six positions: $q(0.05)=q(-0.05)=q(-0.08)=0.952381$; $q(-0.10)=q(-0.12)=0.761905$; $q(-0.15)=0$.

Answer (c). $\delta(B') \in \{0,\ 0.2,\ 1\}$ on the three regions of (124); $q = 0.952381$ at $0.05, -0.05, -0.08$; $q = 0.761905$ at $-0.10, -0.12$; $q = 0$ at -0.15.

Step 4 — (d) Interest rates and spreads. By slide 15, $1 + r(B') = 1/q(B')$. On the risk-free region,

$$1 + r(B') = \frac{1}{1/1.05} = 1.05, \qquad r(B') - r = 0.05 - 0.05 = 0.$$

On the region where only y_L defaults ($B' = -0.10, -0.12$),

$$1 + r(B') = \frac{1}{0.8/1.05} = \frac{1.05}{0.8} = 1.3125, \qquad r(B') - r = 0.3125 - 0.05 = 0.2625.$$

At $B' = -0.15$, $q = 0$: no lender pays anything for a bond that is never repaid, so no interest rate clears the market. Lender check at $B' = -0.10$: the issue costs $q\,b = (0.761905)(0.10) = 0.0761905$ and pays 0.10 with probability 0.8 and nothing in default (the output loss λy_L is destroyed, not transferred), so

$$\frac{E[\text{payoff}]}{\text{price}} = \frac{(0.8)(0.10) + (0.2)(0)}{0.0761905} = \frac{0.08}{0.0761905} = 1.05 = 1 + r.$$

Answer (d). Risk-free positions: $1+r(B') = 1.05$, spread 0. $B' = -0.10, -0.12$: $1+r(B') = 1.3125$, spread 0.2625 (26.25 percentage points). $B' = -0.15$: no market. The lender earns exactly 1.05 per unit at $B' = -0.10$.

Step 5 — (e)(i) Maximum amount raised. With $b = -B'$, the cash raised is $X(b) = -q(B')B' = q\,b$. By (125):

$$X(b) = \begin{cases} b/1.05, & 0 < b \le 0.08, \\ 0.8\,b/1.05, & 0.08 < b \le 0.12, \\ 0, & b > 0.12. \end{cases}$$

On the first step X rises to $0.08/1.05 = 0.076190$; on the second it drops to just above $0.8(0.08)/1.05 = 0.060952$ and rises to $0.8(0.12)/1.05 = 0.091429$. The largest value is at $b = 0.12$: $B^* = -0.12$, raising 0.091429.

Step 6 — (e)(ii) A dominated band. Take $b \in (0.08, 0.10]$; it raises $0.8b/1.05$. A bond of face value $b_s = 0.8b$ raises $b_s/1.05 = 0.8b/1.05$, the same amount, and is risk-free because $b_s = 0.8b \le 0.8(0.10) = 0.08$. What each costs at $t = 1$:

$$\begin{aligned} y_H: &\quad \text{risky pays } b, \quad \text{safe pays } 0.8b < b; \\ y_L: &\quad \text{risky defaults and loses } \lambda y_L = 0.08, \quad \text{safe pays } 0.8b \le 0.08. \end{aligned}$$

So $t = 1$ consumption is at least as high in both states with the safe bond, and strictly higher in y_H: every $B' \in [-0.10, -0.08)$ is dominated. (For $b \in (0.10, 0.12]$ the amount $0.8b/1.05 > 0.076190$ exceeds anything risk-free debt can raise, so no such comparison exists.)

Step 7 — (e)(iii) Never beyond B^*. For $B' < B^* = -0.12$ the price is 0: the country raises nothing at $t = 0$, and at $t = 1$ defaults in both states, losing λy of output. $B' = 0$ gives the same $t = 0$ consumption and no loss, so $B' < B^*$ is strictly dominated. Slide 29's general argument: any amount raised with $B' < B^*$ can also be raised with some $B' \ge B^*$ at a higher price, hence with a smaller repayment.

Answer (e). (i) $B^* = -0.12$, maximum 0.091429. (ii) For $B' \in [-0.10, -0.08)$ the risk-free bond $b_s = 0.8b$ raises the same cash and costs $0.8b$ in both states, against b in y_H and 0.08 of output in y_L. (iii) Beyond B^* the price is zero: nothing is raised and default costs output in every state.

Intuition. A sovereign repays only because default destroys output: debt up to λy is credible in state y. Debt that exceeds the cost in bad times but not in good times is repaid only in good times, so lenders charge a default premium. The price schedule is a staircase, and the cash raised is a "debt Laffer curve": more face value raises more cash until the next default state is crossed, where cash drops. Default is pure waste (lenders get nothing, the country loses λy), which is why risky debt that a safe bond can replicate is never chosen.

Common mistakes.

- Sign of B': on the slides B' is *assets*; debt is $B' < 0$ and the cash raised is $-q(B')B' > 0$.
- Writing the spread as $1/q - 1 = 0.3125$: that is the risky rate $r(B')$; the spread subtracts $r = 0.05$.
- Treating ties as defaults: $D(B)$ uses a strict inequality, so $b = \lambda y$ is repaid (this keeps $q(-0.12) > 0$).

Ref: Prof. Ricca, Topic 5 slides 13–18 (setup, prices, interest rate, zero profit), 21 (default set), 28–29 (borrowing limit B^*). Original exercise; numbers in the check script.

15. The price schedule $q(B', y)$ with persistent output — Original exercise

Statement.

Origin: original exercise on the sovereign bond price schedule, built on Prof. Ricca's Topic 5 slides 15, 17–18, 21, 23–24 and 27–29 (Aula/Topic_5.pdf) · Answer: in the solutions file

A small open economy lives for two periods, $t = 0, 1$, with preferences $u(c_0) + \beta E_0[u(c_1)]$, $\beta \in (0, 1)$, $u' > 0$, $u'' < 0$. Output takes three values, $y \in \{0.8, 1.0, 1.2\}$, and follows a Markov chain: given output y at $t = 0$, output y' at $t = 1$ has the probabilities $\pi(y' \mid y)$ below.

current y	next y'	0.8	1.0	1.2
0.8		0.6	0.3	0.1
1.0		0.2	0.6	0.2
1.2		0.1	0.3	0.6

At $t = 0$ the government has no assets, observes y, and chooses the face value B' of one-period, non-state-contingent bonds ($B' < 0$ is a debt of face value $b = -B'$ due at $t = 1$), raising $-q(B', y)B'$ at $t = 0$. Period $t = 1$ is the last one (no borrowing or saving at $t = 1$). After observing y', the government either repays, consuming $y' + B'$, or defaults, consuming $h(y') = \min\{y', \hat{y}\}$ with $\hat{y} = 0.9$ (the output-cost form used by Arellano 2008: default costs nothing when output is already below $\hat{y}$ and costs $y' - \hat{y}$ when it is above). The government defaults whenever default gives strictly higher utility, and repays when indifferent. Lenders are risk-neutral and competitive, can invest at $r = 0.04$, observe y and know the transition probabilities.

(a) Let $D(B') = \{y' : V^d(y') > V^c(B', y')\}$. Show that $B'_2 \geq B'_1$ implies $D(B'_2) \subseteq D(B'_1)$, and conclude that for every current y the price $q(B', y)$ is nondecreasing in B' (nonincreasing in the debt b). Does $D(B')$ depend on current output y? Does $q(B', y)$?

(b) Find $D(B')$ for every B'.

(c) Compute $q(B', y)$ for $B' \in \{0.10,\ 0,\ -0.05,\ -0.10,\ -0.20,\ -0.30,\ -0.40\}$ and $y \in \{0.8, 1.0, 1.2\}$; present the results as a table.

(d) At $B' = -0.05$, compute the gross interest rate $1 + r(B', y) = 1/q(B', y)$ and the spread $r(B', y) - r$ for each current y. Explain why the same bond pays a higher interest rate when current output is low. What would $q(-0.05, y)$ be, for every current y, if instead y' were i.i.d. with probabilities $(0.3, 0.4, 0.3)$ on $(0.8, 1.0, 1.2)$?

(e) For each current y, find the bond position $B^*(y) \in [-0.5, 0)$ that maximizes the amount raised, $-q(B', y)B'$, and the maximum amount. Interpret the result as an endogenous borrowing limit.

Context needed to solve it *From Prof. Ricca's Topic 5 slides 15, 17–18, 21, 23–24 and 27–29 (Aula/Topic_5.pdf), restated for this exercise's two-period setting:*

- A bond with face value B' costs $q(B',y)B'$; the interest rate is $1+r(B',y)=1/q(B',y)$ (slide 15). Repaying, the country consumes $c=y+B-q(B',y)B'$; defaulting, it consumes $y^{def}=h(y)\le y$ while excluded (slide 17).
- Lenders' zero profit: $q(B',y)=[1-\delta(B',y)]/(1+r)$, with $\delta(B',y)=0$ for $B'>0$ (slide 18). Default set $D(B)=\{y\in Y: V^d(y)>V^c(B,y)\}$; $\delta(B',y)=\Pr[y'\in D(B')\mid y]=\sum_{y'\in D(B')}\pi(y'\mid y)$ (slide 21, with the integral over the density $f(y',y)$ written as a sum for a finite state space).
- Claim 1: for given y, $V^c(B,y)$ is increasing in B. Claim 2: if $B_2\ge B_1$ then $D(B_2)\subseteq D(B_1)$, since V^c increases with B while $V^d(y)$ stays constant. Hence the equilibrium price function $q(B',y)$ is increasing in B' and the interest rate decreasing (slides 23–24).
- In the i.i.d. case the price function $q(B')=[1-F(y^*(B'))]/(1+r)$ does not depend on current y; equilibrium prices still depend on y because the chosen B' does (slide 27).
- The maximum increase in consumption in a period is given by the B^* that solves $\min_{B'} q(B')B'$; this endogenous borrowing limit means that, after a bad shock, the country would like to borrow more but such a contract is not available; choosing $B'<B^*$ is never optimal (slides 28–29).

Original exercise, built on Prof. Ricca's Topic 5 slides 15, 17–18, 21, 23–24 and 27–29.

What is asked. (a) Prove, in this two-period economy, that more assets (less debt) shrink the default set, so the bond price falls with debt for any current output; say what depends on current output. (b) Find the default set at every position. (c) Fill the price table for seven positions and three current outputs. (d) Compute interest rates and spreads on a small debt for each current output, explain the recession premium, and price the same bond under i.i.d. output. (e) For each current output, find the debt that raises the most cash.

Approach. Claims 1–2 (slides 23–24) in their simplest form: at the last date $V^c=u(y'+B')$ rises with B' while $V^d=u(h(y'))$ does not, so default sets are nested. Prices follow from zero profit with Markov probabilities, $q=(1-\sum_{y'\in D(B')}\pi(y'\mid y))/(1+r)$ (slides 18, 21): the default *set* depends only on tomorrow's state, the default *probability* on today's. The Arellano cost $h(y)=\min\{y,\hat{y}\}$ makes default cheap in recessions.

Notation.

- y: output at $t=0$ (current); y': output at $t=1$; $\pi(y'\mid y)$: transition probabilities (table in the statement).
- $h(y')=\min\{y',0.9\}$; $\kappa(y')\equiv y'-h(y')$: output lost by defaulting in state y'.
- B', $b=-B'$, $D(B')$, $\delta(B',y)$, $q(B',y)$, $1+r(B',y)=1/q(B',y)$, $r=0.04$; $B^*(y)$: cash-maximising position at current y.

Step 1 — (a) Nested default sets and a monotone price. At $t=1$ (last date) $V^c(B',y')=u(y'+B')$ and $V^d(y')=u(h(y'))$. Let $B'_2\ge B'_1$ and $y'\in D(B'_2)$. Then

$$\begin{aligned} u(h(y')) &> u(y'+B'_2) && (y'\in D(B'_2))\\ &\ge u(y'+B'_1) && (B'_2\ge B'_1,\ u'>0), \end{aligned}$$

so $y'\in D(B'_1)$: $D(B'_2)\subseteq D(B'_1)$. Since probabilities are nonnegative, summing over a smaller set gives a smaller number:

$$\begin{aligned} \delta(B'_2,y)=\sum_{y'\in D(B'_2)}\pi(y'\mid y)\le\sum_{y'\in D(B'_1)}\pi(y'\mid y)=\delta(B'_1,y)\\ q(B'_2,y)=\frac{1-\delta(B'_2,y)}{1+r}\ge\frac{1-\delta(B'_1,y)}{1+r}=q(B'_1,y). \end{aligned} \tag{126}$$

Answer (a). $D(B'_2)\subseteq D(B'_1)$ for $B'_2\ge B'_1$, hence $q(B',y)$ is nondecreasing in B' (nonincreasing in debt) for every y. $D(B')$ depends only on B' and y', not on current y; $\delta(B',y)$ and $q(B',y)$ depend on y through $\pi(\cdot\mid y)$.

Step 2 — (b) Default sets. As in Exercise 14, default iff $h(y') > y' - b$, i.e. iff $b > \kappa(y')$. The costs are

$$\kappa(0.8) = 0.8 - \min\{0.8, 0.9\} = 0, \qquad \kappa(1.0) = 1.0 - 0.9 = 0.1, \qquad \kappa(1.2) = 1.2 - 0.9 = 0.3.$$

With ties going to repayment,

$$D(B') = \begin{cases} \emptyset & B' \geq 0, \\ \{0.8\} & -0.1 \leq B' < 0, \\ \{0.8,\ 1.0\} & -0.3 \leq B' < -0.1, \\ \{0.8,\ 1.0,\ 1.2\} & B' < -0.3. \end{cases} \tag{127}$$

Answer (b). (127): any debt is defaulted on in the recession state 0.8 (default costs nothing there); debt above 0.1 also in state 1.0; above 0.3 always.

Step 3 — (c) The price table. $\delta(B', y)$ adds the row-y probabilities of the states in (127), and $q = (1 - \delta)/1.04$.

$$\begin{aligned} -0.1 \leq B' < 0: \quad & \delta = \pi(0.8 \mid y) = 0.6,\ 0.2,\ 0.1, \\ & q = \tfrac{0.4}{1.04},\ \tfrac{0.8}{1.04},\ \tfrac{0.9}{1.04} = 0.384615,\ 0.769231,\ 0.865385; \\ -0.3 \leq B' < -0.1: \quad & \delta = \pi(0.8 \mid y) + \pi(1.0 \mid y) = 0.9,\ 0.8,\ 0.4, \\ & q = \tfrac{0.1}{1.04},\ \tfrac{0.2}{1.04},\ \tfrac{0.6}{1.04} = 0.096154,\ 0.192308,\ 0.576923; \end{aligned}$$

for $y = 0.8, 1.0, 1.2$ respectively; $q = 1/1.04 = 0.961538$ for $B' \geq 0$ and $q = 0$ for $B' < -0.3$.

B'	0.10	0	−0.05	−0.10	−0.20	−0.30	−0.40
$q(B', 0.8)$	0.961538	0.961538	0.384615	0.384615	0.096154	0.096154	0
$q(B', 1.0)$	0.961538	0.961538	0.769231	0.769231	0.192308	0.192308	0
$q(B', 1.2)$	0.961538	0.961538	0.865385	0.865385	0.576923	0.576923	0

Answer (c). The table above: each row is a nonincreasing staircase in debt, and each column (for $B' < 0$) rises with current y.

Step 4 — (d) Spreads and the recession premium. At $B' = -0.05$, $1 + r(B', y) = 1/q = 1.04/(1 - \delta)$:

$$\begin{aligned} y = 0.8: \quad & 1 + r(B', y) = \tfrac{1.04}{0.4} = 2.6, && r(B', y) - r = 1.6 - 0.04 = 1.56; \\ y = 1.0: \quad & 1 + r(B', y) = \tfrac{1.04}{0.8} = 1.3, && r(B', y) - r = 0.3 - 0.04 = 0.26; \\ y = 1.2: \quad & 1 + r(B', y) = \tfrac{1.04}{0.9} = 1.155556, && r(B', y) - r = 0.155556 - 0.04 = 0.115556. \end{aligned}$$

The bond is defaulted on exactly when $y' = 0.8$; with persistent output a recession today makes a recession tomorrow likelier (0.6 against 0.2 or 0.1), so lenders demand more. With i.i.d. output $\delta(-0.05) = \Pr(y' = 0.8) = 0.3$ whatever y is, and

$$q(-0.05, y) = \frac{1 - 0.3}{1.04} = \frac{0.7}{1.04} = 0.673077 \quad \text{for every } y,$$

as slide 27 says: the i.i.d. price function does not depend on current output.

Answer (d). Spreads at $B' = -0.05$: 1.56 ($y = 0.8$), 0.26 ($y = 1.0$), 0.115556 ($y = 1.2$). Persistence raises the default probability after a bad shock. Under i.i.d. $(0.3, 0.4, 0.3)$: $q(-0.05, y) = 0.673077$ for all y.

Step 5 — (e) Borrowing limit by current state. Within each price step, $q(B', y)b$ rises with b, so the maximum is at a right end of a step, $b = 0.1$ or $b = 0.3$ (beyond 0.3 the price is 0):

$$\begin{aligned} y = 0.8: &\quad (0.1)(0.384615) = 0.038462 \;>\; (0.3)(0.096154) = 0.028846 &&\Rightarrow B^*(0.8) = -0.1; \\ y = 1.0: &\quad (0.1)(0.769231) = 0.076923 \;>\; (0.3)(0.192308) = 0.057692 &&\Rightarrow B^*(1.0) = -0.1; \\ y = 1.2: &\quad (0.1)(0.865385) = 0.086538 \;<\; (0.3)(0.576923) = 0.173077 &&\Rightarrow B^*(1.2) = -0.3. \end{aligned}$$

Answer (e). $B^*(0.8) = -0.1$ (raises 0.038462), $B^*(1.0) = -0.1$ (0.076923), $B^*(1.2) = -0.3$ (0.173077). The most the country can raise is smallest in the recession, exactly when it would like to borrow to smooth consumption: an endogenous, procyclical borrowing limit (slide 28).

Intuition. The default decision belongs to tomorrow's state, the price to today's information about it. Output costs that are small in recessions (Arellano's $\min\{y, \hat{y}\}$) put the default states at the bottom of the distribution, and persistence makes a bad state today a signal of default tomorrow. So credit is dearest and scarcest when the country needs it most — the countercyclical spreads of emerging markets.

Common mistakes.

- Letting $D(B')$ depend on current y: the default decision is taken at $t = 1$ and depends on y' only; current y enters through $\pi(\cdot \mid y)$.
- Using the stationary or unconditional probabilities with Markov output: lenders condition on current y.
- Looking for the maximum of $q\,b$ by calculus: q jumps, so the candidates are the right ends of the steps.

Ref: Prof. Ricca, Topic 5 slides 15, 17–18, 21 (prices, default set), 23–24 (Claims 1–2, monotone q), 27 (i.i.d. price), 28–29 (borrowing limit). Original exercise.

16. No capital inflows when default is possible (Claim 3) — Original exercise

Statement.

Origin: original exercise on the i.i.d. case with permanent exclusion of the Eaton–Gersovitz model, built on Prof. Ricca's Topic 5 slides 17–22 and 25 (Aula/Topic_5.pdf) · Answer: in the solutions file

Consider the infinite-horizon small open economy with preferences $E_0 \sum_{t=0}^{\infty} \beta^t u(c_t)$, $\beta \in (0,1)$, $u' > 0$, $u'' < 0$. Output y_t is i.i.d. over time. Each period the government enters with assets B ($B < 0$ is debt), observes y, and either repays — choosing next period's assets $B' \geq Z$ ($Z < 0$ is a no-Ponzi bound) and consuming $c = y + B - q(B')B'$ — or defaults. After a default the country is excluded from credit markets and re-enters each period with probability θ; while excluded it consumes $h(y) \leq y$. Lenders are risk-neutral and competitive, with risk-free rate $r > 0$, so $q(B') = [1 - \delta(B')]/(1 + r)$, where $\delta(B') = \Pr[\tilde{y} \in D(B')]$ and $D(B) = \{y : V^d(y) > V^c(B, y)\}$. The value functions are

$$V^o(B, y) = \max\{V^d(y), V^c(B, y)\},$$

$$V^d(y) = u(h(y)) + \beta E\big[\theta V^o(0, \tilde{y}') + (1 - \theta)V^d(\tilde{y}')\big],$$

$$V^c(B, y) = \max_{B' \geq Z} \Big\{u\big(y + B - q(B')B'\big) + \beta E\big[V^o(B', \tilde{y}')\big]\Big\}.$$

In parts (a)–(d), $\theta = 0$ (permanent exclusion) and $h(y) = y$ (no output cost).

(a) Show that $V^d(y) = u(y) + \dfrac{\beta}{1 - \beta} E[u(\tilde{y})]$. Evaluate V^d in both states when $u(c) = \ln c$, $y \in \{0.9, 1.1\}$ with probability $1/2$ each, and $\beta = 0.9$.

(b) Prove: if, for some B, the default set is non-empty, $D(B) \neq \emptyset$, then there is no bond position $\hat{B}$ available with $B - q(\hat{B})\hat{B} > 0$ (no capital inflow at B).

(c) A commentator claims that, with the numbers in (a), a country holding $B = -0.2$ defaults when $y = 0.9$ even though lenders offer to buy bonds $\hat{B} = -0.25$ at the risk-free price $q(\hat{B}) = 1/(1+r)$, with $r = 0.05$. Compute the consumption and the net capital inflow that taking this offer would give, and show — using only $V^o \geq V^d$ — that default at $y = 0.9$ cannot then be optimal.

(d) Show that $D(0) = \emptyset$: a country with no debt never defaults. Does the result in (b) say that the country can never borrow?

(e) Which assumptions does the proof in (b) use? Show that the result still holds (i) for any output cost $h(y) \leq y$, with $\theta = 0$, and (ii) with Markov output, where prices are $q(\hat{B}, y)$.

(f) Now let $\theta > 0$. Show at which step the proof in (b) breaks, and explain the economics.

Context needed to solve it *From Prof. Ricca's Topic 5 slides 17–22 and 25 (Aula/Topic_5.pdf):*

- Repaying, the country's resource constraint is $c = y + B - q(B', y)B'$. Defaulting, it loses access to foreign credit for a stochastic number of periods, regains access next period with probability θ, and consumes $y^{def} = h(y) \leq y$ until it re-enters (slide 17). Lenders: $q(B', y) = [1 - \delta(B', y)]/(1 + r)$, with zero default probability when $B' > 0$ (slide 18).
- $V^o(B, y) = \max\{V^d(B, y), V^c(B, y)\}$ (slide 19); $V^d(y) = u(y^{def}) + \beta \int_{y'} [\theta V^o(0, y') + (1 - \theta)V^d(y')]f(y', y)\, dy'$ and $V^c(B, y) = \max_{B' \geq Z}\{u(y + B - q(B', y)B') + \beta \int_{y'} V^o(B', y')f(y', y)\, dy'\}$, where $Z < 0$ is a lower bound on debt that prevents Ponzi schemes (slide 20). In the i.i.d. case the integrals are expectations over $\tilde{y}'$ and q does not depend on current y.
- Default is optimal if $V^d(y) > V^c(B, y)$; $D(B) = \{y \in Y : V^d(y) > V^c(B, y)\}$; $\delta(B', y) = \Pr[y' \in D(B') \mid y]$ (slide 21). An equilibrium is a set of policy functions $c(s)$, $B'(s)$, default sets $D(B)$ and prices $q(B', y)$ such that consumption satisfies the resource constraint, choices are optimal given prices, and prices reflect default probabilities with zero profits (slide 22).
- Slide 25 states the result of part (b) for the i.i.d. case with $\theta = 0$ and $h(y) = y$ ("Claim 3") and adds: if the country could roll over the debt, that is, borrow an amount large enough to repay the previous debt, it would choose to do so, given that it can still exercise the option to default in the next period.

Original exercise, built on Prof. Ricca's Topic 5 slides 17–22 and 25.

What is asked. (a) Solve the autarky value with permanent exclusion and no output cost, and evaluate it in a two-state log example. (b) Prove slide 25's Claim 3: if the country would default in some state at debt B, no bond on offer can give it a net inflow at B. (c) Apply the argument to a concrete offer. (d) Show a debt-free country never defaults, and say what Claim 3 does *not* say. (e) Find which assumptions the proof needs and extend it to output costs and Markov output. (f) Show where re-entry breaks the proof.

Approach. Revealed preference on the Bellman equations of slides 19–20: V^c is a maximum, so any feasible bond gives a lower bound, and $V^o = \max\{V^d, V^c\} \geq V^d$ everywhere. With permanent exclusion the continuation of default is V^d itself, so "roll over now, keep the option to default later" is always at least as good tomorrow as defaulting today (slide 25's remark); a positive inflow makes it strictly better today.

Notation.

- B: assets entering the period ($B < 0$ debt); $\hat{B}$: a candidate bond position; $B - q(\hat{B})\hat{B}$: net capital inflow (consumption minus output).
- V^o, V^c, V^d: value with the option, of repaying, of defaulting; $D(B)$: default set; θ: re-entry probability; $h(y) \leq y$: output after default.
- $E[\cdot]$: expectation over next period's i.i.d. output $\tilde{y}'$; numerical example: $u = \ln$, $y \in \{0.9, 1.1\}$ equally likely, $\beta = 0.9$, $r = 0.05$.

Step 1 — (a) Autarky value. With $\theta = 0$ and $h(y) = y$, $V^d(y) = u(y) + \beta E[V^d(\tilde{y}')]$. With i.i.d.

output, $m \equiv E[V^d(\tilde{y}')]$ is the same number whatever today's y; taking expectations of both sides,

$$\begin{aligned} m &= E[u(\tilde{y})] + \beta m \\ m &= \frac{E[u(\tilde{y})]}{1-\beta}, \\ V^d(y) &= u(y) + \beta m = u(y) + \frac{\beta}{1-\beta} E[u(\tilde{y})]. \end{aligned} \tag{128}$$

Numbers: $\ln 0.9 = -0.1053605$, $\ln 1.1 = 0.0953102$, so $E[\ln \tilde{y}] = \frac{1}{2}(-0.1053605 + 0.0953102) = -0.0050252$ and, with $\beta/(1-\beta) = 0.9/0.1 = 9$, $9E[\ln \tilde{y}] = -0.0452265$ (products from unrounded logarithms):

$$V^d(0.9) = -0.1053605 - 0.0452265 = -0.1505870, \quad V^d(1.1) = 0.0953102 - 0.0452265 = 0.0500837.$$

Answer (a). $V^d(y) = u(y) + \frac{\beta}{1-\beta} E[u(\tilde{y})]$; $V^d(0.9) = -0.150587$, $V^d(1.1) = 0.050084$.

Step 2 — (b) Proof of Claim 3. Suppose $D(B) \neq \emptyset$; pick $y \in D(B)$, and suppose, to reach a contradiction, that some $\hat{B} \geq Z$ has $B - q(\hat{B})\hat{B} > 0$. Three facts:

$$V^d(y) > V^c(B, y) \qquad (y \in D(B)) \tag{129}$$

$$V^c(B, y) \geq u(y + B - q(\hat{B})\hat{B}) + \beta E[V^o(\hat{B}, \tilde{y}')] \qquad (\hat{B} \text{ feasible; } V^c \text{ is a max}) \tag{130}$$

$$V^d(y) = u(y) + \beta E[V^d(\tilde{y}')] \qquad (\theta = 0,\ h(y) = y) \tag{131}$$

Substituting (131) and (130) into (129),

$$\begin{aligned} u(y) + \beta E[V^d(\tilde{y}')] &> u(y + B - q(\hat{B})\hat{B}) + \beta E[V^o(\hat{B}, \tilde{y}')] \\ u(y) - u(y + B - q(\hat{B})\hat{B}) &> \beta E[V^o(\hat{B}, \tilde{y}') - V^d(\tilde{y}')] \ \geq\ 0, \end{aligned} \tag{132}$$

the last step because $V^o(\hat{B}, y') = \max\{V^d(y'), V^c(\hat{B}, y')\} \geq V^d(y')$ in every state. So $u(y) > u(y + B - q(\hat{B})\hat{B})$, and since u is increasing, $y > y + B - q(\hat{B})\hat{B}$, i.e. $B - q(\hat{B})\hat{B} < 0$ — contradicting the inflow. With i.i.d. output the menu $\{q(\hat{B}), \hat{B}\}$ is the same in every state, so no contract with an inflow is available at B at all.

Answer (b). If $D(B) \neq \emptyset$, then $B - q(\hat{B})\hat{B} \leq 0$ for every available $\hat{B}$: at debt levels where default is possible, the country can only make net repayments.

Step 3 — (c) The commentator's example. Taking the offer at $B = -0.2$, $y = 0.9$:

$$\begin{aligned} c = y + B - q(\hat{B})\hat{B} &= 0.9 - 0.2 - \tfrac{1}{1.05}(-0.25) = 0.7 + 0.238095 = 0.938095, \\ B - q(\hat{B})\hat{B} &= -0.2 + 0.238095 = 0.038095 > 0. \end{aligned}$$

By (130), $V^o \geq V^d$ and (131),

$$\begin{aligned} V^c(-0.2, 0.9) &\geq \ln 0.938095 + \beta E[V^o(-0.25, \tilde{y}')] \\ &\geq \ln 0.938095 + \beta E[V^d(\tilde{y}')] \\ &= V^d(0.9) + \ln \frac{0.938095}{0.9} = V^d(0.9) + 0.041457 > V^d(0.9). \end{aligned}$$

So $0.9 \notin D(-0.2)$; the same steps at $y = 1.1$ give $1.1 \notin D(-0.2)$. If such an offer exists, $D(-0.2) = \emptyset$: the claim is inconsistent.

Answer (c). $c = 0.938095$, inflow $0.038095 > 0$; repaying and rolling over beats default by at least 0.041457 utils, so default at $y = 0.9$ is not optimal.

Step 4 — (d) A debt-free country never defaults. At $B = 0$ the choice $B' = 0$ is feasible and gives $c = y$. Then

$$\begin{aligned} V^c(0, y) &\geq u(y) + \beta E\big[V^o(0, \tilde{y}')\big] && (B' = 0 \text{ feasible}) \\ &\geq u(y) + \beta E\big[V^d(\tilde{y}')\big] && (V^o \geq V^d) \\ &= V^d(y), && \text{by (131)} \end{aligned}$$

so $V^d(y) > V^c(0, y)$ never holds: $D(0) = \emptyset$. Claim 3 therefore does not apply at $B = 0$, and a debt-free country can borrow ($-q(B')B' > 0$ for any $B' < 0$ with $q(B') > 0$). Claim 3 says something narrower: once debt is high enough that default is possible in some state, the market offers only net repayment — default happens when the country "cannot roll over its debt" (slides 25, 27). How much debt is sustained comes only from the value of keeping access to smoothing: with the numbers of (a), a grid solution (check script; Exercise 17(e)) finds a risk-free limit of only 0.01, one percent of mean output.

Answer (d). $V^c(0, y) \geq V^d(y)$, so $D(0) = \emptyset$. No: Claim 3 restricts net inflows only at debt levels where default is possible; positive (if small) borrowing is sustained.

Step 5 — (e) Which assumptions matter. The proof used: $\theta = 0$ in (131) (the continuation of default is V^d, which is below V^o); $h(y) = y$ in (131); and i.i.d. output only to make the menu of contracts the same in every state. (i) With $h(y) \leq y$ and $\theta = 0$, (131) becomes $V^d(y) = u(h(y)) + \beta E[V^d(\tilde{y}')]$ and (132) becomes

$$\begin{aligned} u\big(h(y)\big) - u\big(y + B - q(\hat{B})\hat{B}\big) > \beta E\big[V^o(\hat{B}, \tilde{y}') - V^d(\tilde{y}')\big] \geq 0 \\ \Rightarrow \quad h(y) > y + B - q(\hat{B})\hat{B} \\ \Rightarrow \quad B - q(\hat{B})\hat{B} < -\big[y - h(y)\big] \leq 0, \end{aligned}$$

which is even stronger: in a default state, the net repayment demanded exceeds the output cost $y - h(y)$. (ii) With Markov output, replace E by $E[\cdot \mid y]$ and $q(\hat{B})$ by $q(\hat{B}, y)$; every step holds state by state, so for each $y \in D(B)$ no contract offered *at that* y has an inflow (the menu may differ across states).

Answer (e). Only $\theta = 0$ is essential. The result holds for any $h(y) \leq y$ (with the stronger bound $B - q\hat{B} < -(y - h(y))$) and, state by state, with Markov output.

Step 6 — (f) Re-entry breaks the proof. With $\theta > 0$, $V^d(y) = u(h(y)) + \beta E[\theta V^o(0, \tilde{y}') + (1 - \theta)V^d(\tilde{y}')]$, and the right side of (132) becomes

$$\begin{aligned} &\beta E\big[V^o(\hat{B}, \tilde{y}') - \theta V^o(0, \tilde{y}') - (1 - \theta)V^d(\tilde{y}')\big] \\ &\quad = \beta\theta E\big[V^o(\hat{B}, \tilde{y}') - V^o(0, \tilde{y}')\big] + \beta(1 - \theta)E\big[V^o(\hat{B}, \tilde{y}') - V^d(\tilde{y}')\big]. \end{aligned}$$

The second term is still ≥ 0, but the first is ≤ 0 for $\hat{B} < 0$: V^c rises with B (Claim 1, slide 23) and V^d does not depend on B, so $V^o = \max\{V^d, V^c\}$ is nondecreasing in B and $V^o(\hat{B}, \cdot) \leq V^o(0, \cdot)$. The right side can be negative, so a small inflow no longer contradicts default.

Answer (f). The step "right side ≥ 0" fails: with re-entry, defaulting can lead back to the market with $B = 0$, worth more than returning with debt $\hat{B}$.

Intuition. Rolling over debt is a way of postponing the default decision; with permanent exclusion, postponing is free (the option is kept) and an inflow makes it strictly better today. So a country defaults only when markets demand net repayment. Re-entry turns default into a fresh start: it wipes out the debt, which rolling over does not, and that can be worth more than a small inflow today.

Common mistakes.

- Reading Claim 3 as "no borrowing": it concerns debt levels with $D(B) \neq \emptyset$; at $B = 0$ the default set is empty and the country can borrow.

- Writing (130) as an equality: $\hat{B}$ need not be optimal; the proof only needs the lower bound.
- Thinking output costs break the result: they strengthen it; only re-entry ($\theta > 0$) does.

Ref: Prof. Ricca, Topic 5 slides 17–22 (setup, value functions, default set, equilibrium), 23 (Claim 1), 25 (Claim 3 and the roll-over remark). Original exercise.

17. Infinite horizon with log utility: certifying risk-free debt — Original exercise

Statement.

Origin: original exercise on the risk-free region and the borrowing capacity of the Eaton–Gersovitz model, built on Prof. Ricca's Topic 5 slides 17–24 and 29 (Aula/Topic_5.pdf) · Answer: in the solutions file

Consider the infinite-horizon small open economy of slides 17–22: preferences $E_0 \sum_{t=0}^{\infty} \beta^t \ln c_t$ with $\beta = 0.9$; output i.i.d., $y \in \{y_L, y_H\} = \{0.9, 1.1\}$ with probability $1/2$ each; one-period non-state-contingent bonds priced by risk-neutral competitive lenders at $q(B') = [1 - \delta(B')]/(1 + r)$ with $r = 0.05$; no enforcement. A defaulting country is excluded forever ($\theta = 0$) and consumes $h(y) = (1 - \lambda)y$ every period, with $\lambda = 0.02$. The value functions are

$$V^o(B, y) = \max\{V^d(y), V^c(B, y)\}, \qquad V^d(y) = \ln h(y) + \beta E[V^d(\tilde{y}')],$$

$$V^c(B, y) = \max_{B' \geq Z} \Big\{ \ln\big(y + B - q(B')B'\big) + \beta E\big[V^o(B', \tilde{y}')\big] \Big\},$$

with $Z < 0$ a no-Ponzi bound, and $D(B) = \{y : V^d(y) > V^c(B, y)\}$. Fix a debt level $b > 0$ (assets $B = -b$) and consider the conjecture that bonds with face value b are risk-free: $q(-b) = 1/(1 + r)$.

(a) Compute $V^d(y_L)$ and $V^d(y_H)$.

(b) Under the conjecture, show that servicing the debt forever — repaying and choosing $B' = -b$ every period — gives consumption $y - s$ each period, with $s \equiv rb/(1 + r)$, and that

$$V^c(-b, y) \geq W(b, y) \equiv \ln(y - s) + \frac{\beta}{1 - \beta} E[\ln(\tilde{y} - s)].$$

Conclude that if $W(b, y) \geq V^d(y)$ in both states, then $D(-b) = \emptyset$, so the conjecture $q(-b) = 1/(1 + r)$ confirms itself.

(c) Let $\Delta(s, y) \equiv W(b, y) - V^d(y)$. Show that Δ is increasing in y, so the condition of (b) only needs to be checked at y_L. Show also that with no output cost ($\lambda = 0$) $\Delta(s, y) < 0$ for every $s > 0$, and relate this to Exercise 16.

(d) Find the largest $\bar{s}$ with $\Delta(\bar{s}, y_L) \geq 0$: write the equation that defines it, apply Newton's method from $s_0 = 0$ until six decimals are stable, and verify the sign of $\Delta(s, y_L)$ at $s = 0.019$ and $s = 0.020$. Convert $\bar{s}$ into the certified debt level $\bar{b} = \bar{s}(1 + r)/r$ and compare it with mean output.

(e) A grid solution of the full model (slides 19–22: value iteration on an asset grid of step 0.00375 on $[-0.6, 0.3]$, with prices updated from the default sets until they converge) gives the lowest asset level with an empty default set, $\bar{B} = \inf\{B : D(B) = \emptyset\}$: $\bar{B} = -0.42$ for $(\lambda, \theta) = (0.02, 0)$; $\bar{B} = -0.0375$ for $(\lambda, \theta) = (0.02, 0.5)$; and, on a grid of step 0.0005 on $[-0.05, 0.05]$, $\bar{B} = -0.01$ for $(\lambda, \theta) = (0, 0)$. (i) Explain why $\bar{b}$ from (d) is a lower bound on $-\bar{B}$, and check it against the grid value. (ii) Compute $d\bar{s}/d\lambda$ and $d\bar{b}/d\lambda$ at $\lambda = 0.02$ by implicit differentiation. (iii) Use the grid results to explain how the output cost λ and the re-entry probability θ shape the country's borrowing capacity.

Context needed to solve it *From Prof. Ricca's Topic 5 slides 17–24 and 29 (Aula/Topic_5.pdf):*

- Repaying: $c = y + B - q(B', y)B'$. Defaulting: exclusion from foreign credit, re-entry next period with probability θ, consumption $y^{def} = h(y) \leq y$ until re-entry (slide 17). Lenders: $q(B', y) = [1 - \delta(B', y)]/(1 + r)$ (slide 18).
- $V^o(B, y) = \max\{V^d(B, y), V^c(B, y)\}$ (slide 19); $V^d(y) = u(y^{def}) + \beta \int_{y'} [\theta V^o(0, y') + (1 - \theta)V^d(y')] f(y', y)\, dy'$; $V^c(B, y) = \max_{B' \geq Z}\{u(y + B - q(B', y)B') + \beta \int_{y'} V^o(B', y') f(y', y)\, dy'\}$ with $Z < 0$ a no-Ponzi bound (slide 20); $D(B) = \{y \in Y : V^d(y) > V^c(B, y)\}$, $\delta(B', y) = \Pr[y' \in D(B') \mid y]$ (slide 21); equilibrium: resource constraint, optimal choices given prices, zero-profit prices (slide 22).

- Claim 1: $V^c(y,B)$ is increasing in B; Claim 2: $B_2 \geq B_1 \Rightarrow D(B_2) \subseteq D(B_1)$ (slide 23). $\underline{B} = \sup\{B : D(B) = Y\}$ and $\bar{B} = \inf\{B : D(B) = \emptyset\}$, with $\underline{B} \leq \bar{B} \leq 0$ (slide 24). When $B > \bar{B}$, debt is risk-free and $q(B) = 1/(1+r)$ (slide 29).
- Newton's method for $G(s) = 0$: $s_{k+1} = s_k - G(s_k)/G'(s_k)$. Implicit differentiation of $G(\bar{s}, \lambda) = 0$: $d\bar{s}/d\lambda = -G_\lambda/G_s$.

From Tirole §3.8 "Nonverifiable Income", "Relation to the sovereign debt literature", pp. 143–144 (PDF 156–157), for part (e)(iii):

> Repayment of debt by the sovereign responds to two incentives: international sanctions and the future cost of being shut down from the international capital market after default. A subliterature, starting with Bulow and Rogoff (1989a,b), assumes away sanctions and focuses on the incentives provided by exclusion. [...] Bulow and Rogoff consider an infinite-horizon, symmetric-information model in which (a) the sovereign can decide not to reimburse and (b) the sovereign can save, and (c) the rate of growth of the economy is smaller than the rate of interest. They show that no lending is feasible as the borrower always prefers to default (and save some of the concomitant extra income). [...] Second, an outright exclusion, in which the defaulting sovereign cannot even save, makes it particularly costly for the sovereign to repudiate its debt. Again, some sovereign debt may then be issued in equilibrium (Kehoe and Levine 1993; Kocherlakota 1996).

Original exercise, built on Prof. Ricca's Topic 5 slides 17–24 and 29, with Tirole §3.8, pp. 143–144, for part (e)(iii).

What is asked. (a) Compute the autarky values with a 2% output cost. (b) Show that, if a debt is priced as risk-free, the value of repaying is at least the value of servicing the debt forever, and that this makes a risk-free price self-confirming. (c) Show the test binds in the low state and fails for every debt without output costs. (d) Solve the test's threshold by Newton's method and convert it into a debt level. (e) Compare with a grid solution of the full model, derive the threshold's sensitivity to the cost, and explain how the cost and re-entry drive borrowing capacity.

Approach. A full equilibrium (slides 19–22) needs a computer, but a lower bound on V^c is available by hand: V^c is a maximum, so any feasible plan — here "repay and keep the same debt forever" — bounds it from below. If that plan beats autarky in every state, default never happens at that debt, the debt is risk-free (slide 29: $q = 1/(1+r)$ above $\bar{B}$), and the risk-free price used to compute the plan's value is confirmed. The threshold is a one-dimensional equation in s, solved by Newton's method; implicit differentiation gives its comparative statics.

Notation.

- $y \in \{y_L, y_H\} = \{0.9, 1.1\}$ i.i.d., probability $\frac{1}{2}$ each; $\beta = 0.9$, $K \equiv \beta/(1-\beta) = 9$; $r = 0.05$; $h(y) = (1-\lambda)y$, $\lambda = 0.02$; $\theta = 0$.
- $b > 0$: debt (assets $B = -b$); $s \equiv rb/(1+r)$: per-period net debt service when b is rolled over at the risk-free price.
- $W(b,y)$: value of servicing b forever; $\Delta(s,y) = W - V^d$; $G(s) \equiv \Delta(s, y_L)$; $\bar{s}$, $\bar{b} = \bar{s}(1+r)/r$: certified threshold.
- $\bar{B} = \inf\{B : D(B) = \emptyset\}$: true risk-free limit (slide 24).

Step 1 — (a) Autarky values. Exactly as in (128), with $h(y)$ in place of y: $V^d(y) = \ln[(1-\lambda)y] + K\,E\{\ln[(1-\lambda)\tilde{y}]\}$. With $(0.98)(0.9) = 0.882$ and $(0.98)(1.1) = 1.078$: $\ln 0.882 = -0.1255632$, $\ln 1.078 = 0.0751075$, $E = \frac{1}{2}(-0.1255632 + 0.0751075) = -0.0252279$, $9E = -0.2270509$ (from unrounded logarithms), so

$$V^d(0.9) = -0.1255632 - 0.2270509 = -0.3526141,$$
$$V^d(1.1) = 0.0751075 - 0.2270509 = -0.1519434.$$

Answer (a). $V^d(y_L) = -0.352614$, $V^d(y_H) = -0.151943$.

Step 2 — (b) Servicing the debt forever. Enter with $B = -b$, repay, and buy $B' = -b$ at $q(-b) = 1/(1+r)$:

$$c = y + B - q(B')B' = y - b + \frac{b}{1+r} = y - \frac{(1+r)b - b}{1+r} = y - \frac{rb}{1+r} = y - s.$$

Since this choice is feasible and $V^o \geq V^c$,

$$V^c(-b, y) \geq \ln(y-s) + \beta E\big[V^o(-b, \tilde{y}')\big] \;\geq\; \ln(y-s) + \beta E\big[V^c(-b, \tilde{y}')\big]. \tag{133}$$

Apply (133) to $V^c(-b, \tilde{y}')$ inside the expectation, and again T times (i.i.d.: each future period has the same distribution):

$$V^c(-b, y) \geq \ln(y-s) + \sum_{t=1}^{T} \beta^t E[\ln(\tilde{y} - s)] + \beta^{T+1} E\big[V^c(-b, \tilde{y}_{T+1})\big].$$

Values are bounded (consumption lies in a bounded set), so the last term vanishes as $T \to \infty$, and $\sum_{t \geq 1} \beta^t = \beta/(1-\beta)$:

$$V^c(-b, y) \geq \ln(y-s) + \frac{\beta}{1-\beta} E[\ln(\tilde{y} - s)] = W(b, y). \tag{134}$$

If $W(b, y) \geq V^d(y)$ for $y = y_L, y_H$, then $V^c(-b, y) \geq V^d(y)$, so $V^d > V^c$ never holds: $D(-b) = \emptyset$, $\delta(-b) = 0$, and slide 18 gives $q(-b) = 1/(1+r)$ — the price assumed.

Answer (b). $c = y - rb/(1+r)$; $V^c(-b, y) \geq W(b, y)$; $W \geq V^d$ in both states $\Rightarrow D(-b) = \emptyset \Rightarrow$ $q(-b) = 1/(1+r)$: the conjecture confirms itself.

Step 3 — (c) Only y_L matters; nothing is certified without output costs. Subtract V^d term by term, writing $\ln(y-s) - \ln[(1-\lambda)y] = \ln(1 - s/y) - \ln(1-\lambda)$:

$$\Delta(s, y) = \ln\Big(1 - \frac{s}{y}\Big) - \ln(1-\lambda) + K\, E\Big[\ln\Big(1 - \frac{s}{\tilde{y}}\Big) - \ln(1-\lambda)\Big]. \tag{135}$$

Only the first term depends on current y, and

$$\frac{\partial}{\partial y} \ln\Big(1 - \frac{s}{y}\Big) = \frac{s/y^2}{1 - s/y} = \frac{s}{y(y-s)} > 0 \quad (0 < s < y),$$

so $\Delta(s, y_H) > \Delta(s, y_L)$: if the test passes at y_L it passes at y_H. With $\lambda = 0$ the $\ln(1-\lambda)$ terms vanish and, for $s > 0$, $\ln(1 - s/y) < 0$ in every state, so $\Delta(s, y) < 0$: servicing any debt forever is worse than autarky. Link to Exercise 16: with $h(y) = y$, a constant net outflow s buys nothing that autarky lacks; whatever debt is sustained must rest on the value of smoothing (varying B' with y), which this bound ignores.

Answer (c). $\partial\Delta/\partial y = s/[y(y-s)] > 0$, so test only at y_L. With $\lambda = 0$, $\Delta(s, y) < 0$ for all $s > 0$: the bound certifies no debt.

Step 4 — (d) The threshold by Newton's method. At y_L, (135) with $K = 9$ and probabilities $\frac{1}{2}$ collects to $(1 + \frac{K}{2}) \ln(1 - s/0.9) + \frac{K}{2} \ln(1 - s/1.1) - (1+K)\ln(0.98)$:

$$G(s) = 5.5 \ln\Big(1 - \frac{s}{0.9}\Big) + 4.5 \ln\Big(1 - \frac{s}{1.1}\Big) - 10 \ln 0.98, \tag{136}$$
$$G'(s) = -\frac{5.5}{0.9 - s} - \frac{4.5}{1.1 - s} < 0.$$

G is decreasing, so $\{s : G(s) \geq 0\} = [0, \bar{s}]$ with $G(\bar{s}) = 0$. From $s_0 = 0$: $G(0) = -10 \ln 0.98 = -10(-0.0202027) = 0.202027$ and $G'(0) = -5.5/0.9 - 4.5/1.1 = -6.111111 - 4.090909 = -10.202020$,

so

$$s_1 = 0 - \frac{0.202027}{-10.202020} = 0.019803,$$
$$s_2 = s_1 - \frac{G(s_1)}{G'(s_1)} = 0.019803 - \frac{-0.002089}{-10.414504} = 0.019803 - 0.000201 = 0.019602,$$
$$s_3 = s_2 - \frac{G(s_2)}{G'(s_2)} = 0.019602 - \frac{-0.0000002}{-10.412307} = 0.019602.$$

Six decimals are stable: $\bar{s} = 0.019602$. Signs: $G(0.019) = 5.5(-0.0213371) + 4.5(-0.0174236) + 0.202027 = 0.006266 > 0$ and $G(0.020) = -0.004145 < 0$, bracketing $\bar{s}$. Then

$$\bar{b} = \bar{s}\,\frac{1+r}{r} = 0.019602 \times \frac{1.05}{0.05} = 0.019602 \times 21 = 0.411642,$$

about 41% of mean output ($E[y] = 1$).

Answer (d). $\bar{s} = 0.019602$ (Newton: $0.019803 \to 0.019602 \to 0.019602$); $G(0.019) > 0 > G(0.020)$; $\bar{b} = 0.411642 \approx 41\%$ of mean output.

Step 5 — (e)(i) A lower bound. (134) is an inequality: the country can do better than constant service (borrow more after y_L, repay after y_H), so $V^c(-b, y)$ may exceed $V^d(y)$ even when $W < V^d$. Every $b \le \bar{b}$ passes the test, so $D(-b) = \emptyset$ for all $b \le \bar{b}$ and $\bar{B} \le -\bar{b}$, i.e. $-\bar{B} \ge \bar{b}$. Grid: $-\bar{B} = 0.42 \ge 0.411642$, a gap of $0.42 - 0.411642 = 0.008358$: the bound is within 2% of the grid value.

Step 6 — (e)(ii) Sensitivity to the output cost. Write (136) with general λ: $G(s, \lambda) = 5.5\ln(1 - s/0.9) + 4.5\ln(1 - s/1.1) - 10\ln(1 - \lambda)$. Then $G_\lambda = 10/(1-\lambda) = 10/0.98 = 10.204082$ and $G_s = G'(\bar{s}) = -10.412307$, so

$$\frac{d\bar{s}}{d\lambda} = -\frac{G_\lambda}{G_s} = \frac{10.204082}{10.412307} = 0.980002, \qquad \frac{d\bar{b}}{d\lambda} = 21 \times 0.980002 = 20.580043.$$

One more percentage point of output cost raises the certified debt by about 0.21, a fifth of mean output.

Step 7 — (e)(iii) Output costs and re-entry. Output cost: moving λ from 0 to 0.02 moves the grid limit $-\bar{B}$ from 0.01 to 0.42. Exclusion alone sustains almost nothing here, because losing access to one-period non-contingent bonds costs little when shocks are ±10%; a permanent output loss is a large, certain punishment and sustains most of the debt. Re-entry: moving θ from 0 to 0.5 (with $\lambda = 0.02$) moves $-\bar{B}$ from 0.42 to 0.0375. With $\theta > 0$, V^d puts weight θ on $V^o(0, \cdot) \ge V^d(\cdot)$, so (holding $V^o(0, \cdot)$ fixed) V^d rises with θ; the punishment lasts $1/\theta = 2$ periods on average instead of forever, default sets grow (Exercise 16(f)), and risk-free capacity collapses. This matches Tirole's summary (§3.8, pp. 143–144): if a defaulter could also save, Bulow–Rogoff show no lending is feasible; outright exclusion, under which the defaulter cannot save (autarky here), is what makes some debt possible — and costs of default beyond exclusion make it larger.

Answer (e). (i) $\bar{b} = 0.411642 \le -\bar{B} = 0.42$: the bound holds and is nearly tight. (ii) $d\bar{s}/d\lambda = 0.980002$, $d\bar{b}/d\lambda = 20.580043$. (iii) Output costs raise capacity ($0.01 \to 0.42$ as λ goes $0 \to 0.02$); re-entry lowers it ($0.42 \to 0.0375$ as θ goes $0 \to 0.5$), because default becomes cheaper.

Intuition. A sovereign's debt is worth only what it would lose by walking away. Here that loss has two parts: the output cost, paid every period of autarky, and the lost ability to smooth. "Service the debt forever" prices the first part exactly and ignores the second, which is why it is a lower bound — and a close one, since smoothing is worth little with small i.i.d. shocks. Anything that shortens or softens the punishment (re-entry, saving abroad) shrinks the debt the country can credibly owe.

Common mistakes.

- Using $s = rb$: rolling over raises only $b/(1+r)$ against b repaid, so the net service is $b - b/(1+r) = rb/(1+r)$.
- Testing only y_H, or averaging: the default incentive is strongest at y_L (concavity), so that is where the test binds.
- Reading $\bar{b}$ as the exact risk-free limit: it is a sufficient condition; the full model's limit is weakly larger.

Ref: Prof. Ricca, Topic 5 slides 17–22 (model), 23–24 (Claims 1–2, $\underline{B}$, $\bar{B}$), 29 (risk-free region). Tirole §3.8 "Nonverifiable Income", pp. 143–144 (PDF 156–157). Original exercise; grid solution in the check script (`e17solve`).